BIOLOGY GRADE 10: ANIMAL TRANSPORT

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 3.0: Transport, Gaseous Exchange and Respiration
Sub-Strand	Sub-Strand 3.2: Animal Transport
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– The importance of transport in animals – Structure of the transport system in insects (open circulatory system) – Structure of the transport system in fish (single circulatory system) – Structure of the transport system in amphibians and reptiles (partially double circulatory system) – Structure of the transport system in mammals (double circulatory system) – The pumping mechanism of the mammalian heart (cardiac cycle, heart sounds, pulse rate) – The human lymphatic system and its role in immunity – The immune system and blood clotting mechanism – The ABO blood grouping system in humans – The rhesus factor blood grouping system in humans – The diversity of transport systems in animals (comparative overview) – Project: Learners present final explanations on animal transport systems
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the importance of transport in animals, b) describe the structure of the transport system in insects, fish, amphibians, reptiles and mammals, c) explain the pumping mechanism of the mammalian heart, d) describe the human lymphatic and immune systems and the blood clotting mechanism, e) explain the ABO and rhesus factor blood grouping systems in humans, f) compare the diversity of transport systems in animals, g) appreciate the significance of a functional transport system in maintaining life in animals.
Core Competencies	– Communication and Collaboration: Learners work in groups to investigate and present findings on the diversity of animal transport systems, debating similarities and differences across species such as the tilapia (fish) found in Lake Victoria and the domestic cow. – Critical Thinking and Problem Solving: Learners analyse how structural differences in the hearts of insects, fish, amphibians and mammals relate to each organism's metabolic demands, constructing reasoned arguments supported by evidence. – Imagination and Creativity: Learners design and revise cumulative models (diagrams, flowcharts) of animal circulatory

	systems, creatively representing open vs. closed and single vs. double circuits. – Self-Efficacy: Learners take ownership of building their Driving Question Board and tracking their own conceptual growth across all 12 lessons. – Digital Literacy: Learners use available simulations and videos (where internet is accessible) to observe heart pumping mechanisms and blood flow in real time, evaluating the reliability of digital sources.
Core Values	– Responsibility: Learners handle dissection materials and biological specimens (e.g., a sheep or goat heart obtained from a local Nairobi or Kisumu butchery) with care, following safety protocols and disposing of materials appropriately. – Respect: Learners appreciate diverse cultural views on blood (e.g., blood taboos in some Kenyan communities) during discussions on blood grouping and transfusion, listening without ridicule. – Unity: Learners from different ethnic backgrounds collaborate during group investigations, recognising that the biological principles of transport are universal across all human beings regardless of community or blood group. – Integrity: Learners record experimental observations and data (pulse rate, heart sounds) honestly and accurately, without fabricating results.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate and refine questions for the Driving Question Board in Lesson 1 after observing the anchoring phenomenon of a Kenyan long-distance runner's post-race cardiovascular response. – Planning and Carrying Out Investigations: Learners design a simple investigation to measure resting vs. post-exercise pulse rates using a stopwatch, identifying variables and recording data in tables. – Analysing and Interpreting Data: Learners compare pulse rate data across classmates and relate variation to fitness levels and the efficiency of the double circulatory system. – Developing and Using Models: Learners iteratively build and revise diagrams of circulatory systems across animal groups (insect → fish → amphibian → mammal), annotating each model with evidence gathered per lesson. – Constructing Explanations: Learners construct written and oral explanations of why a marathon runner from Iten in the Rift Valley needs an efficient double circulatory system, using the cardiac cycle and blood composition as evidence. – Obtaining, Evaluating and Communicating Information: Learners read and evaluate information from textbooks, charts and (where available) digital sources on lymphatic and immune system function, then present findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners explore blood grouping, blood transfusion compatibility and the immune system, reinforcing the importance of knowing one's blood group — a critical health literacy skill for Kenyan youth. – Safety and Security: Learners follow strict safety protocols when handling a real mammalian heart (sheep/goat) during dissection, including use of gloves and proper disposal of biological waste. – Environmental Conservation: Learners are guided to use locally and sustainably sourced specimens (e.g., hearts from animals already slaughtered at local butcheries) rather than harvesting live animals, minimising environmental impact. – Life Skills — Decision Making: Discussions on blood donation and the National Blood Transfusion Service of Kenya develop learners' ability to make informed civic health decisions. – Citizenship: Learners appreciate the role of the Kenya National Blood Transfusion Service and discuss why voluntary blood donation is a civic responsibility.
Career Connections	– Medicine (Doctor/Surgeon): Understanding the mammalian heart's pumping mechanism, cardiac cycle and blood grouping directly underpins careers in human medicine and surgery at hospitals such as Kenyatta National Hospital. – Nursing and Clinical Medicine: Knowledge of the lymphatic system, immune response and blood clotting is foundational for nurses and clinical officers managing wound care, transfusions and infections across Kenyan health facilities. – Veterinary Science: Comparative study of transport systems in insects, fish, amphibians and mammals prepares learners for veterinary careers, particularly relevant in Kenya's large livestock and fisheries sectors. – Biomedical Science and Laboratory Technology: Blood grouping (ABO and Rhesus factor) and haematology skills connect to

	careers in blood bank management and clinical laboratory work. – Sports Science and Physiotherapy: Analysis of heart rate, cardiac output and circulatory efficiency relates to sports science careers supporting Kenya's elite athletics community in Iten and Eldoret. – Marine and Freshwater Biology: Study of fish circulatory systems connects to careers in aquaculture and fisheries management at Lake Victoria and other Kenyan water bodies. – Entomology: Understanding the open circulatory system of insects supports careers in pest management and research, critical for Kenya's agricultural sector.
Focus for Lessons	This unit investigates how animals — from the smallest insects to large mammals like cattle on Kenya's Maasai Mara rangelands — move nutrients, gases, hormones and waste products through their bodies, and why the complexity of these systems matches each animal's lifestyle and metabolic needs. Using a Kenyan long-distance runner's extraordinary cardiovascular performance as an anchoring phenomenon, learners move from simple open circulatory systems in insects to the sophisticated double circulatory system of mammals, building cumulative models throughout. The unit integrates hands-on investigation of a real mammalian heart, pulse rate experiments, and collaborative sense-making to develop both deep conceptual understanding and key competencies of critical thinking, communication and self-efficacy.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How does the structure of an animal's transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper? KICD KEY INQUIRY QUESTIONS: 1. Why is transport important in animals? 2. How does the structure of the transport system relate to the complexity of an animal?
Anchoring Phenomenon	ANCHORING PHENOMENON: A video clip (or teacher-narrated account) shows Kenyan elite marathon runner Eliud Kipchoge crossing the finish line of a major race. Moments after finishing, a medical team measures his heart rate — it drops rapidly from over 170 beats per minute back to near his resting rate of approximately 37 beats per minute within minutes. A classmate who attempted to jog just 400 m around the school field took much longer to recover. Learners are shown two sets of data side by side: Kipchoge's recovery heart-rate curve and their own. The puzzle: Why does Kipchoge's heart — a single fist-sized organ — manage to pump enough oxygenated blood to sustain world-record speeds AND recover so fast? And how is it that a grasshopper on the school fence, a tilapia in Lake Victoria, and a frog in the Rift Valley wetlands all survive without anything resembling Kipchoge's heart? This phenomenon is puzzling because it forces learners to confront the question of WHY transport systems look so different across the animal kingdom, and what features of the mammalian heart make it so powerful and efficient.
Storyline Thread	Lesson 1: ANCHORING PHENOMENON + PREDICT PHASE — Learners observe Kipchoge's heart-rate recovery data vs. their own; generate initial predictions and populate the Driving Question Board with "What we notice / What we wonder" cards; teacher introduces the driving question. Lesson 2: IMPORTANCE OF TRANSPORT IN ANIMALS — Learners investigate why large, complex animals cannot rely on diffusion alone; relate surface-area-to-volume ratio to the need for a transport system; connect back to DQB. Lesson 3: TRANSPORT IN INSECTS — Learners examine the open circulatory system of the grasshopper (dissection or diagram); observe haemolymph and the dorsal vessel; update cumulative model; add evidence to DQB. Lesson 4: TRANSPORT IN FISH — Learners study the single circulatory system of the tilapia (Lake Victoria context); trace blood flow through the two-chambered heart; compare to insect system; revise model. Lesson 5: TRANSPORT IN AMPHIBIANS AND REPTILES — Learners compare the three-chambered heart of the frog (Rift Valley wetlands context) and partially divided ventricle of reptiles; analyse the trade-offs of mixed oxygenated/deoxygenated blood; update model. Lesson 6: TRANSPORT IN MAMMALS — OBSERVE PHASE (Heart Dissection) — Learners examine a sheep/goat heart from a

	local butchery; identify four chambers, valves, major vessels; map the pulmonary and systemic circuits; make the biggest model revision yet. Lesson 7: PUMPING MECHANISM OF THE MAMMALIAN HEART — Learners investigate the cardiac cycle (systole/diastole), heart sounds, and the role of the sino-atrial node; conduct a pulse-rate investigation (resting vs. post-exercise); connect to Kipchoge phenomenon. Lesson 8: THE LYMPHATIC SYSTEM — Learners describe the structure and function of the lymphatic system; trace lymph formation and return to blood; relate to oedema and its relevance to Kenyan clinical health contexts. Lesson 9: THE IMMUNE SYSTEM AND BLOOD CLOTTING — Learners investigate specific and non-specific immunity; trace the blood clotting cascade; discuss relevance to wound management and disease (e.g., malaria impact on blood in Kenyan lakeside communities). Lesson 10: ABO AND RHESUS FACTOR BLOOD GROUPING — Learners simulate ABO blood typing using coloured solutions; interpret Kenyan blood group distribution data; discuss transfusion compatibility and the role of the Kenya National Blood Transfusion Service. Lesson 11: DIVERSITY OF TRANSPORT SYSTEMS — EXPLAIN PHASE (Comparative Overview) — Learners construct a full comparative table and annotated model spanning insect → fish → amphibian → reptile → mammal; articulate evolutionary trends in circulatory complexity. Lesson 12: MODEL BUILDING + FINAL PRESENTATIONS — Learners present their final cumulative model and written explanation answering the driving question; class revisits the DQB to confirm which questions are now answered and which remain open; teacher facilitates celebratory sense-making discussion linking back to Kipchoge and the anchoring phenomenon.
Supporting Phenomena	– A sheep or goat heart purchased from a local butchery in Nairobi or Kisumu is placed on the demonstration bench. Learners observe its size, chambers, valves and connecting vessels — they are puzzled by how this muscular organ works as a pump and why it has four chambers when a fish heart has only two. – A live grasshopper (collected from the school compound) is carefully observed under a hand lens; learners note there are no visible blood vessels on its transparent abdomen segments, yet it moves vigorously — this raises the question of how its cells receive nutrients without a closed vessel system. – Learners observe a short video or teacher demonstration showing blood not clotting immediately when a small cut is made on a finger, yet clotting within minutes — puzzling because the blood was liquid a moment ago, raising questions about what triggers this transformation and how it protects the body. – A chart showing ABO and Rhesus blood group distribution statistics among Kenyan population groups is displayed; learners are puzzled by why a person with blood group A cannot receive group B blood, and why this incompatibility can be life-threatening during transfusion.

LESSON 1 (40 minutes): Anchoring Phenomenon: Why Does Kipchoge Recover So Fast?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners launch the sub-strand inquiry by engaging with a puzzling real-world phenomenon involving Kenyan marathon champion Eliud Kipchoge, generating initial predictions about why transport systems differ across animals, and co-creating the Driving Question Board that will anchor all twelve lessons.
Knowledge	Learners will be able to state that animals need transport systems to move substances around their bodies, and recognise that the heart rate data of Kipchoge compared to a classmate raises questions about the efficiency and structure of the mammalian transport system.
Skills	Learners will be able to observe and interpret a heart-rate recovery graph, record initial noticings and wonderings, and formulate a testable prediction about why transport systems look different in different animals.
Attitudes	Learners will demonstrate curiosity and open-mindedness by suspending judgement, valuing the questions of peers, and embracing uncertainty as the starting point of scientific inquiry.
Key Inquiry Question	Why is transport important in animals?
Purpose in Storyline	This is the opening lesson that launches the entire twelve-lesson storyline. Its role is to create cognitive dissonance — learners see data they cannot yet fully explain — and to open the Driving Question Board so that every subsequent lesson is framed as an attempt to answer questions the class itself has generated.
Safety Notes	No hazardous materials are used in this lesson. Learners who find discussion of heart conditions or physical exertion personally sensitive should be given space to participate at a level they are comfortable with. If heart-rate data is collected live, learners with known cardiac conditions should be allowed to use resting data only.

B. LESSON OVERVIEW

Learners begin by watching or listening to a teacher-narrated account of Eliud Kipchoge finishing a major marathon. They are then shown two side-by-side heart-rate recovery curves — Kipchoge's and a typical student's — displayed on the board or as a printed handout. Through structured noticing, wondering, and initial predicting, learners articulate what puzzles them. The teacher introduces the driving question and facilitates the class in co-creating a Driving Question Board using sticky notes or cards under two columns: 'What we notice' and 'What we wonder.' Learners produce a first rough sketch of what they think a transport system might look like, which becomes Iteration 1 of their cumulative model. No answers are given; the lesson ends with productive uncertainty and genuine curiosity.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Circulatory System Structure and Function -	Learners are given a half-page data sheet showing two heart-rate recovery curves side by side. Curve A is labelled 'Elite marathon runner, Berlin 2022' and shows	Teacher distributes the data sheet and says: 'Before I tell you anything, I want you to look at this data the way a scientist would — with fresh eyes. Do not worry	Think-Write-Pair-Share followed by public recording of raw learner predictions. By writing first, every learner commits to an idea before hearing peers, reducing anchoring	Teacher scans written predictions during circulation to identify the range of prior knowledge. Specifically, teacher notes whether any learner spontaneously uses

Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	heart rate dropping from 172 bpm to 38 bpm within approximately 4 minutes. Curve B is labelled 'Average healthy teenager after jogging 400 m' and shows heart rate dropping from 165 bpm to 88 bpm over 15 minutes. Learners spend 3 minutes in silence writing in their exercise books: 'I notice...' (at least two observations) and 'I predict that the difference is caused by...' (one full sentence prediction). They then share predictions with a partner for 2 minutes.	about being right. Write what you genuinely notice and then make your best prediction about why the two curves are so different.' Teacher circulates silently for 3 minutes. After pair-share, teacher cold-calls three or four learners using a name stick. For each response, teacher says: 'Thank you. Can you say more about what in the data made you think that?' WAIT TIME: After each learner responds, teacher waits a full 5 seconds before commenting or calling on another learner, allowing processing time. Teacher does NOT confirm or deny any prediction. Teacher records key prediction words on the board verbatim: for example, 'bigger heart,' 'more practice,' 'stronger muscles,' 'better blood,' 'different lungs.' Teacher says: 'Hold these ideas — we are going to test every single one of them over the next twelve lessons.'	bias and ensuring all voices have a starting point.	terms such as 'circulation,' 'oxygen delivery,' or 'cardiac efficiency' — these learners can be used as intellectual resources in later lessons. Learners who write only surface observations ('Kipchoge is fitter') are identified for targeted probing in Lesson 2.
Observe Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher narrates or plays an audio-visual account of Kipchoge finishing the 2018 Berlin Marathon (or the 2022 race) in world-record time. If a video is not available, the teacher reads a short vivid account: 'Imagine the final 200 metres on Unter den Linden. Kipchoge, a quiet man from Nandi County in Kenya, crosses the line. The crowd erupts. A medical officer presses a pulse oximeter to his finger. Within 90 seconds his heart rate is already below 100 bpm. Four minutes later — 38 bpm. A resting rate most adults never achieve even in deep sleep.' Teacher then introduces the second data point: 'Last week, Omondi from Form 3 jogged once around our school field — 400	Teacher narrates with deliberate pacing, pausing after mentioning each animal. After the grasshopper-tilapia-frog-goat sequence, teacher says: 'Turn to your neighbour and answer this in ten seconds: Does having more heart chambers mean being a better animal?' WAIT TIME: After posing the question, teacher waits 8 seconds before asking for responses — this longer wait signals that the question is genuinely hard. Teacher collects three to four responses, records them on the board without evaluating them, and says: 'These are exactly the right kinds of questions. Science starts with not knowing.' Teacher then holds up a diagram showing a grasshopper, a	Narrative anchoring — the teacher uses a vivid, Kenya-specific story to make the phenomenon emotionally real before introducing comparative data. Learners are then prompted to extend their observations from one animal (human) to four animals, beginning the comparative thinking that will drive the entire sub-strand.	Teacher listens for whether learners connect the observation to the concept of substance movement inside the body. If no learner mentions oxygen, nutrients, or waste removal, teacher notes this as a gap to address explicitly in Lesson 2. Teacher mentally tracks which learners made connections across animals (higher-order) vs. those who focused only on Kipchoge (surface level).

	metres. His heart rate was still at 90 bpm fifteen minutes later.' Learners observe both curves again and spend 2 minutes adding new observations to their 'I notice' list now that they have the narrative context. Teacher then introduces the grasshopper on the fence: 'Right now, there is probably a grasshopper on the fence outside this room. It has no heart like Kipchoge's. A tilapia in Lake Victoria has a two-chambered heart. A frog in the Rift Valley wetlands has three chambers. A goat at Gikomba market has four chambers — just like us. Why? What is going on?'	tilapia, a frog, and a human silhouette side by side and asks: 'What do you predict all four of these animals have in common that keeps them alive?' Teacher waits 5 seconds.		
Explain Phase 📹 **VIDEO:** [Circulatory System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In this opening lesson the Explain phase is intentionally brief and exploratory — it is not about delivering answers but about surfacing what learners already think they can explain. Learners are given a blank box in their exercise book labelled 'My current explanation (Iteration 1)' and are asked to write two to four sentences answering: 'Why do you think animals need a transport system, and why might the transport system of a grasshopper look different from Kipchoge's?' Learners write individually for 3 minutes. Teacher then asks for two volunteers to read their explanation aloud. Teacher responds to each with: 'Interesting — what evidence from the data supports that idea?' This models the habit of linking claims to evidence from the very first lesson.	Teacher frames the task explicitly: 'This explanation is not being marked. It is a scientific starting point. In twelve lessons you will look back at what you wrote today and see how your thinking has changed — that change IS the learning.' After volunteers share, teacher says: 'Notice that two people just gave two different explanations for the same data. That is normal. That is science. Our job this term is to find out which explanation the evidence best supports.' Teacher does not correct misconceptions at this stage — instead, teacher tags them publicly on the board as 'Claims to investigate' and says: 'We will come back to each of these.' WAIT TIME: When asking 'What evidence supports that idea?' teacher waits a full 6 seconds before offering any scaffold.	Individual written explanation followed by public sharing creates a baseline record of learner thinking. Tagging claims publicly on the board transforms potential embarrassment about wrong ideas into collective intellectual resources — the class now has a shared list of things to investigate.	Teacher collects the exercise books of four to five learners at the end of the lesson (rotating across the class across lessons) to read Iteration 1 explanations and use them to calibrate the depth needed in Lesson 2. Teacher looks specifically for: (1) any use of the word 'diffusion' — indicating prior knowledge from Junior Secondary; (2) any mention of cell needs for oxygen or glucose; (3) any comparative thinking across animals.
Driving Question Board (DQB)	Each learner receives two sticky notes (or small cards cut from scrap paper). On the yellow card	Teacher says: 'This board belongs to all of us. Every lesson from now until Lesson 12, we will come back	The DQB externalises curiosity and makes it a shared class resource. By having learners	Teacher photographs the DQB at the end of Lesson 1. The spread and sophistication of wonder-

Creation 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	they write one thing they noticed from the data or the narrative. On the blue card they write one question they genuinely wonder about. Learners approach the board in small groups and place their cards in the correct column under a large display titled: 'Our Driving Question Board — Animal Transport.' The teacher reads the driving question aloud and posts it at the top of the board: 'How does the structure of an animal transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper?' Learners spend 3 minutes reading each other's cards on the board and are invited to place a small tick on any wonder-question that matches their own.	here. Some of your questions will get answered. Some will lead to new questions. A few might never be fully answered — and that is okay, because that is how science actually works.' Teacher reads three to four of the wonder-questions aloud and maps them to future lessons without revealing answers: 'This question — why does the grasshopper not have a pump like ours — we will answer in Lesson 3. This one — what makes Kipchoge's heart so efficient — we will answer in Lesson 7. This one — can a person survive with a hole in their heart — that is a question that connects to Lesson 6.' Teacher then asks: 'Which question on this board do YOU most want answered?' Learners call out or show thumbs. WAIT TIME: After asking which question learners most want answered, teacher allows 4 seconds of silence before taking responses, signalling that personal reflection is valued.	physically place and read each other's cards, the teacher creates a social contract — these are OUR questions and WE will answer them together. Mapping wonder-questions to specific future lessons gives learners a sense of the learning journey without pre-emptying discovery.	questions reveals the class's prior knowledge landscape. Questions that reveal strong prior knowledge (for example, 'Is the lymphatic system part of the circulatory system?') indicate learners who can be stretched. Questions that reveal confusion about basic biology (for example, 'Does blood actually carry food?') reveal learners who will need additional scaffolding in Lesson 2.
Model Building Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, learners open to a new page in their exercise book headed 'My Transport System Model — Iteration 1.' They draw a quick sketch (no more than 2 minutes) answering: 'Draw what you think happens inside an animal to move substances from one place to another.' There is no right answer. Some learners will draw a heart and blood vessels. Some will draw arrows. Some will draw a tube. All are valid starting points. Learners label their sketch with at least two words. Teacher collects a class photograph or has learners share sketches with their tablemates. Teacher says: 'Pin this in your	Teacher circulates and affirms effort, not correctness: 'I love that you drew that — tell me in one word what that arrow represents.' If a learner draws nothing and says they do not know, teacher says: 'Draw what you think might be happening. A guess is a great starting point. Scientists call it a hypothesis.' Teacher does not draw a model on the board during this phase — the point is to elicit, not to show. At the end of 40 minutes, teacher closes with: 'Today you did what every great scientist does at the start of an investigation: you looked carefully, you asked honest questions, and you made a brave first guess. Next	The initial model sketch creates a concrete, personal artefact that will be revised iteratively across all twelve lessons. By naming it 'Iteration 1' and promising a final version in Lesson 12, the teacher builds in metacognitive awareness of learning as a process of model revision rather than accumulation of facts.	Teacher scans sketches for evidence of prior knowledge: closed loops indicate familiarity with circulation; arrows from food to organs indicate understanding of nutrient transport from Junior Secondary. Sketches that show no internal structure (for example, a single dot labelled 'heart') indicate learners who may need a concrete physical anchor — the heart dissection in Lesson 6 will be particularly important for them.

	mind. In Lesson 12 you will draw this again — and the difference between those two drawings will show you everything you have learned.'	lesson we are going to investigate why big animals like us cannot survive on diffusion alone — and that will help us begin answering the questions on our board.' WAIT TIME: When circulating and asking what an arrow represents, teacher waits 5 seconds after the learner speaks before responding.		

D. TEACHER REFLECTION

After the lesson, the teacher should ask: (1) Did every learner produce a written prediction — if not, what barrier existed? (2) Were the wonder-questions on the DQB genuine questions or performed questions — how can I tell? (3) Did I resist the urge to correct misconceptions too early — if I did correct, what was the impact on learner willingness to share? (4) Which learners did not speak at all, and how will I draw them into Lesson 2? (5) Did the Kipchoge phenomenon feel real and relevant to learners, or did it feel like a teacher's story — if the latter, how can I bring in more local data, such as heart-rate data from a Kenyan school athletics meet?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed two heart-rate recovery curves — one from elite marathon runner Eliud Kipchoge and one from a typical teenager after jogging 400 m — and heard a comparative introduction to the transport systems of a grasshopper, a tilapia from Lake Victoria, a frog from the Rift Valley wetlands, and a mammal.
What did I learn?	Learners articulated initial noticings and predictions about why transport systems differ across animals, and began to connect the idea that different animals have different internal structures for moving substances around their bodies.
How does this explain the phenomenon?	Learners produced Iteration 1 of their personal transport system model and generated genuine wonder-questions for the Driving Question Board, establishing the inquiry questions that will be investigated across all twelve lessons of the sub-strand.

LESSON 2 (80 minutes): Why Can't Large Animals Just Use Diffusion? — The Importance of Transport Systems	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To help learners investigate why diffusion alone is insufficient for large, complex animals and to relate surface-area-to-volume (SA:V) ratio to the evolutionary need for a dedicated transport system.
Knowledge	By the end of this lesson, the learner should be able to: (a) explain why diffusion alone is sufficient for small/simple organisms but insufficient for large, complex animals; (b) describe how surface-area-to-volume ratio decreases as organism size increases; (c) state the importance of transport systems in animals — delivery of nutrients and oxygen, removal of waste products, transport of hormones, regulation of body temperature, and immune defence.
Skills	By the end of this lesson, the learner should be able to: (a) calculate and compare surface-area-to-volume ratios for cubes of different sizes; (b) collect, record, and interpret pulse-rate and breathing-rate data before and after exercise; (c) construct and annotate a line graph showing heart-rate recovery; (d) use evidence from data to construct a reasoned argument about transport system necessity.
Attitudes	The learner: (a) appreciates that biological structures are shaped by physical constraints, not chance; (b) shows curiosity in connecting mathematical ratios to living organisms; (c) demonstrates responsibility by accurately recording personal physiological data; (d) values collaborative sensemaking when constructing shared explanations.
Key Inquiry Question	1. Why is transport important in animals? 2. How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	In Lesson 1 learners anchored to the phenomenon of Kipchoge's extraordinary cardiac recovery and posed initial questions on the Driving Question Board. In Lesson 2 learners gather the first critical piece of evidence: that body size creates a diffusion problem that diffusion alone cannot solve. By measuring their own pulse and breathing rates before and after exercise, and by modelling SA:V ratios with agar cubes (or paper cubes), learners build the conceptual foundation for WHY a transport system must exist — connecting their personal physiological data back to Kipchoge and to simpler organisms like the grasshopper on the school fence.
Safety Notes	1. Learners with known cardiac or respiratory conditions should observe and record data rather than participate in the physical exercise component — inform the teacher before the lesson begins. 2. Exercise should be conducted in a ventilated space or outdoors; avoid exercising in direct midday sun. 3. If agar cubes are used, handle the dilute hydrochloric acid (or dilute vinegar alternative) with care; avoid contact with eyes; wash hands after handling. 4. No running near desks or in confined spaces — designate a safe exercise zone.

B. LESSON OVERVIEW

Learners begin by revisiting their Driving Question Board entries from Lesson 1 and making a new prediction: "Could Kipchoge survive if his body relied only on diffusion — no heart, no blood vessels?" They then conduct two evidence-gathering activities: (1) a hands-on SA:V ratio investigation using paper or agar cubes of three sizes to model how increasing body size reduces the efficiency of diffusion; and (2) a pulse-rate and breathing-rate measurement exercise (rest → 2-minute star jumps → recovery at 1 min and 5 min) that generates personal physiological data. In the Explain phase, learners use both datasets to construct a shared class argument: large, metabolically active animals like Kipchoge require a transport system because diffusion cannot move substances fast enough across large distances. The lesson closes with learners adding new evidence and revised questions to the DQB and updating their cumulative model of the animal transport system.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually read a short scenario card: 'Imagine Eliud Kipchoge's body has no heart, no blood vessels — oxygen and glucose must reach every muscle cell in his legs purely by diffusion from the skin surface. He is at the start line of the Nairobi Standard Chartered Marathon. What do you predict will happen when the gun fires?' Learners write a 3-sentence prediction in their science notebooks, then share with a shoulder partner. The class then receives a second prompt: a diagram showing a single-celled amoeba (≈ 0.1 mm) beside a scale outline of a human leg muscle cell cluster (≈ several cm deep from the skin). Learners predict: 'For which organism is diffusion sufficient, and why?' They record predictions before any instruction is given.	1. Display the scenario card on the board and READ it aloud. Say: 'Before I teach you anything new today, I want to know what YOU think. Write your prediction — three sentences — in your notebook. You have 90 seconds. Go.' [Set visible timer for 90 seconds.] 2. After writing, say: 'Turn to your shoulder partner. Partner A speaks first — share your prediction in 30 seconds. Then Partner B responds.' [Allow 60 seconds total.] 3. Cold-call FOUR pairs (8 learners) to share. Use names from the register. Say: 'Amina, tell me what you and your partner predicted.' After each response, ask: 'Does anyone predict something different? Raise your hand.' [WAIT 10 seconds.] 4. Do NOT confirm or correct predictions yet. Say: 'Hold those ideas — we are going to gather evidence today to test them.' Record 3–4 class predictions verbatim on a sticky note on the DQB under the column 'What we PREDICT.' 5. Display the amoeba-vs-muscle diagram. Ask: 'Which of these organisms do you think needs a transport system — and what is your reasoning?' [WAIT 15 seconds.] Cold-call THREE learners.	Think-Write-Pair-Share: Learners first commit predictions to writing (reducing anchoring bias toward the first voice heard), then share with a partner, then with the class. This surfaces prior conceptions about diffusion and transport before instruction, giving the teacher diagnostic information and giving learners a stake in the upcoming evidence.	OBSERVABLE INDICATOR: Each learner has a written 3-sentence prediction in their notebook before partner discussion begins. Teacher scans notebooks during cold-call phase. LOOK FOR: (a) learners who correctly identify that diffusion is too slow for large organisms — these learners can be used as peer explainers later; (b) learners who believe the heart is only about 'pumping hard' not about distance — a key misconception to address in the Explain phase. Red flag: learners who leave the prediction blank — re-prompt individually: 'Just write what your gut tells you — there is no wrong answer yet.'
Observe Phase 📹 **VIDEO:** Circulatory System Structure and Function -	ACTIVITY A — SA:V Ratio Modelling (12 minutes): In groups of four, learners receive three paper cubes (pre-cut nets) or three agar cubes of side lengths 1 cm, 2 cm, and 3 cm. (If agar cubes are	ACTIVITY A SETUP (3 minutes): Distribute cube nets and data tables. Say: 'Your job is to be mathematicians and biologists at the same time. Calculate SA:V for each cube. Use the formula:	Data Collection with Comparative Anchoring: Learners generate THEIR OWN data (pulse, breathing) and then immediately compare it to Kipchoge's published data displayed on the board. This	OBSERVABLE INDICATORS: — Activity A: Each group's data table has correct SA:V values (1 cm^3 cube → SA:V = 6; 2 cm cube → SA:V = 3; 3 cm cube → SA:V = 2). Groups that calculate correctly

Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	used and vinegar/acid is available, cubes are coloured with food dye or phenolphthalein and placed in dilute acid to model diffusion penetration.) For paper cubes: learners calculate surface area, volume, and SA:V ratio and record in a provided data table. They shade a cross-section diagram to show how far diffusion would reach in a set time. Groups record: 'As the cube gets bigger, what happens to the SA:V ratio? What does this mean for a cell trying to get oxygen?' ACTIVITY B — Pulse Rate & Breathing Rate Measurement (13 minutes): Individually, learners (a) find their pulse at the wrist or neck; (b) count beats for 30 seconds and multiply by 2 to get BPM at REST; (c) perform 2 minutes of star jumps in a designated safe space; (d) immediately count pulse for 30 seconds (IMMEDIATELY AFTER); (e) wait 5 minutes and count again (RECOVERY). Learners also count breaths per minute at rest and immediately after exercise. All data is recorded in a structured data table. Learners then plot a simple line graph of pulse rate (y-axis) against time — rest, immediately after, 5 min recovery (x-axis) — and compare their curve shape to Kipchoge's published recovery curve displayed on the board.	Surface Area = 6 × side^2. Volume = side^3. SA:V = Surface Area ÷ Volume. Record every step. You have 10 minutes.' [Circulate. Listen for groups saying 'SA:V gets smaller' — affirm this without over-explaining: 'Interesting — remember that observation.'] Prompt struggling groups: 'Start with the smallest cube. What is the side length? What is 1 squared? What is 6 times that?' Do NOT give answers — scaffold the process. ACTIVITY B SETUP (2 minutes): Say: 'Put your right index and middle fingers gently on the inside of your left wrist, just below the base of your thumb. Can everyone feel a pulse? [WAIT 10 seconds.] Good. If you cannot find it there, try the side of your neck — never press hard on the neck. We will count together for the REST measurement.' [Count aloud with class for 30 seconds.] 'Multiply by two. Write that number under REST in your data table.' Before exercise: 'Star jumps — in the marked zone only. Two minutes. Go.' [Use a timer visible to the class.] Immediately after: 'Stop! Find your pulse NOW. Count for 30 seconds.' [Start timer immediately.] After 5-minute rest: 'Count again.' [Timer.] For graph: Say: 'Plot your three data points and connect them. Then look at the Kipchoge curve on the board. Describe the shape of both curves in one sentence in your notebook.'	creates cognitive dissonance — 'Why does my curve look different from his?' — which drives the need to explain. The SA:V activity provides a physical/mathematical model that makes the abstract diffusion limitation concrete and visible. Together, the two activities give learners two types of evidence: mathematical modelling AND physiological measurement, both pointing toward the same claim.	should be able to state: 'As size increases, SA:V decreases.' Teacher checks at least four group tables during the activity. — Activity B: Every learner has three numeric entries in their pulse-rate data table (rest, after, recovery) and has attempted a line graph with labelled axes. LOOK FOR: learners whose recovery rate is very slow — use this as a positive teaching moment: 'Your heart is working hard — that is what we are investigating today.' Red flag: learners who record identical values for rest and post-exercise — prompt: 'Did you count from the moment you stopped moving? Let us redo together.'
Explain Phase	Learners return to whole-class	1. Open with: 'Let us build a	Claim–Evidence–Reasoning	OBSERVABLE INDICATORS:

📹 **VIDEO:** [Circulatory System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	discussion with their data from both activities. The teacher facilitates a structured argument-building discussion using the Claim–Evidence–Reasoning (CER) framework. Learners are guided to construct the following class claim on the board: 'Large, complex animals like Eliud Kipchoge NEED a dedicated transport system because diffusion alone cannot move substances fast enough across large distances inside the body.' Each group contributes evidence: SA:V data from Activity A, and pulse/breathing data from Activity B. Learners are then introduced to the formal list of functions of a transport system in animals: (1) delivery of oxygen and nutrients (glucose, amino acids, fatty acids) to all cells; (2) removal of metabolic waste products (CO_2, urea) from cells; (3) transport of hormones (e.g., adrenaline during Kipchoge's race); (4) thermoregulation — redistributing heat generated by working muscles; (5) immune defence — circulation of white blood cells. Learners annotate a body-outline diagram connecting each function to a specific Kipchoge marathon scenario (e.g., 'at km 30, his muscles release heat — blood redistributes it to prevent overheating on the Nairobi road'). Learners then revisit their Lesson 1 predictions and explicitly compare: 'What did I predict? What does the evidence show? What do I now think?'	shared explanation. I need a volunteer to read our class claim.' [Display CER frame on board.] 'Now, which group has SA:V evidence to support this claim? Give me a number — what was the SA:V of the 3 cm cube compared to the 1 cm cube?' [Cold-call one group. WAIT 10 seconds.] 2. Say: 'So as the cube — or the animal — gets bigger, what happens to the proportion of surface area available for diffusion?' [WAIT 12 seconds. Cold-call TWO learners who have not yet spoken today.] 3. Connect to phenomenon: 'Think about Kipchoge running at 2:01 per kilometre. His quadriceps muscles are deep inside his thigh — maybe 8 cm from the skin surface. If oxygen had to diffuse from his skin to that muscle cell, how long do you estimate that would take — seconds, minutes, hours?' [WAIT 15 seconds. Accept estimates — affirm the direction: 'Research shows diffusion over 1 mm takes roughly 1 second — over several centimetres it would take hours. Kipchoge cannot wait hours for oxygen.'] 4. Introduce the five transport functions using the board outline. For EACH function, ask: 'Can someone give me a specific Kipchoge marathon example for this function?' [Cold-call; total of five different learners for five functions.] 5. Say: 'Now compare your prediction from the start of the lesson to what the evidence shows. On the next line in your notebook, write: MY PREDICTION WAS ___. THE EVIDENCE	(CER) Framework: This structured argumentation strategy makes scientific reasoning explicit. Learners do not simply accept a textbook statement — they connect their own data (SA:V numbers, their own pulse rates) to a class-constructed claim, and provide reasoning that links evidence to claim. This mirrors actual scientific practice and builds the analytical skill of distinguishing observation from inference. The Kipchoge marathon annotation activity contextualises abstract functions in a compelling, Kenya-relevant narrative.	— Each learner has written a complete CER entry (claim, at least two pieces of evidence, one reasoning sentence) in their notebook. — During cold-calls, learners can state at least THREE of the five transport functions unprompted. — In the prediction-revision writing ('MY PREDICTION WAS / THE EVIDENCE SHOWS / I NOW THINK'), learners who previously believed diffusion was sufficient now write a revised explanation referencing SA:V ratio or diffusion distance. Red flag: Learners who write 'the heart pumps blood' as their entire reasoning — prompt: 'WHY does the blood need to be pumped rather than just diffusing? What does your SA:V data tell you?'

		SHOWS ___. I NOW THINK ___.' [Allow 3 minutes of silent writing.] 6. Close Explain phase: 'The grasshopper on the school fence has a body roughly 4 cm long. A tilapia in Lake Victoria is about 25 cm. Kipchoge is 167 cm tall. Given what we now know about SA:V — does it surprise you that their transport systems look completely different?' [WAIT 10 seconds. Take three responses.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or writes in two designated columns in their DQB notebook section). On sticky note 1 (EVIDENCE colour — e.g., green): they write one piece of evidence from today that helps answer 'Why is transport important in animals?' On sticky note 2 (NEW QUESTIONS colour — e.g., yellow): they write one new question that today's evidence raised — something they still do not understand or want to investigate further. Learners place sticky notes on the class DQB under the appropriate driving question. The teacher reads aloud five of the new questions and the class votes (show of hands) on which three questions feel most urgent to investigate. These three questions are circled on the DQB and will guide Lesson 3. The teacher draws a visible arrow on the DQB connecting today's evidence to the phenomenon: 'Kipchoge's heart drops to 37 BPM at rest — why does HIS transport system recover so fast? We have explained WHY a transport system is needed — next we investigate HOW different systems are structured.'	1. Distribute sticky notes. Say: 'You have 3 minutes. Evidence note: one sentence — what did today's data prove about why transport systems exist? Question note: one sentence — what are you still puzzled about?' [Set visible timer for 3 minutes. Circulate and read notes as they are written.] 2. Direct learners to the DQB: 'Place your green note under Driving Question 1. Place your yellow note under the NEW QUESTIONS column.' [Allow 2 minutes for placement.] 3. Read five yellow notes aloud (choose a mix of factual questions and conceptual puzzles). Say: 'Raise your hand if THIS question is also in your head.' Tally hands. Circle the top three. 4. Say: 'Notice that several of you are asking about HOW blood actually moves through different animals — whether a grasshopper has vessels like we do. That is exactly where we are going in the next lessons. Your questions are driving our investigation.' [Point to the phenomenon image of Kipchoge.] 'Everything we learn must help us answer: what makes HIS system so extraordinary?'	DQB Evidence Mapping with Prioritised Questions: The DQB is not a passive display — it is an active sensemaking tool. By requiring learners to categorise their sticky note as EVIDENCE (what we now know) vs. NEW QUESTION (what we still need to find out), learners practise the metacognitive skill of monitoring their own understanding. Voting on priority questions gives learners genuine agency in the inquiry sequence and increases motivation for subsequent lessons.	OBSERVABLE INDICATORS: — Every learner has placed both a green (evidence) and yellow (question) sticky note on the DQB. — Green notes should reference SA:V ratio, diffusion limitation, or a named transport function (not simply 'the heart is important'). — Yellow notes should be genuine scientific questions, not statements. Red flag: yellow notes that are observations ('diffusion is slow') rather than questions — prompt: 'Turn that into a question — what do you WANT to know?' — Teacher photographs the DQB at the end of this phase to track question evolution across all 12 lessons.

Model Building Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative model notebook page (begun in Lesson 1). They have 8 minutes to revise and annotate their model based on today's learning. Specifically, they must: (1) Add a label or annotation explaining WHY the transport system exists — connecting SA:V ratio to the need for delivery and removal of substances; (2) Add at least TWO arrows to their model — one showing delivery of oxygen/nutrients to a cell, one showing removal of CO_2/waste; (3) Write one sentence at the bottom of their model page: 'Without a transport system, Kipchoge's muscle cells would ________ because ________.' Learners who finish early are challenged to sketch a simple amoeba beside their model and annotate why the amoeba does NOT need a transport system, using today's SA:V evidence. In the final 2 minutes, two volunteers share their revised models under a visualiser or by holding them up, and the class offers one 'star' (something effective) and one 'arrow' (something to improve or add).	1. Say: 'Open your model page from Lesson 1. You are not drawing a new model — you are making your existing one smarter. Add the SA:V idea. Add delivery and removal arrows. Write the Kipchoge sentence at the bottom. You have 8 minutes — go.' [Set timer. Circulate actively — do not sit.] 2. While circulating, use these specific prompts: — 'Show me where diffusion is happening in your model. Now show me why that is not enough for Kipchoge.' — 'Your arrows — which direction is oxygen moving? Which direction is CO_2 moving? Label them.' — 'You have drawn a heart — good. Now write next to it: what is the heart SOLVING?' 3. After 8 minutes: 'I need two volunteers to share.' Select one strong model and one model that is partially correct (to normalise revision and growth). Facilitate peer feedback: 'One star — something this model shows really well. One arrow — one thing to add or clarify.' [WAIT 10 seconds between star and arrow nominations.] 4. Close: 'Every lesson, we will add to this model. By Lesson 12, your model will explain everything from a grasshopper's open system to Kipchoge's four-chambered heart. Today you have added the most fundamental layer: WHY a transport system must exist at all.'	Cumulative Model Revision (Iterative Modelling): Science understanding is built progressively, not all at once. By returning to the same model page each lesson and adding new explanatory layers, learners experience science as a process of refinement rather than a list of facts to memorise. The sentence stem 'Without a transport system, Kipchoge's muscle cells would ________ because ________' forces learners to use causal language — the hallmark of mechanistic scientific explanation.	OBSERVABLE INDICATORS: — Model shows at least two directional arrows (delivery in, waste out) with labels. — SA:V concept appears as an annotation (even if informally worded, e.g., 'big body = less surface for diffusion'). — Kipchoge sentence uses causal language ('because') and references a specific consequence (e.g., 'run out of oxygen,' 'fill up with CO_2,' 'overheat,' 'die'). — Exit indicator: Teacher asks three learners at the door as they leave: 'In one sentence — why does a large animal need a transport system?' Expected answer references either SA:V ratio or diffusion distance/speed. Record names of learners who cannot answer for targeted follow-up at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. SA:V ACTIVITY: Did all groups correctly calculate and interpret the SA:V ratios? Which groups needed the most scaffolding — and was it conceptual (what SA:V means) or procedural (the arithmetic)? How will you address this in Lesson 3?

2. EXERCISE DATA: Did learners' own pulse-rate data connect meaningfully to Kipchoge's recovery curve, or did the comparison feel abstract? If the personal data was too variable or poorly collected, consider introducing a class-average data table in Lesson 3 to strengthen the evidence base.

3. CER QUALITY: Review five learner notebooks. Can learners distinguish between evidence (SA:V numbers, BPM values) and reasoning (the SA:V ratio decreasing means diffusion is insufficient)? If reasoning sentences are missing or circular, plan a brief 5-minute CER mini-lesson at the start of Lesson 3.

4. DQB QUESTION QUALITY: Do the new questions posted on the DQB point forward toward structural differences between animal transport systems (the content of Lessons 3–8)? If most questions are still about Kipchoge's personal physiology rather than comparative animal transport, use the opening of Lesson 3 to explicitly bridge: 'Your questions about Kipchoge's heart lead us to ask — what does a heart even look like in other animals?'

5. INCLUSION CHECK: Were learners who did not participate in the exercise activity (health reasons) still fully engaged through observation and data recording? Adjust Activity B in future iterations to provide an equally rich observer role with a structured observation sheet.

6. PHENOMENON CONNECTION: Could every learner articulate — even approximately — why a grasshopper does not need what Kipchoge has? If not, the SA:V concept needs reinforcement before Lesson 3 introduces open vs. closed circulatory systems.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners measured and recorded their own pulse rate (BPM) and breathing rate at rest, immediately after 2 minutes of star jumps, and after 5 minutes of recovery. They also calculated the surface-area-to-volume (SA:V) ratio for cubes of three different sizes (1 cm, 2 cm, 3 cm sides) and compared their pulse-rate recovery curve to Eliud Kipchoge's published post-race heart-rate data displayed on the board.
What did I learn?	As the size of an organism increases, its surface-area-to-volume ratio decreases — meaning a smaller proportion of its body surface is available for diffusion relative to its volume. Diffusion alone is too slow and too limited in range to supply oxygen and nutrients to — or remove waste from — the deep cells of a large, complex animal. Large animals therefore require a dedicated transport system to: (1) deliver oxygen and nutrients to all cells; (2) remove metabolic wastes (CO_2, urea); (3) transport hormones; (4) regulate body temperature; and (5) circulate immune cells. These needs are dramatically amplified during exercise, as Kipchoge's and learners' own pulse-rate data demonstrate.
How does this explain the phenomenon?	Why Kipchoge cannot survive on diffusion alone: his body is far too large — muscle cells lie centimetres away from any surface, and diffusion over such distances would take hours, far too slow to sustain marathon pace. The data from the SA:V investigation showed that as organisms grow larger, the ratio of surface area to volume falls sharply. A single-celled amoeba (tiny, high SA:V) can exchange all substances directly through its cell membrane; Kipchoge's quadriceps muscles (deep, low SA:V from the skin) require a pumped circulatory system to bridge that distance in seconds. This is why EVERY large, complex animal — fish, amphibian, reptile, bird, mammal — has evolved some form of transport system, even though the structures differ enormously.

LESSON 3 (80 minutes): Transport in Insects: The Open Circulatory System of the Grasshopper	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the structure and function of the open circulatory system in insects, using the grasshopper as a specimen, and to compare it with the closed circulatory system hinted at through Kipchoge's recovery data in Lesson 1.
Knowledge	Learners will be able to: (a) describe the open circulatory system of an insect, including the dorsal vessel, ostia, haemocoel, and haemolymph; (b) explain how haemolymph circulates in an open system and why it does NOT carry oxygen to cells in insects; (c) distinguish between open and closed circulatory systems using structural evidence.
Skills	Learners will be able to: (a) observe and label a diagram or dissection of the grasshopper's transport system; (b) record and interpret observational evidence systematically; (c) construct an updated cumulative model linking insect transport to the anchoring phenomenon.
Attitudes	Learners will: (a) appreciate that structural diversity in transport systems reflects evolutionary adaptation to different environments and body sizes; (b) show curiosity about the relationship between body complexity and transport efficiency; (c) demonstrate care and responsibility when handling biological specimens.
Key Inquiry Question	How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	In Lessons 1 and 2, learners collected baseline evidence — Kipchoge's heart-rate recovery data and their own pulse/breathing-rate data from exercise. They know the mammalian heart is powerful, but they have not yet explained WHY. In Lesson 3, learners encounter the grasshopper — the insect sitting on the school fence in the anchoring phenomenon — and discover it has NO heart in the Kipchoge sense: no closed vessels, no red blood, no oxygen-carrying haemoglobin. This generates productive tension: if an insect survives without a 'real' circulatory system, what does a transport system actually need to do? This evidence is added to the Driving Question Board and the cumulative model, pushing learners closer to answering the driving question.
Safety Notes	If live or freshly killed grasshoppers are dissected: (1) Use blunt-ended dissection scissors and handle pins with care — teacher demonstrates correct technique before learners begin. (2) Learners with known insect allergies must inform the teacher; diagram-based alternatives are provided. (3) Wash hands thoroughly with soap and water after handling any specimen. (4) Dispose of biological material in a sealed bag in the designated waste bin — do NOT leave specimens on desks. (5) If using preserved specimens, ensure the room is ventilated to reduce preservative fume exposure.

B. LESSON OVERVIEW

Learners investigate the open circulatory system of the grasshopper through guided dissection or annotated diagram work. They discover that insect 'blood' (haemolymph) bathes organs directly in an open haemocoel, that the dorsal vessel acts as a simple pulsatile pump, and — critically — that haemolymph does NOT transport oxygen (insects use tracheae for gas exchange). This structural evidence is compared with Kipchoge's closed, high-pressure cardiovascular system from Lessons 1–2. Learners update their cumulative model and add new evidence cards to the Driving Question Board, deepening the central puzzle: why do different animals have such dramatically different transport systems?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a large image of a grasshopper (the same species commonly found on school fields and maize farms across Kenya — Schistocerca gregaria or a local equivalent). They are asked to sketch what they think is INSIDE the grasshopper's body that helps transport substances. Each learner writes a silent, individual prediction in their science notebook under the prompt: 'Draw and label what you think the grasshopper uses to move food, water, and oxygen around its body. Does it have a heart? Does it have blood? Explain your thinking.' After 4 minutes of silent writing, learners share predictions with a partner (Think-Pair-Share). Three pairs are cold-called to share — teacher records key prediction words on the board (e.g., 'heart,' 'blood,' 'tubes,' 'it just absorbs') without correcting them.	1. Display the grasshopper image. Say: 'This is the grasshopper sitting on the school fence in our Kipchoge puzzle. We know Kipchoge has a heart that pumps 37 beats per minute at rest. Before we look inside — predict: does this grasshopper have something like that?' 2. Allow 4 minutes of SILENT individual prediction. Do NOT answer questions during this time — redirect with: 'Write what you think right now — there is no wrong answer yet.' 3. After Think-Pair-Share (2 minutes), cold-call exactly 3 pairs. For each, ask: 'What did you and your partner agree on — and where did you disagree?' 4. Record key prediction vocabulary on the board in two columns: 'Has a heart/blood' vs. 'Does NOT have a heart/blood.' Say: 'We are going to use today's evidence to find out which column is closer to the truth — and then we'll figure out what that means for Kipchoge's puzzle.' 5. Wait time: after each cold-call, wait 10–12 seconds before moving to the next pair.	Strategy: SILENT PREDICTION + THINK-PAIR-SHARE. This activates prior knowledge and creates cognitive dissonance before the observation. By committing predictions to paper before seeing evidence, learners are primed to notice when the evidence surprises them — a key driver of conceptual change. Recording predictions on the board also gives the teacher a rapid formative snapshot of the class's starting mental models.	Observable indicator: Every learner has a written prediction in their notebook with at least one labelled sketch and one sentence of reasoning. Teacher scans notebooks during the pair-share. Red flag: if more than 50% of learners draw a mammalian-style heart with red blood vessels inside the grasshopper, note this as a target misconception to address explicitly during Phase 3.
Observe Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12	Activity A — Dissection or Diagram (groups of 3–4): Each group receives either (a) a freshly killed or preserved grasshopper in a dissection tray, or (b) a printed large-format diagram of the grasshopper's internal anatomy showing the dorsal vessel, ostia, haemocoel, and haemolymph (for schools without specimens). Groups follow a structured observation guide with 5 tasks: (1) Identify the dorsal vessel running along the back — note its colour and position. (2) Locate the ostia	1. Before groups begin, demonstrate safe dissection technique on one specimen at the front (2 minutes). Say: 'I am cutting along the dorsal midline — watch the angle of the scissors. Your job is to observe, not to rush.' 2. Circulate every 4–5 minutes. At each group, ask one probing question — rotate through: 'What colour is this fluid — and why does colour matter?'; 'Can you trace a path from the dorsal vessel to the leg? What do you find?'; 'Look at the Comparison Card — what is	Strategy: STRUCTURED EVIDENCE TABLE with PHENOMENON CONNECTION COLUMN. The final column ('Connection to Kipchoge Puzzle') ensures every observation is anchored to the driving question rather than treated as isolated facts. This prevents the lesson from becoming a mere anatomy exercise and keeps learners building towards the central explanation. The Comparison Card scaffolds the cross-system thinking that is central to the sub-strand.	Observable indicator: Each group's evidence table has all 5 rows completed with specific observations (not vague statements like 'it looks different'). Teacher checks for: (1) correct identification of the dorsal vessel and haemocoel; (2) the observation that haemolymph is NOT red/does not contain haemoglobin; (3) at least one entry in the 'Connection to Kipchoge Puzzle' column that references closed vs. open circulation. Red flag: groups that describe the

🔍 Search ARES for similar readings	(small openings along the dorsal vessel) — count how many they can see. (3) Observe that there are NO distinct blood vessels branching to organs — the body cavity (haemocoel) is an open space. (4) Note the colour of haemolymph (pale yellow-green — NOT red). (5) Record all observations in a structured evidence table with columns: Structure | What I Observed | What I Think It Does | Connection to Kipchoge Puzzle. Activity B — Comparison Data Card: Each group also receives a 'Comparison Card' showing two columns — Grasshopper Circulatory Data vs. Kipchoge Heart Data (from Lessons 1–2). They must add at least two new observations from today to the grasshopper column.	the MOST surprising difference you see compared to Kipchoge?' 3. At the 15-minute mark, pause the class with a hand signal. Cold-call 2 groups: 'Tell me one thing you observed that surprised you.' Wait 12 seconds after each response. 4. If a group finishes early, direct them to extension prompt: 'If haemolymph does not carry oxygen, how do grasshopper cells get oxygen? Write your best hypothesis.' 5. At the end of the observation phase, say: 'Before we explain anything — make sure every row of your evidence table is filled in. We need evidence before explanation.'		dorsal vessel as 'a heart just like Kipchoge's' — target these for direct clarification in Phase 3.
Explain Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class sensemaking discussion anchored by a projected summary diagram showing the open circulatory system of the grasshopper side-by-side with a simplified diagram of the human heart and closed circulatory system. Learners use their evidence tables to contribute to a class explanation. The teacher facilitates a structured discussion using the 'Claim–Evidence–Reasoning' (CER) framework: each contribution must state a claim, cite an observation from the evidence table, and give reasoning connecting it to the driving question. Key conceptual targets: (1) In an open system, haemolymph bathes organs directly — there are no capillaries; pressure is low. (2) Insects do NOT use haemolymph for oxygen transport — the tracheal system handles gas exchange directly. (3)	1. Open the explain phase by returning to the prediction board. Say: 'Look at our predictions from the start of the lesson. What did today's evidence confirm — and what did it overturn?' Cold-call 3 different learners (not the same ones from Phase 1). Wait 10 seconds after each. 2. Introduce CER explicitly: 'We are going to build our explanation using Claim–Evidence–Reasoning. Your claim is your answer, your evidence is what you actually observed today, and your reasoning is the scientific link between the two.' Model one example aloud: 'Claim: The grasshopper does not have a closed circulatory system. Evidence: We observed no blood vessels leading from the dorsal vessel to the legs — haemolymph just filled the haemocoel. Reasoning: Without closed vessels, there is no way to direct	Strategy: CLAIM–EVIDENCE–REASONING (CER) FRAMEWORK. CER structures scientific argumentation by requiring learners to explicitly link their observational evidence to conceptual claims. In this lesson, CER prevents learners from making vague comparative statements ('grasshoppers are simpler') and forces them to cite specific anatomical observations (no closed vessels, pale haemolymph, low-pressure haemocoel) as the basis for their explanations. The Kipchoge scenario provides a concrete, memorable benchmark against which all CER claims are tested.	Observable indicator: Each learner's CER paragraph (1) correctly identifies the open circulatory system as the key structural difference, (2) references at least one specific observation from their evidence table (e.g., 'haemolymph did not appear red,' 'no vessels branched from the dorsal vessel to the legs'), and (3) explicitly connects this to why Kipchoge's system is more efficient for high-demand activity. Teacher collects notebooks at the end of this phase to check CER quality. Red flag: paragraphs that confuse the tracheal gas-exchange system with the circulatory system — use as a teaching point in the next lesson (gaseous exchange).

	This means the grasshopper's transport system is sufficient for a small, low-metabolic-demand body — but would be catastrophically insufficient for a body running a marathon at 2:01/km pace. Learners write a CER paragraph in their notebooks: 'Could the grasshopper's circulatory system support Eliud Kipchoge running a marathon? Use evidence from today's observation to explain.'	high-pressure, oxygen-rich blood to a specific muscle during intense exercise — which is exactly what Kipchoge's heart does.' 3. Ask: 'Using CER — could a grasshopper run a marathon?' Allow 3 minutes of silent writing, then cold-call 4 learners. Press each one: 'What specific observation from your evidence table supports that claim?' 4. Explicitly address the key misconception if it arose: 'Several groups thought haemolymph carries oxygen like our blood. What does the colour tell us? Haemolymph is pale — no haemoglobin — which means...' Wait 12 seconds for learners to complete the reasoning. 5. Close with the phenomenon link: 'Kipchoge's heart dropped from 170 to 37 bpm in minutes. Could the grasshopper's dorsal vessel ever do that job? What specifically makes it impossible?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board (established in Lesson 1) which displays the central driving question and sub-questions. Each group is given two sticky notes of different colours: (1) GREEN — 'Evidence we collected today that helps answer a DQB question' and (2) ORANGE — 'A new question today's evidence raised that we cannot yet answer.' Groups spend 5 minutes writing their sticky notes and placing them on the DQB. The teacher facilitates a 5-minute whole-class gallery walk around the DQB. Learners identify: which DQB questions now have more evidence pointing to them, and which new questions are most important to investigate next. The	1. Direct groups to the DQB. Say: 'Look at the questions we posted in Lesson 1. Today we gathered new evidence. Your job now is to be honest — what does today's evidence actually tell us, and what new questions did it open up?' 2. Circulate during the 5-minute sticky-note writing. Prompt groups that are stuck: 'What was the most surprising thing you observed today? That surprise is probably a new question.' OR: 'Which question on the board did today's dissection give you evidence for? Quote something from your evidence table.' 3. During the gallery walk, say nothing — allow 3 minutes of silent reading. Then facilitate 2 minutes of open comments: 'Who saw a green	Strategy: DRIVING QUESTION BOARD — EVIDENCE MAPPING + NEW QUESTION GENERATION. The DQB is not a display artifact — it is an active knowledge-tracking tool. By requiring learners to place evidence AND new questions, the DQB models how science actually progresses: answering one question generates new ones. The voting mechanism gives learners agency over the inquiry direction and increases investment in subsequent lessons. Connecting the starred question to the next lesson (tilapia/fish transport) maintains narrative continuity across the 12-lesson sequence.	Observable indicator: Every group has posted at least one green and one orange sticky note. Teacher checks that green notes reference specific observations from today's lesson (not vague statements). Teacher checks that orange notes are genuine scientific questions (not statements). Red flag: groups that post green notes with no evidence citation, or orange notes that are not questions — redirect these groups with: 'Can you quote one thing from your evidence table on that green note?' and 'Can you rewrite that as a question starting with HOW or WHY?'

	class votes (show of hands) on the single most important new question generated today. This question is starred on the DQB for future reference.	sticky note that really stood out? Who saw an orange question that we absolutely must answer?' 4. Conduct the vote. After the starred question is chosen, say: 'Write this question in your notebook. As we move through the next lessons — tilapia, frog, mammal — keep asking: does this lesson help answer our starred question?' 5. Explicitly connect to the storyline: 'We now know the grasshopper's transport system. Next we will look at the tilapia in Lake Victoria. Will its system be more like the grasshopper's or more like Kipchoge's? Hold that prediction.'		
Model Building Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative model from Lesson 1 (a diagram/poster they have been building across the sequence showing how different animals' transport systems compare). In Lesson 1, they sketched Kipchoge's human circulatory system. In Lesson 2, they added baseline exercise data. Today, they add a new panel to their cumulative model: a labelled diagram of the grasshopper's open circulatory system, with annotations linking each structure to its function and comparing it explicitly to the human system. Required model annotations: (1) Dorsal vessel — 'Simple pulsatile pump; no chambers like a human heart'; (2) Ostia — 'Entry points for haemolymph; no equivalent in closed systems'; (3) Haemocoel — 'Open body cavity; haemolymph bathes organs directly — no capillaries'; (4) Haemolymph — 'Does NOT carry oxygen; pale/yellow-green; no haemoglobin'; (5) A red arrow pointing from the grasshopper	1. Distribute or direct learners to retrieve their cumulative models. Say: 'Your model is the running record of our investigation. Each lesson we add new evidence. Today's job is to add the grasshopper panel — and then draw the comparison arrow to Kipchoge.' 2. Allow 8 minutes of individual model-building in silence. Circulate and look specifically for: (a) correct labelling of dorsal vessel, ostia, haemocoel; (b) the annotation that haemolymph does NOT carry oxygen; (c) a comparison arrow or annotation linking to the human system. 3. After 8 minutes, do a targeted cold-call — choose 2 learners whose models you noticed had strong comparison arrows. Ask them to read their comparison annotation aloud. Ask the class: 'Do you agree with this comparison? Would you add anything?' Wait 10 seconds. 4. Ask all learners to write their model update statement. Cold-call 3 learners to share theirs. Press each: 'How does what you added	Strategy: CUMULATIVE MODEL REVISION with COMPARISON ANNOTATION. The cumulative model is the central epistemic tool of the 12-lesson sequence. Each lesson's model update forces learners to integrate new evidence with prior knowledge rather than treating each lesson as isolated. The mandatory comparison arrow between the grasshopper and human panels makes the conceptual contrast visible and spatial — learners can literally see the progression from simple to complex transport systems building across their model. The model update statement develops scientific metacognition: learners must articulate what they added, why, and how it advances the driving question.	Observable indicator: Each learner's cumulative model now has a grasshopper panel with all 4 required labelled structures and annotations, PLUS a comparison arrow to the human system panel. The model update statement correctly identifies a structure, links it to a function, and connects it to the driving question. Teacher collects models at the end of the lesson for written feedback. Red flag: models that show the grasshopper with a multi-chambered heart or red blood — these learners need individual follow-up before Lesson 4 to correct the structural misconception before fish transport is introduced.

	panel to the human panel with the label: 'OPEN vs. CLOSED — what difference does this make for Kipchoge?' Learners also write a one-sentence 'model update statement' at the bottom of their cumulative model: 'Today I added [structure] because it shows [function] which helps explain [part of the driving question].'	today bring us closer to answering our driving question?' 5. Close the lesson with the phenomenon: 'We now know that the grasshopper on the school fence survives without anything like Kipchoge's heart. Next lesson we move to the tilapia in Lake Victoria — a more complex animal. Does complexity of the animal predict the complexity of the transport system? Hold that question as your homework thinking task.'		

D. TEACHER REFLECTION
After the lesson, ask yourself: (1) MISCONCEPTION CHECK — Did any learners persist in thinking that the grasshopper's dorsal vessel functions like a mammalian heart with chambers? If so, plan a 5-minute targeted clarification at the start of Lesson 4 before introducing fish transport. (2) CER QUALITY — Were learners able to cite specific observations (colour of haemolymph, absence of branching vessels) in their CER paragraphs, or were explanations vague? If vague, model one more CER example at the start of Lesson 4 using a Kenyan context (e.g., comparing a boda-boda carrying one passenger along a dirt path vs. a Nairobi SGR train delivering goods to multiple stations simultaneously — as an analogy for open vs. closed circulation). (3) DQB ENGAGEMENT — Did the orange 'new question' sticky notes show genuine scientific curiosity, or were they superficial? Review the starred question before Lesson 4 and plan how the fish dissection/diagram evidence will address it. (4) MODEL DEPTH — Did learners make the conceptual leap that the ABSENCE of oxygen-carrying haemolymph is connected to the presence of the TRACHEAL system? If this link is not appearing in models, plan a brief bridge moment in Lesson 4 to revisit it, as it will be critical when learners reach Sub-Strand 3.3 (Gaseous Exchange). (5) EQUITY CHECK — Were diagram-based alternatives equally engaging for learners who did not dissect? Consider pairing diagram learners with dissection learners during the model-building phase in future lessons to allow peer observation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A grasshopper was examined (dissection or diagram). Learners observed: a dorsal vessel running along the grasshopper's back with small openings called ostia; an open body cavity (haemocoel) with no blood vessels branching to individual organs; haemolymph that is pale yellow-green in colour, NOT red.
What did I learn?	Insects have an OPEN circulatory system: haemolymph is pumped by the dorsal vessel through ostia and floods the haemocoel, bathing organs directly. Unlike human blood, haemolymph does NOT carry oxygen — insects use a separate tracheal system for gas exchange. This means an insect's transport system operates at low pressure and cannot direct oxygen-rich fluid to specific muscles on demand.
How does this explain the phenomenon?	The grasshopper survives without a powerful heart because its small body size means diffusion distances are short, and its tracheal system delivers oxygen directly to cells without using the circulatory system. Eliud Kipchoge could NOT survive on a grasshopper's circulatory system because his large body and extreme muscular demands during a marathon require a high-pressure, closed system that can rapidly direct oxygenated blood to working muscles — something an open haemocoel system cannot do. Today's evidence updates our driving question: the STRUCTURE of the transport system (open vs. closed, presence or absence of oxygen-carrying fluid) directly determines the level of physical performance an animal can sustain.

LESSON 4 (80 minutes): Transport in Fish — The Single Circulatory System of the Tilapia	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the single circulatory system of a fish (tilapia) and compare it to the open circulatory system of an insect, in order to advance understanding of how transport system structure relates to animal complexity and survival.
Knowledge	Learners will be able to: (a) describe the structure of the two-chambered heart of a fish; (b) trace the path of blood through the single circulatory system of tilapia; (c) explain why the single circulatory system is less efficient than the double circulatory system of mammals; (d) compare the circulatory systems of insects and fish.
Skills	Learners will be able to: (a) trace and annotate blood flow pathways on a diagram of the fish circulatory system; (b) construct a comparative table of insect and fish transport systems; (c) revise their cumulative model of animal transport systems using new evidence from fish.
Attitudes	Learners will: (a) appreciate how evolutionary adaptations in circulatory systems are linked to the energy demands of different lifestyles; (b) show curiosity about how a tilapia in Lake Victoria sustains life with a much simpler heart than Eliud Kipchoge's.
Key Inquiry Question	How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	In Lessons 1–3, learners established that Kipchoge's mammalian double circulatory system delivers oxygenated blood with high pressure and efficiency, and that the grasshopper's open haemolymph system is far simpler and serves low-energy needs. Lesson 4 adds a critical middle case — the tilapia of Lake Victoria — whose closed but SINGLE circulatory system and two-chambered heart represent an intermediate level of complexity. By tracing blood flow in the fish heart, learners build the comparative evidence they need to answer the driving question and refine their cumulative model.
Safety Notes	No dissection is performed in this lesson. If a preserved or fresh tilapia specimen is brought in for external observation only, ensure hands are washed before and after handling. No sharp instruments are to be used without direct teacher supervision. Learners with fish allergies should be seated away from any specimens and should not handle them.

B. LESSON OVERVIEW

Learners begin by revisiting the DQB and recalling the contrast between Kipchoge's double circulatory system (Lessons 1–2) and the grasshopper's open system (Lesson 3). The teacher introduces the tilapia from Lake Victoria as the new 'witness' — an animal whose circulatory system sits between those two extremes. Through a guided diagram activity and a short reading/video segment, learners trace blood flow through the fish's two-chambered heart (one atrium, one ventricle) and its single circuit (heart → gills → body → heart). They construct a comparative table (insect vs. fish vs. mammal), identify key limitations of the single circulatory system, and connect these limitations back to why a tilapia could never sustain a marathon. The lesson closes with model revision: learners update their cumulative diagrams and add new sticky notes to the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown a large image	Display the tilapia image on the	Predict-before-observe anchoring:	Teacher circulates and notes: (a)

📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	of a Nile tilapia (Oreochromis niloticus) from Lake Victoria alongside a simplified silhouette of its internal organs. They are asked: 'You have already studied the grasshopper's transport system and Kipchoge's circulatory system. Where do you think the tilapia falls — is its system more like the grasshopper's, more like Kipchoge's, or something completely different? Draw or write your prediction in your exercise book. Include: (a) how many chambers you think the fish heart has, (b) whether blood and oxygen travel in the same or separate vessels, and (c) whether you think a tilapia could sustain a 42 km swim at marathon pace.' Learners work individually for 4 minutes, then share with a partner for 2 minutes.	board or printed A3 sheet. Say: 'Before we look at any new information, I want your best scientific guess — use the patterns you already noticed about the grasshopper and Kipchoge.' [WAIT 4 minutes — circulate and observe predictions without correcting.] After partner share, cold-call 3 learners: 'Akinyi, what did you and your partner predict about the number of heart chambers?' [WAIT 10 seconds.] 'Omondi, did your group think the fish system is open or closed — and what was your reasoning?' [WAIT 12 seconds.] 'Wanjiku, tell me one thing about the grasshopper system from last lesson that influenced your prediction today.' [WAIT 10 seconds.] Record the three most common predictions on the board under the heading: 'Our Predictions — Tilapia Heart.' Do NOT confirm or deny. Say: 'Let's gather evidence and see who was closest — and more importantly, WHY.'	By committing predictions to paper before instruction, learners activate prior knowledge from Lessons 1–3 and create a personal cognitive stake in the outcome. The act of comparing grasshopper to fish forces pattern-recognition across systems, which is a key NGSS Practice 6 (Constructing Explanations) precursor skill.	Do learners use evidence from previous lessons (e.g., 'the grasshopper had no heart, so maybe fish has a simple one')? (b) Are predictions structured (chambers + open/closed + energy reasoning) or vague? Red flag: learners who cannot recall whether the grasshopper system was open or closed — these learners need a brief retrieval prompt before moving on.
Observe Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms	ACTIVITY A — Diagram Tracing (25 min): Each learner receives a printed or ARES-platform diagram of the fish circulatory system showing: the two-chambered heart (sinus venosus, atrium, ventricle, conus arteriosus — labelled), the gill capillaries (for oxygenation), the dorsal aorta, and the body tissues. Using a RED pencil/pen for oxygenated blood and a BLUE pencil/pen for deoxygenated blood, learners trace the single circuit: deoxygenated blood → atrium → ventricle → ventral aorta → gill capillaries [oxygenation occurs here] → dorsal aorta → body tissues [oxygen delivered, CO_2 collected] → veins → back to	Distribute diagrams and coloured pencils. Say: 'You are going to become the blood — start at the heart and trace your journey around the body, switching colours as the blood gains or loses oxygen.' [WAIT 2 minutes for learners to orient themselves.] Circulate actively. When you spot a common error (e.g., a learner drawing the blood going directly from gills back to heart without visiting the body), do NOT correct individually — instead, pause the class after 10 minutes and say: 'I am noticing some of us are sending the blood on a shortcut. Remember — the blood has one job before it returns to the heart.	Colour-coded diagram tracing (NGSS Practice 2 — Developing and Using Models): The physical act of switching between red and blue forces learners to attend to the oxygenation state of blood at each stage. The embedded pressure data point creates a direct cognitive bridge to the mammalian double circulatory system studied in Lesson 2, priming the comparison that will occur in the Explain phase.	Collect 5–6 diagrams mid-activity (opportunistic sampling) and scan for: (a) Correct colour-switching at the gills; (b) Blood flowing in the correct direction through the heart chambers; (c) The annotation about lower pressure. Indicator of understanding: Learner correctly shows oxygenated (red) blood leaving gills via dorsal aorta BEFORE reaching body tissues. Indicator of misconception: Learner shows a loop where blood is oxygenated AND delivered to tissues at the same location (gill/body conflation).

🔍 Search ARES for similar readings	atrium. Learners label: (1) where the blood is oxygenated, (2) where oxygen is delivered to tissues, (3) the ONE point where oxygenated and deoxygenated blood are NEVER mixed. ACTIVITY B — Reading + Data (10 min): Learners read a short ARES passage (≤200 words) titled 'The Tilapia's Journey: One Loop Around the Body.' Key data point embedded in the reading: 'After passing through the gill capillaries, the blood pressure in a tilapia drops significantly before it reaches the body tissues — unlike in Kipchoge's body, where the left ventricle re-pressurises the blood before sending it to the muscles.' Learners underline this sentence and annotate: 'This means the tilapia's tissues receive blood at LOWER pressure than a mammal's.'	What is it?' [WAIT 12 seconds.] Cold-call: 'Mwangi — one job. Go.' After diagrams are complete, direct learners to the reading. Say: 'Find the sentence about pressure. Underline it. In your own words, write in the margin WHY lower pressure after the gills might matter for a tilapia trying to move fast.' [WAIT 3 minutes.]		
Explain Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	PART A — Comparative Table Construction (15 min): In pairs, learners complete a three-column comparative table: 'Comparing Animal Transport Systems.' Columns: Insect (Grasshopper) | Fish (Tilapia) | Mammal (Kipchoge). Rows: Type of system (open/closed), Number of heart chambers, Blood vessel types present, Is blood and oxygen always separated?, Pressure of blood reaching body tissues (relative), Estimated activity level the system can support. Learners fill in the Insect and Fish columns using evidence from Lessons 3 and 4; the Mammal column was partially filled in Lessons 1–2. PART B — Explanation Writing (8 min): Each learner independently writes a 3–4 sentence explanation in response to: 'A Nile tilapia in	Project the blank comparative table on the board. Say: 'You are not starting from zero — you already have evidence from the last three lessons. Your job is to organise it.' [WAIT 1 minute for pairs to divide roles.] After 10 minutes, pause and use cold-call structured shares: 'Fatuma — what did your pair write for the fish's blood pressure reaching body tissues compared to the mammal?' [WAIT 10 seconds.] 'Kamau — do you agree with Fatuma's group, or does your evidence say something different?' [WAIT 12 seconds.] Record key contrasts on the board. For Part B, say: 'This is individual. No copying your partner's sentences — I want YOUR scientific voice. You have 8 minutes.' Circulate and prompt struggling learners with: 'Look at	Structured Comparative Table + Evidence-Based Explanation Writing (NGSS Practices 6 & 8 — Constructing Explanations and Obtaining/Evaluating Information): The table externalises the comparative reasoning scaffold, allowing learners to see the progression from simple (open, no chambers) to complex (closed, four-chambered) as a continuum rather than isolated facts. The explanation writing forces learners to select and deploy specific structural evidence — moving from description to mechanistic reasoning.	Collect written explanations. Success criteria: (1) Learner names a specific structural feature (e.g., 'the fish has only one ventricle so blood is not re-pressurised after the gills'); (2) Learner connects structure to function (lower pressure → less efficient oxygen delivery); (3) Learner uses comparative language ('unlike Kipchoge's heart which…'). Flag for follow-up: explanations that say only 'fish have a simpler heart' without explaining the pressure drop mechanism.

	Lake Victoria and Eliud Kipchoge both need oxygen delivered to their muscles. Explain ONE structural reason why Kipchoge's system delivers oxygen more efficiently, using evidence from your diagram and your comparative table.'	your diagram. After the gills, what happens to blood pressure? How is that different from what Kipchoge's left ventricle does?' After writing, cold-call 2 learners to read their explanation aloud. Ask the class: 'Did they use evidence from the diagram? Did they name a specific structure?' [WAIT 10 seconds between each question.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	The class returns to the shared DQB (physical board or ARES digital board). Each learner receives two sticky notes of different colours: YELLOW = 'New Evidence I Have' and GREEN = 'New Question I Now Have.' Learners write and post: On YELLOW: one piece of new evidence from today's lesson that helps answer the driving question (e.g., 'Fish blood loses pressure after the gills — this is why the single circuit is less efficient'). On GREEN: one new question raised by today's lesson (e.g., 'If the fish's blood pressure drops after the gills, how does it still reach all the body tissues?' or 'Is the tilapia's heart rate similar to Kipchoge's — and what does that tell us?'). The teacher reads 4–5 sticky notes aloud and clusters them under two emerging DQB themes: 'Evidence for: System Complexity Matches Energy Demand' and 'Still Wondering About: Pressure and Efficiency.'	Say: 'Our DQB is growing. Every lesson, you are adding pieces to the puzzle. Today, you discovered something that sits BETWEEN the grasshopper and Kipchoge. Let's make that visible.' Distribute sticky notes. [WAIT 4 minutes for individual writing.] Read selected yellow notes aloud, then ask: 'Does this evidence push us closer to answering our driving question? What would we still need to know to fully answer it?' [WAIT 12 seconds.] Point to the driving question displayed at the top of the board: 'What would happen if Eliud Kipchoge had the circulatory system of a grasshopper?' Ask: 'After today — could we add a new version of this question? What if Kipchoge had the heart of a tilapia? Would that be better or worse than a grasshopper's system — and why?' [WAIT 15 seconds.] Cold-call 3 learners for responses. Do not resolve — leave as an open thread for Lesson 5.	Collaborative Driving Question Board update (NGSS Practice 1 — Asking Questions and Defining Problems): The DQB makes the iterative, evidence-building nature of science visible and spatial. The colour-coded sticky note system distinguishes between established evidence and productive uncertainty, which models how real scientists distinguish what is known from what is still under investigation. The teacher's deliberate 'What if Kipchoge had a tilapia's heart?' extension question sustains narrative curiosity across the lesson sequence.	Scan yellow sticky notes for specificity. Success indicator: learner references a named structure (atrium, ventricle, gill capillaries, dorsal aorta) or a named mechanism (pressure drop, single circuit, oxygenation at gills). Weak indicator: generic notes like 'fish hearts are small' — these learners need targeted questioning in Lesson 5's opening retrieval. Green notes reveal productive vs. surface-level confusion; teacher photographs the board to inform Lesson 5 planning.
Model Building Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview	Learners open their cumulative model diagram (started in Lesson 1 and revised each subsequent lesson). Today's revision task: Add the tilapia to the model. Specifically, learners must: (a) Draw and label the two-chambered fish heart (atrium + ventricle) to	Say: 'Your model is your scientific story of how transport systems evolved in complexity. Today you are adding Chapter 3 — the tilapia. Make sure it clearly shows what is the SAME and what is DIFFERENT compared to the grasshopper and Kipchoge.' [WAIT	Cumulative Model Revision (NGSS Practice 2 — Developing and Using Models): The running cumulative model is the central epistemic artefact of the unit. Each revision deepens the learner's ability to use a model as an explanatory tool rather than a	Teacher reviews models for: (a) Correct two-chamber representation (atrium + ventricle — NOT four chambers); (b) Arrows showing blood travelling in a SINGLE loop; (c) Oxygenation correctly located at the gills, NOT the heart; (d) A written efficiency

Source: CK-12 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	scale relative to the four-chambered mammalian heart already on the model; (b) Use red and blue arrows to show the single circuit (heart → gills → body → heart), clearly marking where oxygenation occurs; (c) Write a 1-sentence 'efficiency rating' next to the fish system, using a comparative claim (e.g., 'More efficient than the grasshopper — blood is always in vessels — but less efficient than a mammal because blood pressure drops after the gills'); (d) Add a small 'Lake Victoria tilapia' icon or label to ground the model in the Kenyan context. Learners who finish early add a 'pressure gradient arrow' to both the fish and mammal sections of their model, showing visually where pressure is highest and lowest in each circuit.	12 minutes for model revision.] Circulate and ask targeted questions: 'Show me on your model where the blood is oxygenated in the fish. Now show me where it is oxygenated in Kipchoge's system. What is different about what happens to the blood AFTER oxygenation?' [WAIT 10 seconds.] With 3 minutes remaining, ask one learner to share their model under a visualiser or hold it up. Say to the class: 'Look at Chebet's model. She has drawn the pressure drop with a downward arrow after the gills. Is that consistent with the evidence we gathered today?' [WAIT 12 seconds.] Close by saying: 'Next lesson, we will meet a new animal whose system will challenge your model again — and we will keep moving toward answering our driving question about Kipchoge.'	decorative diagram. The 'efficiency rating' sentence forces learners to commit to a comparative claim — building the scientific habit of making evidence-based relative judgements rather than absolute, isolated descriptions.	claim that uses comparative language and references at least one structural feature. Exit criterion: every learner's model must show red arrows leaving the gills (not the heart) before reaching the body — this is the single most diagnostic indicator of whether the single-circuit concept is understood.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the diagram tracing activity, what was the most common error — did learners struggle with the direction of blood flow, the colour-switching at the gills, or the location of oxygenation? What does this reveal about prior knowledge gaps regarding gas exchange? (2) In the comparative table activity, were learners able to retrieve evidence from Lessons 1–3 independently, or did they need prompting? If retrieval was weak, consider opening Lesson 5 with a structured 'Brain Dump' retrieval exercise before introducing new content. (3) Did the explanation writing (Part B of Explain phase) show evidence of mechanistic reasoning (linking structure → pressure → efficiency), or did learners describe features without explaining function? If the latter, plan a modelling think-aloud at the start of Lesson 5 where you narrate the structural-functional link explicitly. (4) Were there learners who asked genuinely novel questions on the DQB green sticky notes — questions that go beyond the curriculum content (e.g., about evolutionary history, or about whether fish can 'train' their hearts)? These learners may benefit from extension reading on comparative vertebrate physiology. (5) Does the cumulative model now clearly show THREE distinct systems (open insect, single-circuit fish, double-circuit mammal) in a way that could support a comparative argument? If models are still too vague or unlabelled, build in a 5-minute model peer-review protocol at the start of Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners traced blood flow through a colour-coded diagram of the tilapia circulatory system, identifying the two-chambered heart, the location of gill oxygenation, the path of the dorsal aorta to body tissues, and the pressure drop that occurs after blood passes through the gill capillaries.

What did I learn?	Learners learned that the tilapia has a CLOSED but SINGLE circulatory system with a two-chambered heart (one atrium, one ventricle). Blood travels in one loop: heart → gills (oxygenated) → body tissues (oxygen delivered) → back to heart. Because blood passes through TWO sets of capillaries (gills then body) before returning to the heart, pressure drops significantly — making delivery to body tissues less forceful than in a mammal's double circulatory system. This makes the fish system more efficient than an insect's open system, but less efficient than Kipchoge's four-chambered double circulatory system.
How does this explain the phenomenon?	The structural difference between the fish's single circuit and Kipchoge's double circuit explains why the tilapia cannot sustain the high-speed, high-oxygen-demand activity that Kipchoge's cardiovascular system supports. In the double circuit, the left ventricle re-pressurises oxygenated blood before sending it to the body muscles — ensuring high-pressure, high-volume oxygen delivery. In the tilapia, no such re-pressurisation occurs. This is a structural limitation that connects directly to the driving question: transport system structure determines the animal's capacity to survive and thrive at different energy demands.

LESSON 5 (80 minutes): Transport in Amphibians and Reptiles — The Trade-Off Heart	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to analyse the structure and function of the three-chambered heart in amphibians and the partially divided ventricle in reptiles, and evaluate the trade-offs of mixing oxygenated and deoxygenated blood.
Knowledge	Learners will be able to: (i) describe the structure of the three-chambered heart in amphibians (two atria, one ventricle); (ii) explain partial separation in the reptilian ventricle; (iii) define double circulation and identify where it is incomplete in amphibians and reptiles; (iv) explain how mixing of oxygenated and deoxygenated blood affects metabolic efficiency.
Skills	Learners will be able to: (i) construct and annotate a diagram of the amphibian/reptilian heart; (ii) compare and contrast circulatory efficiency across fish, amphibians, and reptiles using evidence gathered in previous lessons; (iii) analyse data on oxygen delivery and link it to organism activity levels; (iv) revise their cumulative transport system model.
Attitudes	Learners will: (i) appreciate that 'simpler' does not mean 'inferior' — each heart design is a trade-off suited to the animal's environment and energy demands; (ii) show curiosity about how Rift Valley frogs and Kenyan monitor lizards survive with mixed blood; (iii) value collaborative model revision as a sign of growing scientific understanding.
Key Inquiry Question	How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	In Lessons 1–4, learners anchored on Kipchoge's heart, investigated open circulatory systems (grasshopper), single circulation (tilapia in Lake Victoria), and began building a comparative model. Lesson 5 advances the storyline by introducing the intermediate step — the Rift Valley frog's three-chambered heart and the reptilian partial septum — revealing that evolution of the heart is a series of trade-offs between efficiency and metabolic cost. This directly feeds the driving question: without full separation of blood, could Kipchoge sustain world-record pace?
Safety Notes	No live dissection in this lesson. If preserved frog specimens are used for observation only, learners must wear disposable gloves and wash hands thoroughly afterwards. Formalin-preserved specimens must be used in a well-ventilated space. Learners with known latex allergies should use nitrile gloves. All specimens must be handled respectfully and disposed of according to school policy.

B. LESSON OVERVIEW
Learners open by revisiting their Driving Question Board entries and the comparative heart-efficiency data gathered in Lessons 1–4. They make a prediction about what a frog's heart must look like to allow it to survive both in water and on land. They then observe labelled diagrams, a teacher-guided dissection demonstration (or printed specimen card), and a short narrated animation of blood flow through the three-chambered amphibian heart and the partially divided reptilian heart. Using a structured data-analysis activity, they calculate and compare the estimated percentage of oxygenated blood reaching body tissues in fish, frogs, and reptiles. They apply this to explain why a Rift Valley frog cannot run a marathon, then revise their cumulative circulatory system model to add the amphibian and reptilian tiers. The lesson closes with a Driving Question Board update and explicit connection back to Kipchoge's four-chambered heart.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** BIOLOGY-FORM-THREE Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners are shown a large projected image: on the left, Eliud Kipchoge's heart-rate recovery curve from Lesson 1; on the right, a silhouette of a frog sitting on a rock at a Rift Valley wetland. They read two statements on the board and, in their science notebooks, write a prediction answering: 'If the frog has a heart, what do you think it looks like compared to Kipchoge's? Draw a quick sketch and write one sentence explaining your reasoning.' Learners then do a Think-Pair-Share: 60 seconds solo writing, 60 seconds sharing with a partner, then selected pairs share with the class. Learners also revisit their Lesson 4 model and write one 'carry-forward question' — something they still wonder about — on a sticky note to add to the Driving Question Board.	Teacher opens with: 'Last lesson we saw that a tilapia in Lake Victoria survives with a two-chambered heart and single circulation. Today we move to a frog in the Rift Valley wetlands. Before I show you anything — I want your brain to work first.' Teacher displays the split image (Kipchoge / Rift Valley frog). After 60 seconds of silent writing, teacher says: 'Turn to your partner. You have 60 seconds — go.' Teacher circulates, listening for: (a) learners who predict three chambers, (b) learners who predict a heart identical to a fish, (c) learners who predict a heart identical to Kipchoge's. After partner talk, teacher cold-calls EXACTLY 3 pairs using name sticks. Teacher annotates predictions live on the board under columns: 'Simpler than fish / Same as fish / More complex than fish / Same as Kipchoge.' Teacher does NOT correct yet — says: 'Hold those predictions. We will come back and score them honestly at the end.' Teacher gives WAIT TIME of 12 seconds after each cold-call before moving on.	Think-Pair-Share with Prediction Tracking. This strategy activates prior knowledge from Lessons 1–4, surfaces misconceptions (e.g., 'all vertebrates have four-chambered hearts'), and creates cognitive dissonance that motivates observation. By recording predictions publicly, learners have a personal stake in the evidence phase.	Teacher scans notebook sketches during circulation — looking for whether learners draw at least two chambers (showing carry-forward knowledge from fish lesson) or default to a four-chambered model (misconception to address). Teacher notes on a clipboard which 3 cold-called students predicted closest to a three-chambered model — these students will be revisited during the Explain phase to track conceptual change.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** BIOLOGY-FORM-THREE Source: Kenya Curriculum Tools 2025 🔍 Search ARES	PART A — Diagram Observation (15 min): Learners receive the 'Heart Comparison Resource Card' (printed A4 sheet) showing four clearly labelled diagrams side by side: (1) Grasshopper open system, (2) Tilapia two-chambered heart, (3) Frog three-chambered heart (right atrium, left atrium, single ventricle), (4) Nile monitor lizard heart with partially divided ventricle. Blood flow arrows are colour-coded: red = oxygenated, blue = deoxygenated, purple = mixed. Learners complete an	Teacher distributes Resource Cards and says: 'You have 3 minutes to study these diagrams silently — do not write yet. Just look.' (Sets a visible 3-minute timer.) After 3 minutes: 'Now complete your Observation Table — you have 5 minutes.' Teacher circulates and uses targeted prompts: 'What does the purple arrow tell you about the blood in that chamber?' and 'Where does the frog get oxygen — only from lungs?' (probing prior knowledge that frogs also exchange gas	Multi-modal Observation with Colour-Coding and Data Graphing. Using colour-coded diagrams (red/blue/purple) makes the abstract concept of blood mixing visible and concrete. The blood-cell trace forces learners to follow sequential logic rather than passively view a diagram. The bar graph externalises patterns in data, making the efficiency trade-off visible before it is explained verbally. This multi-modal approach supports diverse learners including visual and	Teacher checks Observation Tables during circulation — specifically the 'Does blood mix?' column. A correct response for the frog is 'Yes' and for the monitor lizard is 'Partial.' Misconception flag: if learners write 'No' for the frog, teacher pauses and asks the group nearest that learner: 'Look at the purple arrow in the ventricle — what colour is purple made from?' This re-directs without singling out the learner. Bar graphs are checked for correct axis labelling and relative bar heights — teacher

for similar readings	'Observation Table' in their notebooks with columns: Animal | Number of chambers | Does blood mix? (Yes/No/Partial) | One thing I notice. PART B — Blood Flow Tracing Activity (10 min): Using the frog diagram, learners trace the journey of one red blood cell using a coloured pencil, starting from the lungs/skin, through the heart, to the leg muscle, and back. They annotate with arrows and labels. PART C — Data Table Analysis (10 min): Learners are given a simple data table showing estimated % oxygenation of blood reaching body tissues: Grasshopper ~60%, Tilapia ~80%, Frog ~65%, Monitor Lizard ~75%, Kipchoge (human) ~98%. Learners plot a bar graph (hand-drawn) of this data and circle the two animals with the lowest oxygen delivery.	through moist skin). For Part B, teacher models the first step of the blood-cell trace on the board using a red marker: 'I am starting my red blood cell here — at the lung. Watch — which atrium does it enter?' Teacher then says: 'Now YOU continue the trace in your own diagram — go.' WAIT TIME: 10 seconds of silence before circulating. For Part C, teacher says: 'Plot your bar graph. Put animals on the x-axis, % oxygenation on the y-axis. Title it: Oxygen Delivery Efficiency Across Animals.' After graphs are drawn, teacher cold-calls 2 students to share which two animals they circled and why — WAIT TIME 12 seconds each.	kinaesthetic processors.	notes any learner who inverts the graph (animals on y-axis) and corrects individually.
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** BIOLOGY-FORM-THREE Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	PART A — Structured Notes with Gaps (10 min): Learners complete a guided notes sheet with key blanks: 'The frog has ___ atria and ___ ventricle(s). Because there is only one ventricle, oxygenated blood from the ___ and deoxygenated blood from the ___ mix in the ventricle. This means the blood delivered to body tissues is only ___ oxygenated. In reptiles, a partial ___ in the ventricle reduces mixing but does not eliminate it.' Teacher reads through the notes aloud, pausing at each blank and cold-calling students to fill them in (cold-calls 5 students, uses WAIT TIME of 10 seconds each). PART B — Trade-Off Discussion (10 min): Teacher poses the question on the board: 'If mixing blood is less efficient, why haven't frogs evolved a four-	For Part A, teacher says: 'These notes summarise what you just observed — but I have left the most important words blank. YOUR job is to fill them in from evidence, not guesswork.' After each cold-call, teacher affirms correct answers with: 'Exactly — and why does that matter for the animal's survival?' (secondary probe). For the trade-off discussion, teacher resists giving the answer immediately — says: 'I am going to give your groups 3 minutes. I should hear the word ENERGY in at least one group's sentence.' After group sentences are shared, teacher does a 'Sentence Harvest' — writes 3–4 group sentences on the board and asks the class to vote (thumbs up/sideways/down) on which is most complete. Teacher then	Guided Notes with Socratic Cold-Calling, followed by Collaborative Trade-Off Argumentation and Independent Application. The guided notes lower cognitive load for recall while cold-calling keeps all learners accountable. The trade-off discussion uses argumentation from evidence (NGSS SEP 7) — learners must justify a claim using biological reasoning. The independent application question assesses transfer — can learners apply the concept to a novel but familiar context (marathon running)?	The independently written application question (Part C) is the primary formative assessment artefact for this phase. Teacher collects notebooks and uses a 3-point rubric at lesson close: 3 = correctly uses all four terms and links mixing to insufficient oxygen delivery; 2 = uses 2–3 terms correctly but reasoning is incomplete; 1 = mentions heart structure but does not connect to oxygen demand. Learners who score 1 are identified for targeted support in Lesson 6. During the trade-off discussion, teacher listens for groups that conflate ectotherm/endotherm — a common misconception — and addresses it whole-class before moving on.

	chambered heart like Kipchoge's?' Learners discuss in groups of 4. Each group must produce ONE written sentence that starts: 'A frog does not need a four-chambered heart because...' Groups share. Teacher facilitates by connecting answers to the Rift Valley wetlands context: 'A frog in the Rift Valley is ectothermic — it does not need to generate its own body heat. Lower energy demand means lower oxygen demand. The three-chambered heart is enough.' PART C — Application Question (5 min): Learners answer independently in notebooks: 'Explain why a Rift Valley frog could not complete a 42 km marathon, even if its legs were strong enough. Use the terms: ventricle, oxygenated, mixed blood, oxygen demand.' This is collected at the end of the lesson.	synthesises: 'All of you are partly right. Here is the full picture...' and delivers the ectotherm/endotherm explanation using the Rift Valley frog vs. Kipchoge contrast explicitly. For Part C, teacher says: 'This is your independent thinking — no partner, no neighbour. You have 4 minutes. Go.' WAIT TIME: 15 seconds of silence before learners begin writing.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** BIOLOGY-FORM-THREE Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Each learner receives two sticky notes (different colours). On the FIRST (e.g., green): they write one thing they now understand about the driving question that they did not understand before this lesson — starting with 'I now know that...' On the SECOND (e.g., orange): they write one new question that today's lesson has raised — starting with 'I still wonder...' or 'This makes me ask...' Learners walk to the Driving Question Board and physically place their sticky notes in the correct columns: 'Evidence We Have Gathered' and 'Questions We Still Have.' The class then does a 2-minute Gallery Walk around the board, reading each other's new questions silently. Teacher reads aloud 2–3 of the 'I still wonder' notes and says: 'These are exactly the	Teacher says: 'Before you write your sticky notes, look back at your prediction from Phase 1. Were you right? Be honest with yourself.' (30 seconds silent reflection.) Teacher then distributes sticky notes and gives 3 minutes for writing — enforces silence during writing. During the Gallery Walk, teacher circulates and marks 2–3 'I still wonder' notes with a small star — these will be read aloud. Teacher ends this phase by pointing to the driving question displayed prominently in the room: 'Eliud Kipchoge's heart drops from 170 to 37 BPM in minutes. A frog's heart mixes its blood. Write this in your notes: the four-chambered heart is not just more complex — it is more EFFICIENT. We will prove exactly how efficient in Lesson 6.' Teacher	Driving Question Board (DQB) Sticky-Note Protocol with Gallery Walk. The DQB is a visible, cumulative record of the class's growing understanding — a core NGSS sensemaking tool. Requiring two different-coloured notes separates 'evidence gained' from 'new questions raised,' mirroring how scientists track the knowledge frontier. The Gallery Walk builds scientific community norms — learners read each other's thinking and recognise shared wonderings, reducing isolation and increasing collective intellectual momentum.	Teacher photographs the DQB at the end of each lesson for tracking. After Lesson 5, teacher reviews the 'I now know that...' notes to check: Do learners correctly reference mixing of blood and the three-chambered heart? Are any notes still referencing only the fish (showing failure to update their model)? 'I still wonder' notes that ask about the four-chambered heart signal readiness for Lesson 6. Notes that ask unrelated questions (e.g., about reproduction) indicate a learner who may be disengaged — teacher follows up individually.

	questions scientists ask. Some of them — you will answer by Lesson 9.' Teacher explicitly connects today's additions to the overarching driving question: 'So — could Kipchoge survive with a frog's heart? What does today's evidence say?'	cold-calls 1 student: 'Based on today, give me one reason why Kipchoge cannot have a frog's heart.' WAIT TIME: 12 seconds.		
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** BIOLOGY-FORM-THREE Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners open their cumulative 'Animal Transport System Model' — a large diagram they have been building since Lesson 1. Today they add two new rows: AMPHIBIANS and REPTILES. Using the Resource Card from Phase 2, they draw and annotate the frog's three-chambered heart (labelling: right atrium, left atrium, ventricle, pulmonary circuit, systemic circuit, site of mixing). They draw the reptilian heart beside it and annotate the partial septum. They then complete the 'Efficiency Meter' column they have been filling in across lessons — drawing a simple arrow from 0% to 100% and marking where each animal sits: Grasshopper (~60%), Fish (~80%), Frog (~65%), Lizard (~75%). Finally, learners write a 'Model Explanation Sentence' below their diagram: 'As the heart becomes more chambered and blood mixing decreases, the efficiency of oxygen delivery to tissues [increases/decreases] because ___.' They share their sentence with a partner and refine it before writing the final version.	Teacher says: 'Your model is a living document — it grows every lesson. Today it gets two new animals. Open your model to the next blank row.' Teacher displays a model template on the board (projected) showing where the new rows go, and annotates the frog heart diagram live as learners draw. Teacher narrates: 'Right atrium receives deoxygenated blood from the body — I am drawing a blue arrow in. Left atrium receives oxygenated blood from the lungs and skin — I draw a red arrow. They both empty into the ONE ventricle — and here is where the purple mixing happens.' After independent drawing (8 minutes), teacher says: 'Now write your Model Explanation Sentence. Use the word BECAUSE — that word forces you to give a reason, not just a description.' After partner sharing, teacher cold-calls 2 students to read their final sentences aloud. WAIT TIME: 12 seconds each. Teacher closes the lesson by saying: 'Your model now covers five animals. Next lesson, we will add the most efficient heart on Earth — and connect it directly back to Kipchoge crossing that finish line.'	Cumulative Annotated Model Revision with Efficiency Meter. The ongoing model is the central sensemaking artefact of the 12-lesson sequence — each revision forces learners to integrate new evidence with prior knowledge rather than treat lessons as isolated. The Efficiency Meter provides a visual quantitative scaffold that makes the progression from open systems to closed four-chambered hearts feel like a coherent story rather than disconnected facts. The Model Explanation Sentence uses the Because Protocol to push beyond description into causal reasoning (NGSS SEP 6: Constructing Explanations).	Teacher circulates during model drawing and checks: (a) Are all three chambers of the frog heart correctly labelled? (b) Is the partial septum drawn and labelled in the reptile heart? (c) Is the Efficiency Meter populated with values from at least 4 animals? (d) Does the Model Explanation Sentence use causal language ('because,' 'therefore,' 'which means')? Learners who only draw without annotating are prompted: 'Labels are evidence — a diagram without labels is just a picture.' The two cold-called sentences are assessed live against the criterion: does the sentence correctly state the direction of change (increases) and provide a mechanism (less mixing = more oxygenated blood to tissues)?

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. PREDICTION ACCURACY: How many of the 3 cold-called learners in Phase 1 predicted a three-chambered heart? If fewer than 1 out of 3 did, consider adding a 'heart complexity teaser' at the close of Lesson 4 next time you teach this sequence.

2. MISCONCEPTION TRACKING: Did any learners persistently conflate ectotherm/endotherm during the trade-off discussion? If yes, consider building a dedicated 5-minute ectotherm/endotherm warm-up into the start of Lesson 6.

3. APPLICATION QUESTION (Part C): Review the collected notebooks. What percentage of learners used all four required terms correctly? If fewer than 60% scored 3/3, the concept of 'mixing reducing efficiency' needs reinforcement — plan a 10-minute retrieval activity at the start of Lesson 6 before introducing the four-chambered heart.

4. MODEL QUALITY: Are learners' cumulative models growing in complexity and accuracy, or are they becoming cluttered and hard to read? If the latter, consider providing a structured model template for Lesson 6 onward.

5. DQB MOMENTUM: Read the 'I still wonder' sticky notes tonight. Do learners' new questions anticipate the content of Lesson 6 (four-chambered heart, double circulation)? If yes, the storyline is working. If learners are asking questions far outside the scope of the sequence, consider re-anchoring at the start of Lesson 6 with a direct reference to the driving question.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed colour-coded diagrams of the frog's three-chambered heart (two atria, one ventricle) and the reptilian heart with a partially divided ventricle. They traced the path of a red blood cell through the frog's circulatory system using coloured pencils, noting where oxygenated and deoxygenated blood mix in the single ventricle. They also analysed a comparative data table showing estimated % oxygenation of blood reaching body tissues across five animal groups and plotted a bar graph to visualise the efficiency differences.
What did I learn?	Learners learned that the frog has a three-chambered heart in which oxygenated blood (from lungs and skin) and deoxygenated blood (from body tissues) mix in the single ventricle, reducing the efficiency of oxygen delivery to tissues. Reptiles have a partial septum in the ventricle that reduces — but does not eliminate — this mixing. This structural difference is a trade-off: frogs and reptiles are ectotherms with lower energy demands, so a fully separated circulation is not necessary for their survival. The four-chambered heart (as in Kipchoge) is needed only when oxygen demand is very high and sustained.
How does this explain the phenomenon?	Learners explained that the mixing of oxygenated and deoxygenated blood in the frog's single ventricle means body tissues receive blood that is only partially oxygenated, limiting the animal's sustained aerobic capacity. This is why a Rift Valley frog cannot sustain high-intensity activity like marathon running. They connected this to the driving question by reasoning that Kipchoge's heart must have complete separation of the two blood types — a feature that allows near-100% oxygenated blood to reach his muscles at world-record pace. Learners updated their cumulative Animal Transport System Model to include the amphibian and reptilian tiers and revised their Efficiency Meter to show the relative oxygen delivery performance of each animal group studied so far.

LESSON 6 (40 minutes): Transport in Mammals — The Four-Chambered Heart (Heart Dissection and Circuit Mapping)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners examine a sheep or goat heart obtained from a local butchery (Nairobi, Kisumu, Eldoret, or nearest town market) and use direct observation to identify the structural features of the four-chambered mammalian heart, then map the complete double circulatory system to explain why this design supports the extreme oxygen demands of a mammal like Kipchoge.
Knowledge	Learners will be able to: (1) identify and label the four chambers (right atrium, left atrium, right ventricle, left ventricle), the septum, the atrioventricular valves (bicuspid and tricuspid), the semilunar valves, the aorta, vena cava, pulmonary artery, and pulmonary vein on a real or diagrammatic heart; (2) trace the pathway of blood through the pulmonary circuit (right side of heart to lungs and back) and the systemic circuit (left side of heart to body and back); (3) explain how complete separation of oxygenated and deoxygenated blood by the septum maximises oxygen delivery to tissues; (4) compare left ventricular wall thickness to right ventricular wall thickness and relate this to the greater pressure needed to pump blood around the systemic circuit.
Skills	Dissection observation skills (or guided diagram annotation if live specimen is not used); recording labelled diagrams; tracing fluid pathways using colour-coding; comparing structural features across specimens examined in lessons 3 to 6; constructing an updated cumulative model.
Attitudes	Curiosity and careful observation during dissection; respect for the specimen as a learning resource; appreciation that structures seen in a butchery heart are the same as those sustaining every mammal, including Kipchoge; willingness to revise prior model based on new evidence.
Key Inquiry Question	How does the structure of the mammalian heart — specifically its four chambers and double circuit — allow it to deliver oxygen far more efficiently than the hearts of insects, fish, frogs, or reptiles?
Purpose in Storyline	This is the pivotal OBSERVE lesson in the evidence-gathering sequence. Learners have built up a progression from open insect circulation (Lesson 3) through fish single circulation (Lesson 4) to the partially mixed amphibian and reptile hearts (Lesson 5). Now they encounter the fully separated, double-circuit mammalian heart. This lesson provides the structural evidence needed to explain WHY Kipchoge can sustain world-record pace and recover so rapidly. It is the largest single model revision of the sequence and directly sets up Lesson 7 (the pumping mechanism and cardiac cycle) and the final comparative explanation in Lessons 11 and 12.
Safety Notes	(1) Use gloves — one pair per learner if budget allows, or shared group gloves. (2) The specimen should be fresh or recently refrigerated; inspect for unusual smell before use. (3) Dissection instruments (scalpel or razor blade) must be handled blade-down and passed handle-first. (4) Learners with known blood-related anxieties should be allowed to observe from a partner group. (5) Wash hands with soap and water after handling specimen. (6) Dispose of the specimen in a sealed bag and place in the school refuse — do not leave on benches. (7) If a real heart is not available, a large detailed diagram printed on A3 paper or a labelled model is an acceptable substitute for all learning objectives.

B. LESSON OVERVIEW
Learners begin by revisiting their cumulative model from Lesson 5 and predicting what new features they expect the mammalian heart to have. They then conduct a guided observation of a sheep or goat heart (procured from a local butchery at low cost), identifying all four chambers, the septum, the valves, and the major vessels. Using coloured pens, they trace the pulmonary and systemic circuits on a diagram. In the Explain phase, they articulate why complete separation of blood circuits is an advantage over the frog and reptile hearts studied in Lesson 5. They update their cumulative model and post new evidence cards to the Driving Question Board, bringing

the class closer to a full answer to the driving question about Kipchoge.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Tissue Specific Gene..:Image of four cells Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners open their workbooks to their cumulative circulatory model (updated through Lesson 5). Working in pairs for 3 minutes, they respond to the prompt written on the board: 'You have now seen the hearts of a grasshopper, a tilapia, and a frog. Write two predictions: (1) What new structural features do you expect the mammalian heart to have? (2) How do you think these features will help a mammal like Kipchoge run at 21 km/h for over two hours?' Learners write individually first (1 minute), then share with a partner (1 minute), then two or three pairs share with the class.	Teacher points to the progression chart on the wall showing the hearts studied so far (open vessel, two chambers, three chambers) and says: 'We have been following a pattern. Each time we moved to a more complex animal, the heart changed. Today we open a heart that came from the same local butchery where your family might buy nyama. What do you predict you will find inside?' [WAIT 60 seconds in silence — do not fill the silence.] After pairs share, teacher says: 'Hold those predictions in your mind. We are about to test them with our own hands.' Teacher does NOT confirm or deny predictions at this stage.	Prediction anchoring — learners commit to a written prediction before observing, which creates cognitive tension when observations confirm or contradict expectations. This activates prior knowledge from Lessons 3 to 5 and primes attention during dissection.	Teacher circulates and reads two or three prediction cards without comment; listens for whether learners are drawing on the progression built in earlier lessons (e.g., referencing partial septum from Lesson 5). Misconceptions to note: some learners may predict only one extra chamber; some may not yet connect wall thickness to pumping pressure.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Tissue Specific Gene..:Image of four cells Source: MIT Blossoms 🔍 Search ARES for similar readings	Groups of four receive one sheep or goat heart (or a large A3 printed diagram if hearts are unavailable). A printed observation guide with labelled blanks is distributed. Step 1 (3 min): Learners observe the external surface — identify the apex, base, coronary vessels, and distinguish left from right by wall thickness when squeezed gently. Step 2 (4 min): Teacher or a designated learner cuts the heart open along the coronal plane (or learners view the pre-cut specimen). Learners locate the right atrium, right ventricle, left atrium, left ventricle, and the septum. They record and sketch in their guides. Step 3 (3 min): Learners probe the atrioventricular	Before cutting, teacher says: 'This heart kept a sheep alive. It beat approximately 70 to 80 times per minute. Kipchoge is a human with a heart the same size. What does that tell you?' [WAIT 30 seconds.] During observation, teacher circulates asking targeted questions: 'Squeeze the left wall and then the right wall. What do you feel? Why might that difference exist?' and 'Where does the blood go when it leaves this side? Where does it come back from?' When learners find the valves, teacher says: 'These valves exist to prevent blood from going backwards. What would happen to Kipchoge if one of these valves leaked during a race?'	Direct observation of a real specimen makes structural features concrete and memorable. Colour-coded arrow tracing forces learners to actively construct the pathway rather than passively copy it. The squeeze test for wall thickness creates a tactile anchor for the concept that the left ventricle generates higher pressure than the right.	Teacher checks that every learner has correctly colour-coded both circuits on their diagram before the class moves on. A quick show-of-hands check: 'Hold up your diagram — red arrows on the left side, blue on the right?' Groups with reversed colour-coding are gently redirected: 'Check — which side receives blood coming back from the lungs, already loaded with oxygen?'

	valves (tricuspid on the right, bicuspid/mitral on the left) and the chordae tendineae using a pencil or thin probe. Step 4 (2 min): Learners locate and compare the pulmonary artery and aorta at the top of the heart, noting wall thickness differences. Step 5 (2 min): Using a red pen and a blue pen, learners draw arrows on their diagram tracing: (red) oxygenated blood from left ventricle through aorta to body; (blue) deoxygenated blood from right ventricle through pulmonary artery to lungs; and completing both loops back to the heart.	[WAIT 45 seconds before taking responses.] Teacher uses anatomical names consistently: right atrium, left ventricle, bicuspid valve — not just 'this chamber' or 'that flap.'		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Tissue Specific Gene..:Image of four cells Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners individually write a three-sentence 'circuit summary' in their workbook using this sentence scaffold: (1) 'In the pulmonary circuit, blood travels from the [blank] to the [blank] to pick up oxygen, and returns to the [blank].' (2) 'In the systemic circuit, blood travels from the [blank] through the [blank] to deliver oxygen to [blank], and returns through the [blank].' (3) 'Because the septum completely separates oxygenated and deoxygenated blood, the mammalian heart is more efficient than the frog heart because [blank].' After writing, groups discuss and compare their explanations (3 min). Class shares: teacher cold-calls two groups to read their third sentence aloud. Teacher then poses the comparison question: 'In Lesson 5 we said the frog has a trade-off — mixed blood is acceptable because frogs have low energy demands. What is different about Kipchoge?' Learners respond in pairs (think-pair-share, 2 min).	Teacher writes the word EFFICIENCY on the board and says: 'Efficiency here means: how much oxygen arrives at the muscles per heartbeat. Does the frog get the maximum possible oxygen in each pump? Does Kipchoge? Why is that difference important when you are running at 21 kilometres per hour for two hours?' [WAIT 60 seconds — this is a high-order question and silence is productive.] If learners struggle, teacher offers a bridging hint: 'Think about what happens when oxygenated and deoxygenated blood mix in the ventricle — what is the oxygen concentration of the blood that eventually reaches the muscles compared to pure oxygenated blood?' After learners explain, teacher confirms and extends: 'Kipchoge does not have a special heart because he is a champion. He has a mammalian heart. The difference is that his heart has been trained — we will explore that in Lesson 7. But the structural advantage — the four chambers	Sentence scaffolds lower the cognitive load of written explanation while ensuring learners use correct anatomical vocabulary. The comparative question directly links back to Lesson 5 learning and advances the storyline. Cold-calling after pair work ensures accountability.	Teacher listens for whether learners can articulate the efficiency argument (separated circuits mean tissues receive fully oxygenated blood) rather than simply listing the chambers. Learners who say only 'four chambers is better than three' are prompted: 'Better in what way, specifically? What does that extra chamber actually change about the blood that arrives at Kipchoge's leg muscles?'

		and double circuit — that belongs to every mammal in this room.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Tissue Specific Gene..:Image of four cells Source: MIT Blossoms 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the first (green — new evidence), learners write one key fact from today that advances the driving question. On the second (orange — new sub-question), learners write one new question today's lesson raised. Examples expected: green — 'The left ventricle wall is thicker because it pumps blood the full length of the body'; orange — 'How does the heart know when to beat faster, and how does Kipchoge control his recovery?' Learners post their cards on the DQB in the designated sections. Teacher reads five cards aloud and places them in conversation with earlier cards already on the board from Lessons 3 to 5.	Teacher stands at the DQB and says: 'Three lessons ago, someone wrote on this board that they wondered why a grasshopper can survive without a proper heart. Let us look at what we know now.' Teacher physically traces the progression on the board from the insect card cluster to the mammal card cluster. 'We have been building one big answer for six lessons. Today we made the biggest jump. What is now answered that was not answered in Lesson 5?' [WAIT 45 seconds.] Teacher then previews: 'There are still questions on this board about HOW the heart beats and WHY Kipchoge recovers so fast. Those are exactly what we investigate in Lesson 7.'	The DQB update makes the cumulative nature of the evidence-gathering sequence visible. By physically connecting today's cards to earlier ones, the teacher reinforces that science knowledge builds progressively. The preview creates anticipatory motivation for Lesson 7.	Teacher scans green cards for accuracy and orange cards for sophistication. Misconceptions visible on green cards (e.g., 'mammals have more blood than frogs') are addressed briefly: 'Volume of blood is not the key — it is what happens to the blood as it travels. Check your circuit diagram.'
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Tissue Specific Gene..:Image of four cells Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners open their cumulative model page (maintained since Lesson 1). Using the model revision protocol established in earlier lessons, they add the mammalian heart to their comparative diagram. Specifically: (1) Draw the four-chambered heart with the septum clearly shown and both circuits colour-coded; (2) Annotate with the names of the four chambers, at least four named vessels (aorta, vena cava, pulmonary artery, pulmonary vein), and the two valves (tricuspid, bicuspid); (3) Add a comparative efficiency arrow alongside the insect, fish, frog, and reptile hearts already in the model, indicating relative separation of oxygenated and deoxygenated blood (none / partial / complete); (4) Write one 'model revision sentence' stating	Teacher circulates and looks specifically for correct placement of the pulmonary vein entering the LEFT side of the heart (a common reversal error). If found, teacher says: 'The pulmonary vein brings blood FROM the lungs — blood that has just picked up oxygen. Which side of the heart should receive that oxygenated blood, ready to pump it to the whole body?' [WAIT 30 seconds.] Teacher also prompts the efficiency comparison: 'In your model, can you see the trend from Lesson 3 to today? What is the direction of change in terms of blood separation?' With 2 minutes remaining, teacher says: 'Write your model revision sentence now. This is the most important sentence in your model — it explains why today changed your	The cumulative model is the primary artifact of the 12-lesson sequence. Each revision makes prior thinking visible alongside updated thinking, supporting metacognitive awareness of how understanding has developed. The revision sentence requires learners to explicitly compare new evidence to prior belief — a core scientific practice.	Exit check: teacher asks three learners to read their model revision sentence aloud. Criteria for quality: does the sentence reference a specific structural feature (not just 'four chambers'), does it explain a functional consequence (not just 'more efficient'), and does it connect to the phenomenon (Kipchoge or another mammal context)? Sentence quality informs how much recapping is needed at the start of Lesson 7.

	what they changed from their Lesson 5 model and why. Target time: 5 minutes. Learners who finish early add a Kipchoge connection note: 'This structure helps Kipchoge because...'	thinking more than any previous lesson.'		

D. TEACHER REFLECTION

After the lesson, consider: (1) Were all learners able to correctly trace both circuits, or did some consistently reverse the pulmonary vein connection? If so, plan a 3-minute clarification at the start of Lesson 7 using the diagram only — no re-dissection needed. (2) Did the dissection (or diagram) generate genuine curiosity, or did some learners disengage? If engagement was low, consider whether the specimen was large and clear enough — a larger goat heart from a butchery in towns like Nakuru, Nyeri, or Kisumu is often more visually accessible than a smaller sheep heart. (3) Were learners able to articulate the efficiency argument in their Explain phase, or did they only list structural features? The efficiency concept (fully oxygenated blood reaches tissues because there is no mixing) is the conceptual core of this lesson and must be secure before Lesson 7. (4) Review the DQB orange cards — are learners asking questions about the pumping mechanism and heart rate control? These are exactly the questions Lesson 7 is designed to answer. If no learner raised them, seed the question at the start of Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a real sheep or goat heart (or detailed diagram): four distinct chambers separated by a thick muscular septum; the left ventricular wall noticeably thicker than the right ventricular wall; valves between the atria and ventricles preventing backflow; the aorta and pulmonary artery leaving from the top; the vena cava and pulmonary veins entering at the top; coronary vessels on the outer surface supplying the heart muscle itself.
What did I learn?	The four-chambered double circulation of the mammalian heart explains the efficiency advantage over the three-chambered frog heart and the two-chambered fish heart. Full separation of oxygenated and deoxygenated blood means muscles receive blood with the maximum possible oxygen concentration at every heartbeat. This is the structural basis for the high metabolic activity of endothermic mammals. The question of HOW the heart beats rhythmically and how Kipchoge controls his heart rate during and after a race remains open — this is the focus of Lesson 7.
How does this explain the phenomenon?	The jump from the reptile partial septum (Lesson 5) to the complete mammalian septum (Lesson 6) explains why mammals can sustain oxygen demands far beyond what a frog or reptile could support. The driving question advances: we now have the structural evidence to explain why Kipchoge does not run out of oxygen at world-record pace. What remains is to explain the pumping mechanism, the role of the sinoatrial node in controlling heart rate, and why his recovery is so rapid — which Lesson 7 will address directly.

LESSON 7 (2 lessons (80 minutes)): Pumping Mechanism of the Mammalian Heart: The Cardiac Cycle, Heart Sounds, and the SAN	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the pumping mechanism of the mammalian heart, including the cardiac cycle (systole and diastole), heart sounds, and the role of the sino-atrial node (SAN), and to connect these mechanisms to the phenomenon of Eliud Kipchoge's extraordinary cardiac recovery.
Knowledge	Learners will be able to: (a) describe the stages of the cardiac cycle (atrial systole, ventricular systole, and diastole); (b) explain the origin and significance of heart sounds ('lub-dub'); (c) explain the role of the sino-atrial node (SAN) as the heart's pacemaker; (d) relate cardiac output to heart rate and stroke volume.
Skills	Learners will be able to: (a) measure resting and post-exercise pulse rate accurately using the radial and carotid arteries; (b) record and analyse pulse-rate data in a table and graph; (c) conduct a fair investigation comparing resting versus post-exercise heart rate recovery; (d) interpret heart-rate recovery curves.
Attitudes	Learners will: (a) appreciate the elegance of the heart's self-regulating pumping mechanism; (b) show curiosity and rigour in collecting and recording pulse-rate data; (c) value physical fitness as a factor that improves cardiac efficiency, inspired by Kipchoge's example.
Key Inquiry Question	Why is transport important in animals? How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	In Lessons 1–6 learners established WHY closed, double circulatory systems evolved and how blood vessels and blood components are structured. Lesson 7 now opens the 'engine room': HOW does the heart actually pump? Learners gather direct, first-hand evidence from their own bodies — pulse rate data before and after exercise — and map those numbers onto the cardiac cycle model. This is the pivotal evidence-gathering lesson that allows learners to explain why Kipchoge's heart rate drops so rapidly: a trained heart has a higher stroke volume, a powerful SAN rhythm, and efficient diastolic refilling, so it needs fewer beats per minute to deliver the same cardiac output.
Safety Notes	Ensure no learner with a known cardiac condition, asthma, or injury participates in the exercise component without teacher clearance. Exercise must be done on a flat, uncluttered surface. Stop any learner who feels dizzy or unwell immediately. Wash hands before taking pulse measurements to avoid cross-contamination if shared stopwatches are used.

B. LESSON OVERVIEW

Learners begin by revisiting the Driving Question Board (DQB) to surface what they already know about the heart's pumping action. They predict what happens inside the heart during one heartbeat. They then conduct a structured pulse-rate investigation (resting, immediately post-exercise, and 5-minute recovery), recording data in a table and sketching a recovery curve. The teacher guides sensemaking through a cardiac cycle diagram, auscultation simulation (tapping desk to mimic 'lub-dub'), and an explanation of the SAN. Learners connect their own recovery data to Kipchoge's recovery curve introduced in Lesson 1, building a mechanistic explanation. The lesson closes with model revision on the DQB and individual exit-ticket responses to a focused question about the SAN's role.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Anatomy of a skeletal muscle fiber Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Cycle (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a still image of Kipchoge crossing the finish line at the Berlin Marathon alongside the heart-rate recovery graph introduced in Lesson 1. The teacher poses the prompt: 'Inside Kipchoge's chest right now, his heart is beating about 170 times per minute. What do you think is physically happening inside the heart with EACH of those beats? In your exercise book, draw a simple sketch of the heart and annotate what you think happens during ONE beat — where does blood go, what opens or closes, what squeezes?' Learners work silently for 4 minutes on their annotated sketch. Pairs then share predictions for 2 minutes (Think-Pair-Share). The teacher cold-calls 4 pairs to share their ideas, recording key prediction words on the board (e.g., 'squeeze,' 'pump,' 'valves open,' 'blood pushed out') without correcting yet. The teacher notes any misconceptions (e.g., 'the heart sucks blood in') on a sticky-note strip labelled 'To Investigate.'	Display the Kipchoge recovery graph on the board or read the data aloud if no projector is available. Say: 'Look at this curve — Kipchoge's heart rate falls from 172 to 37 beats per minute in under 4 minutes. Before we can explain that, we need to understand what happens in a SINGLE heartbeat. In your book, sketch the heart and show me what you think happens in one beat.' [Allow 4 minutes silent work — do not prompt further.] After pairs share, cold-call exactly 4 pairs using name cards or a class list. Record their words verbatim on the board. Say: 'These are our predictions — we are going to test every single one of them today.' Circle any misconception gently and write it on the 'To Investigate' strip. Apply 10–15 seconds wait time after each cold-call before accepting responses.	Think-Pair-Share Prediction Sketch: Learners externalise prior knowledge as a labelled diagram before instruction. This activates schema about the heart's structure (from Lesson 6) and creates cognitive dissonance when the cardiac cycle model is introduced — the gap between their sketch and the actual mechanism drives curiosity and attention during the Observe phase.	Teacher circulates during the 4-minute silent sketch phase and scans 6–8 books. Observable indicator: Can the learner correctly place arrows showing blood entering and leaving the heart? Are valves mentioned? If fewer than half the class include valves in their sketch, the teacher flags this as a prerequisite gap to address explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** Anatomy of a skeletal muscle fiber Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Cycle (Flexbook) Source: CK-12 🔍 Search ARES for similar	ACTIVITY 1 — Pulse Rate Investigation (Resting vs. Post-Exercise Recovery): Learners locate their radial pulse (wrist) or carotid pulse (neck) with guidance. Each learner counts beats for 30 seconds and doubles the value to get beats per minute (BPM) — this is their RESTING heart rate. Data is entered into a shared class table on the board and into individual exercise books. Learners then perform 2 minutes of vigorous star jumps in the open area adjacent to the classroom (or in the corridor). Immediately upon stopping, each learner counts pulse for 30	Before Activity 1, demonstrate correct finger placement for the radial pulse on your own wrist. Say: 'Use two fingers — not your thumb, which has its own pulse — press lightly on the inside of your wrist just below the base of your thumb.' Walk the room during the count; if a learner cannot find their pulse after 30 seconds, direct them to the carotid artery on the side of the neck. For the exercise phase, stand at the door to supervise. Count down: 'Star jumps — begin! … 2 minutes — stop! Find your pulse NOW — you have 30 seconds starting … now!'	Multimodal Evidence Triangulation: Learners gather evidence through three complementary modes — kinaesthetic (their own pulse data), auditory (lub-dub rhythm tapping), and visual (cardiac cycle diagram). Triangulating across these three modes builds a richer, more durable mental model of the cardiac cycle than any single mode alone. The personal pulse-rate data is the anchor: it is real, variable, and directly comparable to the Kipchoge phenomenon data, making the subsequent explanation personally meaningful.	Observable indicator 1: Each learner's data table has four pulse readings and a plotted recovery curve with axes labelled. Teacher checks 8 books during the 5-minute recovery wait. Observable indicator 2: During the lub-dub tapping, the teacher watches whether learners can name the correct valves without looking at notes when cold-called — target: 3 out of 3 cold-called learners correctly identify at least one valve pair. If not, the teacher re-teaches valve function before moving to the Explain phase.

readings	seconds (BPM recorded as 'Post-exercise: 0 min'). They sit, and counts are repeated at 2 minutes and 5 minutes post-exercise. Each learner draws a recovery curve (time on x-axis, BPM on y-axis) in their exercise books using their four data points. ACTIVITY 2 — Heart Sounds Simulation ('Lub-Dub'): With learners seated, the teacher demonstrates a 'lub-dub' rhythm by tapping the desk twice in rapid succession, pausing, then repeating: tap-TAP … pause … tap-TAP. Learners echo the rhythm. The teacher then reveals: 'The first sound — lub — is the closing of the bicuspid and tricuspid valves at the START of ventricular systole. The second sound — dub — is the closing of the aortic and pulmonary semi-lunar valves at the END of ventricular systole. The pause is diastole.' Learners tap the rhythm on their desks while the teacher narrates each stage of the cardiac cycle aloud. ACTIVITY 3 — Cardiac Cycle Diagram Reading: Learners are given (or copy) a four-panel diagram showing: (i) Atrial systole, (ii) Ventricular systole, (iii) Complete diastole, (iv) Pressure-volume axes. Learners annotate which valves are open/closed in each panel and draw arrows showing blood flow direction.	Record the class median resting BPM and post-exercise BPM on the board next to Kipchoge's values (37 BPM resting; ~170 BPM racing). Say: 'Notice anything interesting about our resting rates versus Kipchoge's?' [10 seconds wait — cold-call 2 learners.] For Activity 2, slow the lub-dub rhythm deliberately: 'I want you to FEEL the two sounds as events — something is CLOSING each time.' For Activity 3, circulate and ask targeted questions: 'Which valves are open during ventricular systole? Point to them on your diagram.' Apply 10–15 seconds wait time. Cold-call 3 learners to narrate one panel each aloud to the class.		
Explain Phase 📹 **VIDEO:** Anatomy of a skeletal muscle fiber Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	The teacher draws a simplified cardiac cycle timeline on the board, marking three phases: Atrial Systole (0.1 s) → Ventricular Systole (0.3 s) → Diastole (0.4 s) = 0.8 s per beat = 75 BPM at rest. Learners calculate: if the whole cycle takes 0.8 seconds, how many cycles happen per minute?	Write the timeline on the board slowly, narrating: 'At rest, one complete heartbeat takes 0.8 seconds — most of that is REST (diastole), not contraction.' Say: 'Calculate for me — at 170 beats per minute, how long is one cardiac cycle? Work it out in your books.' [Wait 90 seconds.] Cold-	Quantitative Reasoning with Anchored Data: Learners use real numbers from the Kipchoge phenomenon (37 BPM resting, 170 BPM racing, ~5 L/min cardiac output) to derive stroke volume through simple arithmetic. This strategy transforms an abstract concept (cardiac efficiency) into a	Observable indicator: Each learner correctly calculates stroke volume for both Kipchoge (≈135 mL) and an average student (≈67 mL) and writes one sentence explaining the difference. Teacher collects 5 random exercise books to scan calculations. If more than 30% of learners make an arithmetic error

📖 **HTML:** [Cell Cycle (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	(Answer: 60 ÷ 0.8 = 75 BPM.) They then calculate: at Kipchoge's racing rate of 170 BPM, how long does one cardiac cycle last? (60 ÷ 170 ≈ 0.35 s.) Learners discuss in pairs: 'Which phase of the cardiac cycle gets shorter when heart rate increases? What are the consequences of a very short diastole?' The teacher introduces the SINO-ATRIAL NODE (SAN): a patch of specialised muscle in the wall of the right atrium that generates its own electrical impulse, setting the rhythm for the whole heart. The teacher uses the analogy of a conductor at a Nairobi Symphony Orchestra concert — the SAN gives the beat; the rest of the heart follows. Learners annotate the SAN onto their cardiac cycle diagram. The teacher then introduces CARDIAC OUTPUT = Heart Rate × Stroke Volume, and poses: 'Kipchoge's resting heart rate is only 37 BPM. Yet his resting cardiac output is NORMAL (about 5 litres per minute). How is this possible?' Learners calculate: if cardiac output = 5 L/min and heart rate = 37 BPM, then stroke volume = 5000 mL ÷ 37 ≈ 135 mL per beat. They compare this to an average untrained person: 5000 ÷ 75 ≈ 67 mL per beat. The class discusses why a bigger stroke volume (due to a larger, stronger ventricle) means fewer beats are needed — directly explaining Kipchoge's low resting rate and fast recovery.	call 2 learners to share their working. When introducing the SAN, say: 'Has anyone been to a live music performance — maybe at the Carnivore grounds in Nairobi, or a school concert? Who sets the rhythm for everyone else? [Wait 10 seconds.] The SAN is the conductor of your heart — it fires an electrical signal that spreads across the atria, down to the AV node, and into the ventricles. Every beat you just counted on your wrist started in the SAN.' For the cardiac output calculation, write the formula on the board and walk through the Kipchoge example step by step. Then say: 'Now YOU calculate the stroke volume for an average Form 3 student whose resting heart rate is 75 BPM and cardiac output is 5 litres per minute.' [Wait 2 minutes.] Cold-call 3 learners for their answers and working. Pose the synthesis question: 'So why does Kipchoge's heart recover faster than ours?' [Wait 15 seconds — cold-call 2 learners.]	concrete, personally meaningful calculation. By comparing their own pulse recovery curves to Kipchoge's, learners construct a causal explanation — not just a description — of what makes a trained heart different. This is a core NGSS Science and Engineering Practice: Using Mathematics and Computational Thinking.	or cannot write the explanatory sentence, the teacher pauses and re-models the calculation before proceeding to the DQB phase.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Anatomy of a	Learners return to the class DQB (a wall display or shared chart paper running since Lesson 1). In pairs, they are given two sticky notes each: one GREEN ('Evidence I now have') and one	Distribute sticky notes (or torn paper strips if sticky notes unavailable). Say: 'You have 3 minutes — GREEN note: what EVIDENCE did we collect today that helps answer our driving	Driving Question Board Evidence Mapping: The DQB is a living, public record of the class's growing understanding. By requiring learners to explicitly categorise today's findings as 'evidence' and	Observable indicator: Each pair's GREEN sticky note contains a specific, accurate piece of evidence (not a vague statement like 'we learned about the heart'). Teacher scans all GREEN notes

skeletal muscle fiber Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Cycle (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ORANGE ('New question this lesson raised'). On the GREEN note, each pair writes one piece of evidence from today's lesson that helps answer either DQB question: 'Why is transport important in animals?' or 'How does structure relate to complexity?' On the ORANGE note, they write one new question — for example: 'What happens to the SAN if the heart is diseased?' or 'Can training actually change stroke volume over time?' or 'Why does adrenaline speed up the heart?' Pairs post their notes on the DQB. The teacher reads 3–4 new ORANGE questions aloud and says: 'These are excellent scientific questions. Some of them we will answer in the next lessons; some of them are real research questions that Kenyan cardiologists at Kenyatta National Hospital are still investigating.' The teacher updates the DQB 'Storyline Progress Bar' (a simple horizontal arrow) to mark Lesson 7 as complete, noting: 'We now understand HOW the mammalian heart pumps — this is the engine behind Kipchoge's performance.'	question? ORANGE note: what NEW question do you now have?' Circulate and read over shoulders. After posting, read 3 GREEN notes aloud and affirm the evidence: 'Yes — the fact that Kipchoge's stroke volume is twice ours at rest IS direct evidence for why his system is more efficient.' Then read 3 ORANGE notes and explicitly validate them as scientifically interesting. Update the Storyline Progress Bar visibly. Say: 'Every lesson we add to this board, we get closer to fully explaining the Kipchoge phenomenon — and to answering whether a grasshopper could ever run a marathon.'	to generate new questions, this strategy develops the NGSS practice of Asking Questions and Defining Problems, and makes metacognitive progress visible to the whole class. The contrast between GREEN (answers) and ORANGE (new questions) reinforces that science is an iterative process — knowledge generates new questions rather than closing inquiry.	after posting. If more than 3 pairs have vague or incorrect GREEN notes, the teacher addresses these at the start of the next lesson as a brief 'evidence audit.'
Model Building Phase 📹 **VIDEO:** Anatomy of a skeletal muscle fiber Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Cycle (Flexbook) Source: CK-12	Learners open the cumulative 'Transport System Model' they have been building since Lesson 1 (a labelled diagram in their exercise books showing the full circulatory pathway, blood composition, and vessel types added in prior lessons). Today they add TWO new layers to their model: (1) They annotate the heart on their diagram with the three phases of the cardiac cycle (atrial systole → ventricular systole → diastole) and mark the SAN's location with a star and the label 'pacemaker — generates electrical	Say: 'Open your cumulative model — the diagram you have been building since Lesson 1. Today you are going to make it more powerful by adding the ENGINE: the cardiac cycle and the SAN.' Display or draw a reference model on the board with the SAN starred in the right atrial wall. Walk around and check that learners are adding both layers. With 8 minutes remaining, say: 'Now write your 3-sentence Model Narration. These three sentences must be in YOUR OWN WORDS — they are your exit ticket. I will collect them at the	Cumulative Model Revision (Progressive Diagrammatic Modelling): Learners do not build a new model each lesson — they revise and enrich the SAME model, making their growing understanding visible as accumulating layers of annotation. This mirrors how scientists refine models as new evidence arrives, and directly supports the NGSS practice of Developing and Using Models. The 3-sentence Model Narration forces learners to translate visual/diagrammatic understanding into precise	Observable indicator: The 3-sentence exit ticket correctly names the SAN as the pacemaker (Sentence 1), correctly identifies valve closure as the cause of heart sounds (Sentence 2), and correctly attributes Kipchoge's faster recovery to higher stroke volume / lower resting heart rate (Sentence 3). Success threshold: 70% of collected exit tickets meet all three criteria. Where a sentence is incorrect, the teacher uses this as a springboard for a targeted 5-minute re-explanation at the opening of Lesson 8.

🔍 Search ARES for similar readings	rhythm.' (2) They add a speech bubble or annotation box next to Kipchoge's silhouette (drawn or pasted on their model in Lesson 1) that reads: 'My SAN fires at only 37 times/minute at rest because each beat pumps ~135 mL — my ventricles are large and strong from training.' Learners also add a comparison annotation for an average Grade 10 student: '~75 BPM at rest; ~67 mL per beat.' Finally, learners write a 3-sentence 'Model Narration' below their diagram: Sentence 1 — what the SAN does; Sentence 2 — how the cardiac cycle produces the 'lub-dub' sounds; Sentence 3 — why Kipchoge's heart recovers faster than ours. These three sentences serve as the exit ticket for the lesson.	door.' Collect the exit slips (learners can tear the page corner or write on a slip) as learners leave. Review 6–8 exit tickets before the next lesson to identify persistent misconceptions about the SAN or cardiac cycle timing.	language, surfacing any remaining gaps in mechanistic reasoning.	

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners' pulse-rate recovery curves show enough variation across the class to generate meaningful discussion about fitness and cardiac efficiency — or were the data too similar to be interesting? If so, consider inviting the school's athletics coach (or a learner who trains regularly) to share their resting heart rate in a future lesson. (2) Were learners able to make the conceptual leap from 'my heart beats faster after exercise' to 'my heart beats faster because stroke volume has not increased enough to meet demand'? If this causal chain was unclear, plan a brief 5-minute re-teach at the start of Lesson 8 using the cardiac output equation as an anchor. (3) Did the lub-dub tapping simulation adequately convey the link between valve closure and sound — or did learners treat it as a mnemonic without mechanistic understanding? If so, consider using a stethoscope (if available from the school clinic) or a recorded heart sound in Lesson 8. (4) Review the 3-sentence exit tickets: what proportion of learners correctly identified the SAN as the origin of the heart's rhythm rather than the brain? This is a common misconception that must be resolved before Lesson 8 (regulation of heart rate by the autonomic nervous system). (5) Consider whether the calculation activity (cardiac output = heart rate × stroke volume) was appropriately scaffolded for learners with weaker numeracy — if many made arithmetic errors, provide a worked example template in the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured their own resting and post-exercise pulse rates, plotted a heart-rate recovery curve, and compared it to Kipchoge's recovery data. They also observed the lub-dub rhythm through a desk-tapping simulation and annotated a four-panel cardiac cycle diagram showing atrial systole, ventricular systole, and diastole.
What did I learn?	The heart pumps through a repeating three-phase cardiac cycle (atrial systole → ventricular systole → diastole). Heart sounds ('lub-dub') are caused by the closing of valves — bicuspid/tricuspid valves at the start of ventricular systole, and semi-lunar valves

	at its end. The sino-atrial node (SAN) in the right atrial wall generates the electrical impulse that initiates each cycle, acting as the heart's intrinsic pacemaker. Cardiac output equals heart rate multiplied by stroke volume; Kipchoge's trained heart achieves normal cardiac output at only 37 BPM because each beat pumps approximately 135 mL — nearly double that of an untrained heart.
How does this explain the phenomenon?	Kipchoge's heart recovers so rapidly from 170 BPM to 37 BPM because his trained ventricles have a much larger stroke volume (~135 mL per beat vs. ~67 mL for an average person). His SAN only needs to fire 37 times per minute at rest to maintain adequate cardiac output, so once exercise demand drops, his heart rate falls quickly back to this efficient baseline. This is why a structurally superior mammalian heart — with its four chambers, powerful ventricular walls, and intrinsic SAN pacemaker — enables elite athletic performance in a way that the open circulatory system of a grasshopper, with no true heart chambers and no SAN, never could.

LESSON 8 (2 lessons (80 minutes)): The Lymphatic System: The Hidden Highway of Animal Transport

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To describe the structure and function of the lymphatic system, trace the pathway of lymph formation and its return to the blood, and relate this to oedema in Kenyan clinical health contexts.
Knowledge	Learners will be able to: (a) describe the structure of the lymphatic system including lymph capillaries, lymph vessels, lymph nodes, and lymphatic ducts; (b) explain how tissue fluid (lymph) is formed from blood plasma and returned to the bloodstream; (c) define oedema and explain its causes in terms of lymphatic or circulatory dysfunction.
Skills	Learners will be able to: (a) construct and annotate a diagram tracing lymph formation and return; (b) analyse clinical data to identify signs of oedema; (c) develop and revise a model of the body's fluid transport network to incorporate the lymphatic system.
Attitudes	Learners will: (a) appreciate the lymphatic system as an essential, often overlooked partner to the cardiovascular system; (b) show empathy and awareness toward community members affected by lymphatic conditions such as elephantiasis (filariasis), which is present in coastal Kenya.
Key Inquiry Question	Why is transport important in animals? How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	In Lessons 1–7, learners built understanding of how blood is pumped, how vessels distribute it, and how the heart's double circulatory system supports Kipchoge's extraordinary performance. But blood capillaries constantly leak plasma into surrounding tissues — so how does the body prevent dangerous fluid build-up? Lesson 8 reveals the lymphatic system as the essential 'return drainage highway' that keeps tissue fluid in balance. This directly advances the driving question: even Kipchoge's world-class circulatory system would fail without the lymphatic system silently maintaining fluid equilibrium. It also deepens comparison with simpler animals — invertebrates like the grasshopper lack a lymphatic system entirely, raising the question of how they manage fluid balance.
Safety Notes	No hazardous materials are used in this lesson. If using any body fluid analogies with water and salt demonstrations, ensure clean water is used. Sensitivity must be observed when discussing oedema and elephantiasis — some learners may have family members affected by these conditions, particularly in coastal counties. The teacher should frame all clinical examples with dignity and without stigma.

B. LESSON OVERVIEW

Learners begin by confronting a puzzling clinical image: a photograph (or teacher description) of a patient with severely swollen legs — a condition called oedema, seen in some hospitals along the Kenyan coast due to lymphatic filariasis. They connect this to the question: if Kipchoge's cardiovascular system is so efficient, what stops his leg tissues from swelling during a marathon? Through a structured inquiry cycle, learners discover that the lymphatic system is a one-way drainage network running parallel to the blood circulatory system. They trace how plasma leaks from blood capillaries into tissue spaces to form tissue fluid (lymph), how lymph capillaries absorb this fluid, and how it travels through lymph vessels and nodes before rejoining the blood via the thoracic duct and right lymphatic duct. Learners construct annotated pathway diagrams, analyse a simple filtration-and-reabsorption model using local materials, and update their cumulative anchor model. They close by explaining why oedema occurs when the lymphatic system is blocked or overwhelmed.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Lymphatic System - VHSG Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two images side by side on the board (or described vividly by the teacher): Image A — Eliud Kipchoge's legs, lean and muscular, crossing a finish line. Image B — a clinical photograph (or teacher sketch) of a patient with severely swollen lower legs consistent with oedema/elephantiasis, labelled as a case from Coast General Hospital, Mombasa. The teacher poses the Predict prompt: 'We know blood is pumped around the body under pressure. Some fluid from the blood leaks out of capillaries into the spaces between cells. In your exercise book, write your prediction: Where does that leaked fluid go? What would happen if it could never return to the blood? How might this connect to the swollen legs in Image B?' Learners write individual predictions silently for 4 minutes, then share with a shoulder partner for 2 minutes.	Display or describe both images clearly. Say: 'Do not worry about being right — your prediction is your current best thinking, and we will return to it at the end of the lesson.' After 4 minutes of silent writing, say: 'Turn to your shoulder partner. Share your prediction. You have 2 minutes.' After partner sharing, cold-call 3 learners using sticks or a class list. Ask each: 'What did you predict, and what evidence from our earlier lessons made you think that?' [WAIT TIME: 12 seconds after each cold-call before accepting the answer.] Record 3–4 key prediction phrases on the Driving Question Board under the column 'What we think we know.' Highlight any learner who mentions 'fluid balance' or 'drainage' — affirm this as strong scientific thinking without confirming the answer yet.	Predict-Share-Record: By externalising predictions before instruction, learners activate prior knowledge about blood pressure and capillary function from Lessons 5–7. Recording predictions publicly on the DQB creates accountability and gives learners a concrete claim to revise by the end of the lesson.	Observable indicator: Each learner has written a prediction of at least 2 sentences in their exercise book before sharing. Teacher scans written predictions during partner talk. Look for evidence that learners connect fluid leakage to pressure concepts from previous lessons. Flag any learners who write 'the fluid just stays there' — these learners will need additional scaffolding during the Observe phase.
Observe Phase 📹 **VIDEO:** Lymphatic System - VHSG Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — THE LEAKY PIPE MODEL (Concrete Analogy, 10 min): The teacher sets up a simple demonstration at the front using two transparent containers connected by a tube (representing an artery/vein circuit). A small sponge or cloth placed beside the tube represents body tissues. Teacher uses a syringe to push coloured water (representing blood plasma) through the tube, and demonstrates that small amounts 'leak' into the sponge. Learners observe and record: 'What happens to the fluid in the sponge? How does it get back into the tube?' This creates the	For Activity A: Perform the leaky pipe demonstration slowly. Ask: 'What do you notice about the sponge?' [WAIT 10 seconds.] Cold-call 2 learners. Then ask: 'So if the sponge keeps absorbing fluid and nothing drains it — what eventually happens?' [WAIT 12 seconds.] Cold-call 2 learners. For Activity B: Draw the diagram step by step on the board, pausing at each structure to say: 'Copy this label. What do you think this structure does based on its position?' [WAIT 10 seconds per structure, cold-call 1–2 learners each time.] When reaching lymph nodes, say: 'Hands up if you have	Concrete-to-Abstract Bridging with Annotated Diagram Construction: The physical leaky-pipe model gives learners a tangible anchor for an abstract process (plasma leakage and drainage). Constructing the annotated diagram — rather than simply copying one — requires learners to process each structure's function actively. Connecting the completed diagram back to Kipchoge's marathon closes the evidence loop and keeps the anchoring phenomenon central.	Observable indicator: Learners' completed diagrams correctly label all 6 structures (lymph capillary, lymph vessel, lymph node, thoracic duct, right lymphatic duct, subclavian vein) and include a directional arrow showing one-way flow. Teacher circulates and uses a checklist to note which learners have mislabelled or omitted the subclavian vein connection — this is the most commonly missed step. Cold-call responses during Activity C should include the phrase 'draining excess fluid' or equivalent — if fewer than 50% of cold-called learners achieve this, pause and re-explain capillary

	cognitive need for a drainage system. ACTIVITY B — TRACING THE LYMPHATIC PATHWAY (Structured Reading + Diagram, 20 min): Learners receive a one-page ARES information card (or the teacher draws a detailed labelled diagram on the board) showing: (1) Blood capillary leaking plasma → tissue fluid forms in intercellular spaces. (2) Blind-ended lymph capillaries absorbing tissue fluid → now called lymph. (3) Lymph vessels (with valves — like veins) carrying lymph toward the chest. (4) Lymph nodes (found in armpits, groin, neck — teacher notes: 'When you have a throat infection, you may feel swollen lumps under your jaw — those are your lymph nodes fighting infection'). (5) Thoracic duct (draining the left side of the body) and right lymphatic duct (draining the right side of the head, neck, and arm) both emptying into subclavian veins near the heart. Learners individually trace and annotate the pathway on a blank outline diagram, labelling: lymph capillary, lymph vessel, lymph node, thoracic duct, right lymphatic duct, subclavian vein. ACTIVITY C — CONNECTING TO THE PHENOMENON (Data Observation, 5 min): Teacher poses: 'During the Berlin Marathon, Kipchoge's capillaries were under enormous pressure. His muscles were flooded with blood. Enormous volumes of plasma were being forced into his muscle tissues. How did his legs NOT swell up during the race?' Learners annotate their diagram with a small arrow and note: 'Lymphatic system draining excess	ever felt a swollen lump under your jaw when you had tonsillitis or a bad throat infection.' Acknowledge hands, then say: 'That swelling is your lymph nodes — they are white blood cell factories that filter pathogens from the lymph. We will study this more in immunology.' For Activity C: After learners annotate their diagrams, cold-call 2 learners: 'Complete this sentence aloud: During the Berlin Marathon, Kipchoge's lymphatic system prevented oedema by...' [WAIT 12 seconds each.] Affirm responses that link high capillary pressure to increased lymph drainage.		pressure before proceeding.

	fluid rapidly.'			
Explain Phase 📹 **VIDEO:** Lymphatic System - VHSG Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — OEDEMA CASE STUDY (15 min): Learners read a short clinical vignette: 'Mama Zawadi is a 52-year-old fish trader from Kilifi, on the Kenyan coast. She has noticed that her lower legs swell badly every evening, and the swelling only partially goes down overnight. Her doctor at Kilifi County Referral Hospital suspects lymphatic filariasis — a parasitic worm infection transmitted by mosquitoes that blocks lymph vessels.' In pairs, learners answer three structured questions: (1) Using your lymphatic pathway diagram, explain WHY Mama Zawadi's legs swell when her lymph vessels are blocked. (2) Why does the swelling reduce (but not disappear) when she lies down overnight? (Hint: think about gravity and capillary pressure.) (3) What would you tell a Form 1 student about what the lymphatic system does, in simple Swahili-influenced language? PART B — COMPARE WITH SIMPLER ANIMALS (10 min): Teacher poses: 'We have now described the lymphatic system as part of the closed circulatory system of mammals. Remember the grasshopper — it has an open circulatory system where haemolymph just bathes all the organs. Does the grasshopper need a lymphatic system? Why or why not?' Learners think individually for 3 minutes, then share in groups of 4. Groups appoint a spokesperson to share one key idea. PART C — INDIVIDUAL WRITTEN EXPLANATION (5 min): Each learner writes a 3–5 sentence	For Part A: Distribute the vignette (or read it aloud clearly). Say: 'You have 8 minutes in pairs to answer the three questions. Write full sentences — not bullet points. I will cold-call pairs to share.' After 8 minutes, cold-call 3 pairs (one per question). For Question 2, if no pair mentions gravity, prompt: 'Think about what changes when she lies down — what happens to the pressure in her capillaries?' [WAIT 12 seconds.] For Part B: Say: 'This is a thinking question. There is no single right answer — I want to see your reasoning.' [WAIT 15 seconds of silent thinking before pairs begin.] Cold-call 2 group spokespersons. Listen for reasoning that links 'open system = no separate plasma compartment = no tissue fluid build-up problem.' If not raised, ask: 'In an open system, where is the haemolymph? Is it ever separated from the tissues by a capillary wall?' [WAIT 12 seconds.] For Part C: Say: 'This is your individual evidence. Write it in your own words — imagine you are explaining this to Mama Zawadi's daughter who is in Form 2.' Circulate and read 4–5 paragraphs over the writing period.	Case-Based Reasoning + Cross-Species Comparison: The Kilifi clinical case grounds abstract physiology in a real Kenyan health context, making the function of the lymphatic system personally meaningful. The grasshopper comparison (returning to the anchoring phenomenon's cross-species thread) requires learners to apply their understanding contrastively — the deepest form of conceptual consolidation. Writing the individual paragraph forces learners to synthesise both threads into a coherent explanation.	Observable indicator: In written paragraphs, look for three key ideas: (1) lymph drainage prevents tissue fluid accumulation; (2) blockage of lymph vessels leads to oedema; (3) grasshoppers/open-system animals do not have a separate plasma-tissue boundary requiring lymphatic drainage. Learners who omit idea (3) need a follow-up prompt connecting back to Lesson 3 (open vs. closed systems). Collect 6–8 exercise books at the end of the lesson for written evidence review.

	paragraph in their exercise book: 'Explain, using the lymphatic system, why Eliud Kipchoge's legs do not swell during a marathon, and why a grasshopper does not need a lymphatic system.'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Lymphatic System - VHSG Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class gathers around the Driving Question Board. The teacher reads aloud the two standing key inquiry questions: 'Why is transport important in animals?' and 'How does the structure of the transport system relate to the complexity of an animal?' The teacher then reads the driving question aloud: 'How does the structure of an animal's transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper?' Learners are given two sticky notes (or write in two DQB columns on the board). On the first sticky note, they write one thing their evidence from today CONFIRMS or ADDS to what they already knew. On the second sticky note, they write one NEW question that today's lesson raised for them. The teacher collects and sorts the sticky notes into two columns: 'What We Now Know' and 'New Questions We Have.' The teacher reads 3–4 new questions aloud and places them on the DQB. The class votes (thumbs up) on which 2 questions are most important for future lessons to answer.	Say: 'Before you write, look at the predictions we put on the DQB at the start of the lesson. Did the evidence confirm, challenge, or extend your prediction? Take 30 seconds to compare.' [WAIT 30 seconds.] Then: 'Now write your two sticky notes. You have 3 minutes.' After collecting notes, read 4–5 aloud (mix of strong and developing responses). For new questions, prompt the class: 'Which of these questions do you think connects most directly to the driving question about Kipchoge and the grasshopper?' [WAIT 12 seconds.] Cold-call 2 learners to justify their vote. Explicitly update the DQB by adding a new evidence card: 'Lesson 8: The lymphatic system drains excess tissue fluid and returns it to the blood — without this, oedema develops. Simpler animals with open circulatory systems do not require a separate lymphatic system.'	Claim-Evidence-Question Mapping on DQB: By requiring learners to make a specific claim supported by today's evidence AND generate a new question, the DQB update becomes a tool for metacognitive monitoring — learners see their own understanding growing across lessons. The voting ritual gives all learners agency in directing the inquiry storyline.	Observable indicator: Each learner has produced two legible sticky note entries — one knowledge claim and one question. Quality indicator: knowledge claims should reference the lymphatic system specifically (not just 'transport is important'). Flag any learner whose 'new question' is actually a factual recall question (e.g., 'What is a lymph node?') — redirect them: 'That is a knowledge question. Can you turn it into a WHY or HOW question that connects to the driving question?'
Model Building Phase 📹 **VIDEO:** Lymphatic	Learners retrieve their cumulative transport model — a diagram they have been building since Lesson 1 showing how transport systems work in different animals. In	Say: 'Take out your cumulative model from your portfolio. Find a space — you may need to extend your diagram onto the next page. You are adding the lymphatic	Cumulative Model Revision with Colour-Coded Layering: The use of a distinct colour for each lesson's addition makes the growth of the model visually explicit —	Observable indicator: Learners' updated models include all 5 labelled lymphatic components in a new colour, a directional arrow showing one-way flow, and the

System - VHSG Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Lessons 1–7, this model included: heart chambers, blood vessels, blood components, the double circulatory system, and comparisons across animal groups. Today, learners ADD a new layer: the lymphatic system. Using a coloured pen (different from colours used previously — suggested: green), learners draw and label: (1) Lymph capillaries branching from the tissue spaces near blood capillaries. (2) A lymph vessel with valves running parallel to a vein. (3) A lymph node (as a small oval on the vessel). (4) The thoracic duct and right lymphatic duct connecting back to subclavian veins near the heart. (5) A small annotation box next to the grasshopper section of their model reading: 'No lymphatic system — open circulatory system. Haemolymph bathes tissues directly. No separate tissue fluid compartment.' Learners write a one-sentence 'Model Update Statement' at the bottom of their model page: 'Today I added the lymphatic system because...'	system in GREEN.' [Circulate immediately — some learners may not have their model.] After 8 minutes of drawing, say: 'Hold up your model. I want to see green lines.' Do a visual scan of the room. Then cold-call 2 learners: 'Walk me through your addition — where does the lymph start, and where does it end up?' [WAIT 12 seconds each.] After both share, say: 'Compare your model with your partner's. Find one thing they included that you missed, and add it now.' [WAIT 3 minutes.] Close by asking: 'If Kipchoge had the circulatory system of a grasshopper — which means no lymphatic system — what would happen to his legs during the Berlin Marathon?' [WAIT 15 seconds.] Cold-call 3 learners. Accept answers that include: massive oedema, fluid flooding muscle tissues, inability to sustain exertion.	learners can see, literally, how their understanding has accumulated across 8 lessons. The comparative annotation for the grasshopper reinforces cross-species thinking. The closing driving-question probe ('what would happen to Kipchoge's legs') requires learners to use the model as a predictive tool, not just a recording device.	grasshopper annotation box. The 'Model Update Statement' at the bottom should include a causal explanation (e.g., '...because tissue fluid must be drained back to the blood to prevent oedema') rather than a descriptive statement (e.g., '...because we learned about it today'). Collect 5 models to review for completeness before Lesson 9.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before Lesson 9: (1) Could learners articulate the difference between blood plasma, tissue fluid, and lymph — or did they treat these as synonyms? If confusion persisted, plan a 5-minute recap at the start of Lesson 9 using the three-term comparison written on the board. (2) Did the Mama Zawadi (Kilifi) case study generate genuine engagement? Did learners connect it to lived experience? If not, consider inviting a school nurse or community health worker to speak briefly in a future lesson about conditions they see locally. (3) When asked about the grasshopper comparison, did learners draw confidently on their Lessons 3–4 knowledge of open circulatory systems? If not, the open vs. closed circulatory system concept may need reinforcement — consider adding a quick comparative table as a starter activity in Lesson 9. (4) Review the 6–8 exercise books collected: are learners writing causal explanations ('because...') or only descriptive statements ('the lymph goes to...')? If mostly descriptive, introduce the sentence stem 'This matters because...' as a scaffold in Lesson 9. (5) Were all learners represented in cold-calls? Check your tally — ensure girls and learners from quieter areas of the room are being called on equitably. (6) The driving question progress: learners should now be able to explain both the cardiovascular AND lymphatic dimensions of why Kipchoge's body manages fluid so effectively. If fewer than two-thirds of the class can do this in their written paragraph, revisit the phenomenon data at the start of Lesson 9 before moving to the next topic.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A demonstration showing coloured water leaking from a tube into a sponge (representing plasma leaking from capillaries into tissue spaces), and an annotated diagram tracing the lymphatic pathway from lymph capillaries through lymph nodes and ducts back to the subclavian veins near the heart.
What did I learn?	The lymphatic system is a one-way drainage network that collects excess tissue fluid (lymph) from intercellular spaces and returns it to the bloodstream via the thoracic duct and right lymphatic duct. Lymph nodes filter pathogens from lymph. When lymph vessels are blocked (e.g., by parasitic worms in filariasis), fluid accumulates in tissues causing oedema. Simpler animals with open circulatory systems, like grasshoppers, do not require a lymphatic system because their haemolymph bathes tissues directly without a separate capillary-tissue boundary.
How does this explain the phenomenon?	During the Berlin Marathon, the enormous blood pressure in Kipchoge's capillaries forces large volumes of plasma into his muscle tissues. The lymphatic system drains this fluid rapidly back into the bloodstream, preventing oedema and keeping his muscles functioning at peak capacity. Without a functioning lymphatic system — as seen in patients with lymphatic filariasis in coastal Kenya — fluid accumulates in the legs causing chronic swelling. A grasshopper with an open circulatory system has no such fluid-balance problem because there is no separate plasma compartment: haemolymph simply fills all body spaces freely.

LESSON 9 (2 lessons (80 minutes total)): The Immune System and Blood Clotting: The Body's Defence and Repair Network	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate non-specific and specific immunity, trace the blood clotting cascade, and connect these defence mechanisms to the efficiency of the mammalian transport system — using Kipchoge's athletic performance, wound management, and the impact of malaria on blood in Kenyan lakeside communities as anchoring contexts.
Knowledge	Learners identify the components of blood involved in immunity (white blood cells: phagocytes, lymphocytes) and clotting (platelets, fibrinogen, thrombin, fibrin); distinguish between non-specific and specific immune responses; describe the sequence of events in blood clotting; explain how malaria (Plasmodium) disrupts red blood cells and compromises the transport system.
Skills	Learners construct annotated flow diagrams of the clotting cascade; analyse case-study data comparing healthy blood counts with those of a malaria patient from a Kisumu lakeside community; evaluate how a compromised immune/clotting system would affect an athlete like Kipchoge.
Attitudes	Learners appreciate the body's integrated defence and transport systems; demonstrate empathy for communities affected by malaria near Lake Victoria; show responsibility in applying wound-management knowledge in daily life.
Key Inquiry Question	1. Why is transport important in animals? 2. How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	Lessons 1–8 established how blood is structured, how the mammalian heart pumps it, and how open vs. closed circulatory systems compare across animals. Lesson 9 deepens the story: blood is not merely a delivery vehicle — it is also a defence and repair system. Learners now ask: what happens to Kipchoge's performance if his blood cannot defend itself or seal a wound? And why are children in lakeside Kisumu communities especially vulnerable — because Plasmodium destroys the very red blood cells the transport system depends on?
Safety Notes	No blood or bodily fluids are to be used. All clotting and immunity activities use diagrams, case-study data cards, and modelling materials only. Remind learners that discussions about malaria and disease should be handled with sensitivity and respect for affected communities. Wash hands after handling any modelling materials (yarn, beads, clay).

B. LESSON OVERVIEW

Learners begin by revisiting Kipchoge's recovery data and a new provocation: what would happen if he suffered a small cut during a race — would his blood clot fast enough? This launches an investigation into the blood clotting cascade (non-cellular and cellular components) and the two branches of immunity. Using a structured case study of a 12-year-old child from Kisumu diagnosed with malaria, learners analyse how Plasmodium falciparum destroys red blood cells and suppresses immune function, directly impairing the transport system. Learners build a physical flow-diagram model of the clotting cascade using yarn and beads, annotate an immunity comparison table, and update their Driving Question Board. The lesson closes with learners explicitly connecting a healthy, intact blood system to Kipchoge's extraordinary cardiovascular efficiency — and contrasting it with the grasshopper's open system, which has no such defence mechanism.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Biodiversity and extinction, then and now Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Life Cycles of Non-flowering Plants - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are presented with a two-part provocation card. PART A: 'Kipchoge is 38 km into a marathon. A competitor's spike scratches his leg — a small but open wound. His blood must clot while his heart is beating at 160 bpm and his blood is flowing fast. Predict: Will it clot faster, slower, or the same speed as when he is resting? Give a reason.' PART B: 'A child in Kisumu near Lake Victoria is diagnosed with malaria. The disease destroys red blood cells. Predict: How might this affect the child's ability to transport oxygen? Could this child run like Kipchoge? Why or why not?' Learners write individual predictions in their science journals (3 minutes silent writing). They then share with a shoulder partner for 2 minutes. The teacher cold-calls 4 learners — 2 per question — to share predictions aloud. All predictions (including wrong ones) are recorded on the class Prediction Wall under the headings 'CLOTTING' and 'MALARIA & TRANSPORT.' Learners are told: 'We are not correcting these yet — we will return to them at the end of the lesson.'	Distribute the provocation card face-down. Say: 'Do NOT turn over until I say go.' After setting context, say: 'Turn over. You have 3 minutes to write your own prediction — no talking yet.' Set a visible 3-minute timer. After timer: 'Now share with your shoulder partner. You have 2 minutes — each person must speak.' After pair share, say: 'I am going to cold-call four people. I want to hear your prediction AND your reason — not the right answer.' Cold-call 4 learners (choose across gender and seating zones). After each response, say: 'Thank you — I am writing that exactly as you said it on our Prediction Wall. We will test it.' Do NOT affirm or correct — maintain cognitive tension. Point to the Prediction Wall and say: 'Every one of these ideas is now a hypothesis. Evidence will judge them — not me.'	Prediction Wall (Hypothesis Anchoring): By publicly recording all predictions — including misconceptions — the teacher creates cognitive dissonance that will be resolved only through evidence. This strategy activates prior knowledge from Lessons 1–8 (blood composition, heart function) and gives the teacher real-time formative data on learner understanding of blood function and disease.	Observable indicator: Learner predictions reference at least one blood component from previous lessons (e.g., 'red blood cells carry oxygen', 'the heart pumps faster during exercise'). Red-flag misconception to listen for: 'Malaria only affects the stomach' or 'clotting happens in the heart' — if heard, note it and address in Phase 3 without singling out the learner.
Observe Phase 📹 **VIDEO:** Biodiversity and extinction, then and now Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Life Cycles of	ACTIVITY A — THE CLOTTING CASCADE MODEL (Groups of 4, 20 minutes): Each group receives a 'Blood Clotting Construction Kit': a length of red yarn (~80 cm), 10 small yellow beads (platelets), 5 white beads (clotting factors), a strip of loose cotton wool (fibrinogen), and a strip of tightly woven cotton wool (fibrin). Groups also receive an Observation Guide card with the following sequence printed as a partially-labelled flow	For Activity A: Distribute kits and say: 'Your job is to BE the clotting cascade. Each material represents a real component. Read your Observation Guide and physically build what happens — step by step.' Circulate actively. At 8 minutes, pause groups: 'Hold up your clot model. Can I see platelets aggregated? Can I see your fibrin mesh?' At 15 minutes, ask one group to narrate their model aloud while others listen	Physical Modelling (Enactive Representation): Manipulating yarn, beads, and cotton wool to enact the clotting cascade transforms an abstract biochemical sequence into a spatial, kinesthetic experience. Research shows that enactive representations help learners retain sequential processes far better than diagrams alone. The Malaria Case Study uses real-format data (haemoglobin levels, RBC counts)	Observable indicators: (1) Clotting model correctly sequences platelets → prothrombin/thrombin → fibrinogen/fibrin with no major steps reversed or omitted. (2) Comparison table correctly identifies that specific immunity involves lymphocytes and forms memory cells; non-specific does not. (3) Learners can articulate the causal chain: Plasmodium destroys RBCs → haemoglobin drops → oxygen delivery falls →

[Non-flowering Plants - Advanced (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	diagram with gaps: INJURY → blood vessel wall damaged → [GAP 1] aggregate at wound site → clotting factors activated → [GAP 2] converted to thrombin → thrombin converts [GAP 3] to fibrin → fibrin mesh traps red blood cells → CLOT FORMS. Groups must: (1) use their materials to physically enact each step (e.g., cluster the yellow beads at a pinched point of yarn, then wrap cotton wool around the cluster), (2) fill in the three gaps on their Observation Guide (answers: platelets; prothrombin; fibrinogen), and (3) annotate each stage with one sentence explaining what is happening and why it matters. ACTIVITY B — IMMUNITY COMPARISON DATA CARDS (Individual then group, 10 minutes): Each learner receives two data cards. Card 1 describes a non-specific immune response: 'A thorn from a fence near a farm in Kitale pierces Amina's finger. Within minutes, the area turns red, swells, and feels warm. Phagocytes arrive and engulf bacteria.' Card 2 describes a specific immune response: 'Three weeks later, the same bacteria enter Amina's bloodstream. B-lymphocytes that encountered the bacteria last time produce antibodies rapidly. The infection is cleared in 24 hours — much faster than the first time.' Learners complete a two-column comparison table: NON-SPECIFIC vs. SPECIFIC — filling in: speed of response, cells involved, whether memory is formed, and whether it targets a specific pathogen. ACTIVITY C — MALARIA CASE STUDY (Whole class, 10 minutes): The teacher narrates (or reads	and check their own. For Activity B: Say: 'Read both cards silently. Then fill in the comparison table. You have 5 minutes.' After 5 minutes, cold-call 2 learners: 'Tell me one difference between non-specific and specific immunity — use evidence from the cards.' For Activity C: Before reading the case study, say: 'As I read, I want you to think about Kipchoge. Every time you hear a number that would matter to an athlete, circle it.' After reading, give 10 seconds of WAIT TIME before asking: 'Why is Achieng's heart beating faster than normal even at rest?' (Expected answer: her heart is compensating for fewer red blood cells by pumping faster.) Say: 'Connect this to our phenomenon. If Kipchoge had Achieng's blood, what would his marathon time look like?'	to build scientific data literacy while grounding immunity in a Kenyan community context learners may personally know.	heart rate increases to compensate. Teacher listens for the phrase 'compensate' or equivalent — if absent, probe with: 'Why would the heart beat faster if there are fewer red blood cells?'

	aloud) a structured case study: 'Achieng, age 12, lives in Mbita on Lake Victoria. She develops malaria caused by Plasmodium falciparum. A blood smear shows that 18% of her red blood cells are infected and being destroyed. Her haemoglobin level has dropped from a normal 12 g/dL to 7 g/dL. She feels tired, cannot concentrate in school, and her heart beats faster than normal even at rest.' A data table is displayed showing: Normal RBC count (4.5 million/µL) vs. Achieng's count (2.8 million/µL); Normal haemoglobin (12 g/dL) vs. Achieng's (7 g/dL); Normal resting heart rate (75 bpm) vs. Achieng's (98 bpm). Learners annotate the data table with arrows and labels explaining the chain of cause and effect.			
Explain Phase 📹 **VIDEO:** [Biodiversity and extinction, then and now] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Life Cycles of Non-flowering Plants - Advanced (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	PART 1 — CLOTTING CONSOLIDATION (10 minutes): Learners individually write a narrated flow diagram of the complete blood clotting cascade in their science journals — in full sentences, not just arrows. They must include: the role of platelets (aggregation and plug formation), the conversion of prothrombin to thrombin (enzyme activation), the conversion of fibrinogen to fibrin (mesh formation), and the final clot. They then swap journals with a partner and use a green pen to add one improvement or correction. PART 2 — IMMUNITY EXPLANATION CARD (10 minutes): Each learner writes a 'Science Explanation Card' for a fictional younger sibling: 'Explain to a Standard 6 pupil why getting malaria a second time is sometimes less deadly than the first time.' This explanation must	For Part 1: Say: 'Close all cards and guides. In your journal, narrate the clotting cascade as if you are a science journalist reporting from inside a blood vessel. Full sentences. 8 minutes.' Set timer. For peer review: 'Swap with your partner. Green pen only. Add one thing they missed or correct one error. You have 2 minutes.' For Part 2: Say: 'Your little sibling asks you: why am I not as sick the second time I get malaria? Write them a card. You MUST use these four words.' Write on board: lymphocytes / antibodies / memory cells / specific immunity. For Part 3: After posing the synthesis question, say: 'I want 15 seconds of thinking before anyone speaks — no hands yet.' Enforce WAIT TIME visibly (count silently, look around room). Cold-call: 'Wanjiku — what is your answer?' After 3 responses, synthesise on the	Science Explanation Card (Audience-Targeted Writing): Writing for a younger audience forces learners to translate technical vocabulary into genuine understanding — they cannot hide behind copied definitions. This strategy surfaces whether learners have conceptual understanding or merely surface recall. The Prediction Wall Revisit is a metacognitive closure strategy: by publicly revising their own earlier thinking, learners experience the process of science as self-correcting, not as a body of fixed facts to be memorised.	Observable indicators: (1) Narrated flow diagram includes all four key stages in correct order and uses correct vocabulary (platelets, prothrombin, thrombin, fibrinogen, fibrin). (2) Explanation Card correctly identifies memory cells as the mechanism for faster second-response immunity. (3) During the synthesis discussion, learners can articulate that Kipchoge's efficiency depends on blood being intact — healthy RBC count, functioning immune system, effective clotting. Red-flag: if learners cannot connect Achieng's case to Kipchoge's performance, the teacher pauses and asks: 'If you took 40% of Kipchoge's red blood cells away, what happens to his oxygen delivery?' to scaffold the connection.

	use the words: lymphocytes, antibodies, memory cells, and specific immunity. PART 3 — CONNECTING TO THE PHENOMENON (5 minutes): The teacher poses the synthesis question aloud: 'We said Kipchoge's heart is remarkably efficient. But today we learned that blood does more than carry oxygen — it defends and repairs. How does having a healthy, fully-functioning immune and clotting system contribute to Kipchoge being able to run at his best?' Learners think silently for 15 seconds (WAIT TIME enforced). The teacher then cold-calls 3 learners and builds a class consensus answer on the board, writing key phrases verbatim. The teacher then pivots: 'Does a grasshopper have any of this? Phagocytes? Clotting cascades? Fibrin?' Learners recall from Lesson 3 (open circulatory system) that insects have haemolymph — not blood — with limited cellular immunity and no fibrin-based clotting. The teacher asks: 'So what is the trade-off? Why is the grasshopper's simpler system enough for it?' (Expected insight: lower metabolic demand, smaller body, no need for high-pressure oxygen delivery to muscles performing sustained work.) PART 4 — PREDICTION WALL REVISIT (5 minutes): Return to the Prediction Wall from Phase 1. The teacher reads each prediction aloud and asks: 'Now that we have evidence — do we confirm, revise, or reject this prediction?' Learners vote with thumbs (up = confirm, sideways = revise, down = reject). For each revised or rejected prediction, one learner explains	board. For the grasshopper pivot, say: 'Turn back to your notes from Lesson 3. What do you remember about the grasshopper's circulatory system?' Give 10 seconds. Cold-call 2 learners. For Prediction Wall: Read each prediction. 'Thumbs ready — on three. One, two, three.' For any 'down' thumb, say: 'Who can correct this in one sentence? Raise your hand.'		

	the correction in one sentence.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Biodiversity and extinction, then and now] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Life Cycles of Non-flowering Plants - Advanced (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (different colours — e.g., green and orange). On the GREEN note, they write one NEW thing they now understand about the driving question: 'How does the structure of an animal's transport system determine its ability to survive and thrive?' — specifically referencing immunity or clotting. On the ORANGE note, they write one NEW question that this lesson has raised for them. Examples of expected green notes: 'Kipchoge's transport system works well partly because his blood can also defend itself and seal wounds quickly'; 'A grasshopper cannot do this — its open system has no fibrin clotting.' Examples of expected orange notes: 'Can malaria permanently damage the heart if it goes untreated?'; 'Do fish like tilapia have immune cells in their blood?' Learners post both notes on the class DQB. The teacher reads 3–4 notes aloud and clusters them into emerging themes on the board: DEFENCE, REPAIR, DISEASE, SYSTEM TRADE-OFFS. The teacher points to the cluster labelled DISEASE and says: 'Look at how many of you are asking about what happens when the transport system breaks down. That is exactly what real biologists study — pathophysiology. Keep those questions — Lesson 10 will need them.'	Distribute sticky notes. Say: 'Two notes, two minutes. Green: what do you NOW understand that you did not before? Orange: what new question does this lesson leave you with?' After posting, read selected notes aloud without identifying authors. Cluster visibly on the DQB. Point to orange (question) cluster and say: 'Count the questions in this cluster. That is your brain working like a scientist. Questions are not a sign of confusion — they are a sign of thinking.' Specifically reference the malaria-related questions: 'Several of you asked about malaria and the heart. Hold that thought — we are going to need it.'	Driving Question Board Update (Claim-Evidence-Question Tracking): The DQB is a cumulative public record of the class's growing understanding across all 12 lessons. Green notes make partial claims visible, helping learners see their own intellectual progress. Orange notes maintain productive inquiry tension — the driving question is never fully 'answered' until Lesson 12. Clustering notes into themes helps learners see patterns across their peers' thinking, a key scientific practice.	Observable indicators: Green notes explicitly reference a component or process from today's lesson (immunity, clotting, malaria, phagocytes, fibrin) AND connect it to either Kipchoge or the driving question. Orange notes frame a genuine biological question (not a yes/no question). If green notes are vague ('I learned about blood'), the teacher prompts: 'Which part of blood? What does it do? How does it connect to Kipchoge?'
Model Building Phase 📹 **VIDEO:** [Biodiversity and extinction, then	Learners return to their cumulative 'Transport System Model' — a annotated diagram they have been building since Lesson 1 (which began as a simple outline of Kipchoge crossing the finish line	Distribute learners' cumulative model portfolios (stored in class folders). Say: 'Open your model to the mammalian blood section. Today you are adding a third dimension — blood is not just a	Cumulative Model Revision (Progressive Diagrammatic Modelling): The layered model — built across all 12 lessons — is the primary artefact of learning in this storyline. Each lesson adds	Observable indicators: (1) Model includes all three functional roles of blood, colour-coded correctly. (2) At least three labels from Lesson 9 vocabulary appear on the mammalian section. (3)

and now Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Life Cycles of Non-flowering Plants - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and has grown to include: heart chambers, blood vessels, blood composition, comparison with grasshopper/tilapia/frog circulatory systems, and gas exchange links). In Lesson 9, they ADD a new layer to their model: a colour-coded legend showing the three functional roles of blood — (1) TRANSPORT (red: RBCs carrying O_2/CO_2, nutrients, waste), (2) DEFENCE (blue: WBCs — phagocytes and lymphocytes), (3) REPAIR (yellow: platelets and clotting cascade). Learners annotate the mammalian blood section of their model with at least three labels from today's lesson. They then add a CONTRAST BOX beside the grasshopper diagram with the text: 'Haemolymph — no RBCs, no fibrin clotting, limited cellular immunity — sufficient for low-metabolic-demand insect body.' Finally, learners add a RED FLAG symbol (⚠️) anywhere on the model where malaria could disrupt function, and write one sentence explaining the disruption at each flag. Minimum: 2 red flags (RBC destruction site; immune suppression site).	delivery truck. It is also an army and a repair crew.' Write on board: TRANSPORT = red | DEFENCE = blue | REPAIR = yellow. Say: 'Add the colour-coded legend first, then annotate — you have 8 minutes.' Circulate and check: Are labels from today's lesson present (fibrin, lymphocytes, phagocytes, platelets)? After 8 minutes: 'Now add your CONTRAST BOX for the grasshopper. One minute.' After contrast box: 'Final task — place your red flags. Where does malaria attack this model? Mark it and write one sentence.' Cold-call 2 learners to share their red flag placements before lesson closes. Close by saying: 'Your model now shows that blood is a system within a system. Kipchoge does not just have a powerful heart — he has powerful blood. And now you can explain exactly why that matters.'	conceptual depth without replacing prior understanding, mirroring how scientists build models iteratively. The red-flag annotation is a targeted strategy for applying knowledge to disease contexts, developing both scientific reasoning and health literacy relevant to Kenyan learners in malaria-endemic regions.	Grasshopper contrast box correctly identifies the absence of fibrin clotting and limited immunity. (4) At least 2 red flags placed at biologically correct sites (RBC destruction by Plasmodium; immune cell suppression) with accurate one-sentence explanations. Teacher collects 6 models at random for brief review before Lesson 10 — note any persistent misconception about clotting location (e.g., learners placing clotting inside the heart rather than at wound sites).

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your planning journal:

1. EVIDENCE QUALITY: Were learners able to correctly sequence all four stages of the clotting cascade in their narrated flow diagrams — or did they confuse the order of prothrombin/thrombin and fibrinogen/fibrin? If errors were common, consider opening Lesson 10 with a 3-minute 'cascade relay' where groups race to arrange cut-up steps in the correct order.

2. MALARIA CONNECTION: Did learners make the causal chain (Plasmodium → RBC destruction → haemoglobin drop → compensatory heart rate increase) independently, or did they need scaffolding? If most needed the scaffold question ('If you took 40% of Kipchoge's red blood cells away…'), consider whether the case-study data cards need more explicit guiding questions in future iterations.

3. COMMUNITY SENSITIVITY: Malaria is a lived reality for many learners, especially those with family near Lake Victoria. Were there any learners who appeared emotionally affected by the Achieng case study? If so, acknowledge after class privately and consider whether the case study should be introduced with a brief note: 'This case study represents real experiences for many Kenyan families — we discuss it with respect.'

4. GRASSHOPPER CONTRAST: Did learners successfully connect back to the open circulatory system from Lesson 3 when asked about the grasshopper's limitations? If the connection was weak, this signals that vertical integration across the lesson sequence needs reinforcement — consider adding a 'Looking Back' prompt to the DQB template for Lesson 10.

5. MODEL DEPTH: Were the cumulative models gaining genuine analytical depth (multi-labelled, colour-coded, red-flagged) or becoming rushed and superficial? If models look thin, reduce the Explain phase by 5 minutes and allocate that time to the model-building phase in Lesson 10.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners built a physical model of the blood clotting cascade using yarn, beads, and cotton wool; analysed a data table comparing blood counts of a healthy child versus Achieng, a 12-year-old malaria patient from Mbita on Lake Victoria; and completed a non-specific vs. specific immunity comparison table using data cards set in Kitale.
What did I learn?	Blood performs three integrated functions within the transport system: delivery (RBCs transporting O_2 and CO_2), defence (phagocytes and lymphocytes providing non-specific and specific immunity), and repair (platelets and the fibrinogen-to-fibrin clotting cascade). Malaria caused by Plasmodium falciparum destroys red blood cells, lowers haemoglobin levels, and forces the heart to compensate by increasing resting heart rate — directly impairing the transport system's efficiency.
How does this explain the phenomenon?	Eliud Kipchoge's extraordinary athletic performance depends not only on a powerful heart but on fully functional blood — rich in healthy red blood cells, equipped with an effective immune defence, and capable of rapid clotting at injury sites. A grasshopper's haemolymph, by contrast, lacks fibrin-based clotting and has only limited cellular immunity — sufficient for a low-metabolic-demand insect but entirely inadequate for sustaining world-record marathon performance. The mammalian transport system's complexity is inseparable from the complexity of the demands placed on it.

LESSON 10 (2 lessons (80 minutes)): ABO and Rhesus Factor Blood Grouping	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To understand the ABO and Rhesus Factor blood group systems, their genetic basis, and their clinical importance in blood transfusion compatibility.
Knowledge	Learners will be able to: (a) explain the ABO blood group system including antigen and antibody relationships; (b) describe the Rhesus factor (Rh) and its significance; (c) outline the rules of blood transfusion compatibility; (d) describe the role of the Kenya National Blood Transfusion Service (KNBTS) in saving lives.
Skills	Learners will be able to: (a) simulate ABO blood typing using colour-coded indicator solutions and interpret results; (b) analyse Kenyan blood group distribution data and draw conclusions; (c) construct a transfusion compatibility table; (d) communicate findings using scientific reasoning linked to the anchoring phenomenon.
Attitudes	Learners will: (a) appreciate the importance of voluntary blood donation to the Kenyan community; (b) develop empathy and social responsibility by connecting blood compatibility to real-life emergency scenarios; (c) value evidence-based medical practice.
Key Inquiry Question	Why is transport important in animals? How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	In previous lessons learners established that Kipchoge's closed, double circulatory system delivers large volumes of oxygenated blood rapidly to working muscles. But blood is not simply a carrier of oxygen — its identity matters. Lesson 10 zooms into the composition of blood itself, asking: what IS the fluid being pumped, and why can't just any blood be swapped between people? By understanding ABO and Rhesus grouping, learners deepen their appreciation of why the mammalian transport system is exquisitely matched to the body's needs — right down to the molecular identity tags on each red blood cell. This advances the driving question by revealing that the efficiency of Kipchoge's system depends not just on heart structure and vessel design, but on the precise biochemistry of the blood itself.
Safety Notes	No real blood is to be used at any point in this lesson. All blood-typing simulations use coloured water or food-colouring solutions prepared by the teacher in advance. Learners must wash hands before and after the activity. If any learner has a known needle phobia or anxiety around medical topics (e.g. transfusion trauma), the teacher should be sensitive and allow that learner to take the role of recorder rather than simulator. Dispose of all used cups/droppers responsibly. Remind learners that actual blood grouping must only be done by trained medical personnel.

B. LESSON OVERVIEW

In this lesson learners simulate ABO blood typing using coloured indicator solutions (food colouring in water standing in for Anti-A and Anti-B sera), observe agglutination-analogue reactions, and determine simulated blood groups. They then analyse real Kenyan blood group distribution data from the Kenya National Blood Transfusion Service (KNBTS) and construct a transfusion compatibility matrix. The lesson culminates in a sensemaking discussion that ties blood biochemistry back to Kipchoge's circulatory efficiency and the broader driving question about transport system design in animals. Learners update their Driving Question Board and revise their cumulative transport system models to include blood composition as a functional layer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Proof: Segments tangent to circle from outside point are congruent Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are presented with a short scenario read aloud by the teacher: 'After the Nairobi Marathon, a runner collapses near the finish line at Uhuru Park. The medical team determines she needs an emergency blood transfusion. Three donors are available. The medic says: we cannot just use any blood — we need to know her blood group first.' Learners are then given a PREDICT slip with two questions: (1) Why do you think not all human blood is the same? What do you think makes one person's blood different from another's? (2) Predict: if Kipchoge donated blood to a patient, could that blood go to absolutely anyone? Why or why not? Learners write individual predictions silently for 4 minutes, then share with a shoulder partner for 2 minutes.	Teacher reads the Nairobi Marathon scenario aloud with energy and pauses for effect after 'we need to know her blood group first.' Teacher says: 'Before I tell you anything about blood groups, I want to know what YOU think. Write your honest prediction — there is no wrong answer right now.' Allow 4 minutes of silent individual writing (WAIT TIME enforced — teacher does not speak or prompt). After partner sharing, cold-call 3 learners using name cards: ask each one 'What did you predict, and what evidence from our earlier lessons made you think that?' Teacher records 3–4 key prediction phrases on the board under the heading WHAT WE THINK WE KNOW. Teacher says: 'Hold onto these predictions — by the end of this lesson you will revisit them and decide which ones were correct, which were partially correct, and which need to change.'	Predict–Share–Record: activates prior knowledge from Lessons 7–9 on blood composition and circulation, surfaces misconceptions (e.g. 'all blood looks the same so it must be the same'), and creates cognitive dissonance that motivates the evidence-gathering phase. Connecting to the phenomenon immediately grounds the abstract topic of blood groups in a real Kenyan emergency context.	Teacher scans written predict slips during partner sharing (circulate and read over shoulders). Observable indicator: learner predictions should reference blood cells, oxygen, or the heart — if most predictions are purely superficial ('blood is different colours'), teacher flags this as a misconception to address explicitly in Phase 3. Cold-call responses reveal depth of prior understanding from Lessons 7–9.
Observe Phase 📹 **VIDEO:** Proof: Segments tangent to circle from outside point are congruent Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar	PART A — Blood Typing Simulation (35 minutes): Groups of 4 receive a simulation kit prepared by the teacher: 4 small transparent cups labelled 'Patient Sample A', 'Patient Sample B', 'Patient Sample AB', 'Patient Sample O'; two dropper bottles labelled 'Anti-A Serum' (blue food colouring in water) and 'Anti-B Serum' (red food colouring in water); a white paper backing sheet; a results recording table. The 'patient blood samples' are pre-prepared by the teacher using a colour-mixing code: Sample A = slightly yellow water; Sample B = slightly green water; Sample AB = slightly	Teacher introduces Part A by demonstrating the first test with 'Patient Sample A' at the front: 'Watch what I do — I am NOT doing your thinking for you, I am showing you the procedure only.' Add Anti-A drops, show the colour change, say: 'Record what you SEE, not what you think should happen.' Release groups to work. During group work, teacher circulates and asks targeted questions to each group — rotate through all groups at least once: 'What do you observe when Anti-A meets this sample?' (WAIT 10 seconds before accepting answer). 'What does NO colour change tell	Simulation-Based Observation with Data Integration: the colour-change simulation gives learners a concrete, visual analogue for the antigen–antibody agglutination reaction without using real blood. Immediately following with real Kenyan KNBTS distribution data bridges the simulation to authentic science, reinforcing that what they just 'tested' in cups has life-or-death implications in Kenyan hospitals every day. The three-part structure (simulation → statistics → reading) mirrors how scientists triangulate evidence from multiple sources.	Teacher checks each group's completed results table during circulation — observable indicator: all 4 simulated samples correctly identified (A, B, AB, O) by matching colour-change patterns to the ABO key. Bar charts should have correctly labelled axes and proportional bars. Annotation of Rh reading: teacher spot-checks 4 books — indicator: learners identify (1) Rh+ majority in Kenya ~94% and (2) Rh− is rare and critical for blood banks. Any group that misidentifies a sample is redirected with: 'Look again at which serum caused the change — Anti-A or Anti-B or both or

readings	orange water; Sample O = clear water. The Anti-A serum (blue) reacts visibly only with Sample A and AB (turns them a distinct blue-green or blue-orange). The Anti-B serum (red) reacts visibly only with Sample B and AB (turns them red-green or red-orange). Sample O shows no colour change with either serum. Learners follow a step-by-step observation card: add 3 drops of Anti-A to each sample, observe and record; then add 3 drops of Anti-B to each sample, observe and record. They use their results table to determine the blood group of each sample. PART B — Kenyan Blood Group Distribution Data (10 minutes): Each group receives a data card showing estimated Kenyan blood group distribution (Blood Group O: ~46%, A: ~27%, B: ~20%, AB: ~7%, sourced from KNBTS public data). Learners draw a bar chart of the distribution on graph paper provided. PART C — Rhesus Factor Mini-Reading (5 minutes): Learners read a 150-word structured passage (on their observation card) explaining Rh+ and Rh− designations, that approximately 94% of Kenyans are Rh+, and that Rh− blood is rare and critical to stock in blood banks. They annotate two key facts.	you?' (WAIT 12 seconds). Cold-call 2 groups to share their blood group determinations before moving to Part B. For Part B, teacher says: 'Look at this data from the Kenya National Blood Transfusion Service — the real organisation that manages blood donations across Kenya, from Mombasa to Kisumu to Eldoret. What do you notice about which group is most common?' Allow 3 minutes of quiet graph work then cold-call 2 learners: 'What does this distribution mean for hospitals in Nairobi during an emergency?' (WAIT 15 seconds). For Part C, teacher reads the first two sentences of the Rhesus passage aloud, then says: 'Finish the rest silently and circle the two facts you think are most important.'		neither?'
Explain Phase 📹 **VIDEO:** Proof: Segments tangent to circle from outside point are congruent Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	PART A — Vocabulary Anchor (5 minutes): Learners complete a Frayer Model for the term 'agglutination' in their notebooks: Definition in own words; Characteristics; Example from today's simulation; Non-example. PART B — Transfusion Compatibility Matrix (15 minutes): Learners are given an empty 4×4 compatibility grid (Donor groups	For the Frayer Model, teacher writes 'agglutination' on the board and says: 'This word is the key to everything you just observed. Build your Frayer Model before I say anything more about it — use your simulation results as your evidence.' Allow 5 minutes. Cold-call 2 learners to share their 'example' cell — listen for whether they correctly connect it to the	Frayer Model + Rule-Based Matrix Construction: the Frayer Model forces precise vocabulary ownership before applying the concept. The compatibility matrix uses logical rule application (not memorisation) to build understanding of why transfusion errors are dangerous — this is constructivist learning. The phenomenon connection	Frayer Models collected or photographed — observable indicator: 'Example' cell must reference the simulation (colour change with Anti-A or Anti-B), not a generic statement. Compatibility matrices: teacher checks that Group O is marked as compatible donor for all recipients, and AB as compatible recipient from all donors. Observable indicator for

📖 **HTML:** [Evolution of Simple Cells (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	across the top: O, A, B, AB; Recipient groups down the side: O, A, B, AB). Using the rule cards provided ('A recipient has Anti-B antibodies and will reject B or AB donor blood'; 'O donor has no antigens — universal donor'; 'AB recipient has no antibodies — universal recipient'), learners fill in each cell with a tick (compatible) or cross (incompatible). They also add a row/column for Rh+ and Rh−. PART C — Phenomenon Connection Discussion (10 minutes): Teacher leads a whole-class discussion using the question: 'We have spent nine lessons understanding how Kipchoge's heart and blood vessels are built to deliver blood faster and more efficiently than any other animal we have studied. Now we have learned that blood has a molecular identity. How does this add another layer to our understanding of why a mammalian transport system is so sophisticated?' Learners share ideas in a Think–Pair–Share: think 2 minutes silently, pair 2 minutes, then 3 cold-called shares to the class.	Anti-A/Anti-B colour change. For the compatibility matrix, teacher circulates and pauses at groups that are confused, asking: 'Read the rule card again — what antigens does a TYPE A person have on their red blood cells? What antibodies do they carry in their plasma?' (WAIT 12 seconds). Do NOT give the answer — redirect to the rule card. After matrices are complete, teacher draws the grid on the board and cold-calls 6 different learners to fill in one cell each and justify their choice in one sentence. For the phenomenon discussion, teacher opens with: 'Kipchoge's red blood cells carry specific antigen markers — just like every human's do. His blood group is a fixed feature of his biology. Why does that matter for his career as an elite athlete — think about injuries, surgeries, altitude training camps in Iten?' (WAIT 15 seconds before accepting responses). After 3 cold-called responses, teacher synthesises: 'Blood group is not just a medical curiosity — it is part of the total biological system that keeps an athlete like Kipchoge performing at the highest level. The transport system works because EVERY component — the heart, the vessels, the blood volume, AND the molecular identity of the blood cells — is precisely matched.'	discussion uses the 'zooming out' strategy: learners move from the molecular scale (antigens on red blood cells) back to the whole-organism scale (Kipchoge's athletic career), reinforcing the cross-scale thinking that is central to the driving question.	phenomenon discussion: cold-called learners can articulate at least one connection between blood group identity and the functioning of a complete mammalian transport system. Red-flag misconception to watch for: learners who think 'universal donor means O blood is best' — redirect by asking 'Best for whom? What about the Rh factor?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Proof: Segments tangent to circle from outside point]	Each learner receives two sticky notes. On the YELLOW note they write one piece of evidence from today's lesson that helps answer the driving question: 'How does the structure of an animal's transport system determine its ability to survive and thrive?' On the PINK	Teacher says: 'You have two minutes — write your yellow evidence note first. Be specific: do not just write 'blood groups are important.' Write HOW what you learned today connects to the driving question about transport system structure.' After posting,	DQB Evidence Mapping: by physically posting evidence on a shared board organised by lesson, learners can visually trace the progression of their understanding across the 12-lesson sequence. This metacognitive strategy helps learners see that science	Teacher reads all yellow sticky notes before the next lesson — observable indicator: notes should make an explicit connection between blood group compatibility and transport system function (not just restate facts about ABO groups). Pink notes are reviewed

[are congruent](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Evolution of Simple Cells (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	note they write one new question that today's lesson has raised for them. Learners post their notes on the class Driving Question Board under the column 'BLOOD COMPOSITION & IDENTITY.' The teacher then reads out 3–4 yellow notes and 2–3 pink notes to the class, facilitating a 4-minute discussion on how today's evidence builds on evidence from Lessons 7–9 (heart structure, double circulation, blood vessels).	teacher reads yellow notes aloud and asks: 'How does knowing that Kipchoge has a specific blood group — let's say he is O+ — change or deepen your answer to our driving question?' (WAIT 12 seconds, cold-call 2 learners). Then reads 2 pink question notes: 'These new questions are exactly what scientists feel after every experiment. We will carry these into Lessons 11 and 12.' Teacher points to the DQB and says: 'Look at the board from Lesson 7 to today — do you see how our understanding of the transport system has grown from the whole heart, to vessels, to the fluid itself?'	knowledge builds incrementally — each lesson adds a layer — mirroring how the transport system itself is a layered, integrated structure.	to identify misconceptions or genuine curiosity questions that can be addressed in Lessons 11–12. A yellow note that only says 'I learned about blood groups' is flagged as insufficient — teacher follows up with that learner individually.
Model Building Phase 📹 **VIDEO:** [Proof: Segments tangent to circle from outside point are congruent](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Evolution of Simple Cells (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative transport system model (begun in Lesson 7 and revised in Lessons 8 and 9). Today's revision task: Add a zoomed-in inset panel to the model showing a cross-section of a blood vessel containing red blood cells. On each red blood cell, learners draw and label the relevant antigens (A, B, both, or none) and note the corresponding antibodies in the plasma. They also add a small legend box: 'Blood Group O = No antigens, Anti-A and Anti-B antibodies in plasma — universal donor.' They write a one-sentence model annotation: 'The molecular identity of Kipchoge's red blood cells (his blood group) is a fixed feature of his transport system that must be matched in any medical intervention.' Finally, learners add a comparison note: 'Does a grasshopper have blood groups? Why or why not?' (Answer: insects have haemolymph, not true blood with red blood cells — no ABO system exists.) This comparison	Teacher displays a projected example of the inset panel on the board (hand-drawn is fine — teacher draws it live in 2 minutes): 'I am drawing this in front of you so you can see how a scientist adds detail to an existing model without redrawing everything from scratch. I add an inset — a zoomed-in window.' Teacher circulates during model revision (8 minutes), asking: 'Where in your model does this inset belong? Near the heart? Near a capillary?' (WAIT 10 seconds). Cold-call 2 learners at the end to hold up their models and explain their inset panel to the class in 30 seconds each. Teacher closes with the driving question: 'We started this unit asking what would happen if Kipchoge had the circulatory system of a grasshopper. We have now gone all the way from the grasshopper's open system with haemolymph to Kipchoge's double circulation — and today we zoomed into the molecular identity of his blood cells. What is your best answer	Cumulative Model Revision with Cross-Scale Annotation: adding a molecular-scale inset to an organ/organism-scale model is a key science practice (NGSS SEP 2: Developing and Using Models). The grasshopper comparison note forces learners to apply cross-kingdom contrast — the backbone of the driving question — ensuring that blood group biochemistry is not learned in isolation but as one distinguishing feature of the sophisticated closed circulatory system. The model now spans: whole organism → organ (heart) → vessel → cell → molecule.	Teacher examines models during circulation — observable indicators: (1) inset panel present and positioned near a blood vessel; (2) antigens correctly labelled on red blood cells matching the blood group shown; (3) one-sentence annotation references Kipchoge or the phenomenon; (4) grasshopper comparison note attempts a scientific reason (haemolymph, no RBCs). Models without the grasshopper comparison note are flagged — teacher says: 'Your model is not yet complete — what did we say about what grasshopper blood does NOT have?'

	reconnects the lesson to the full driving question about comparative animal transport systems.	now to our driving question?' Allow 3 cold-called responses. Teacher says: 'In Lessons 11 and 12 we will look at how the heart can go wrong — diseases of the transport system — and what Kenyan medical science is doing about it. Your models will need one more update.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did learners successfully distinguish between antigens and antibodies in the ABO system, or did they confuse which is on the cell and which is in the plasma? If confusion persisted, plan a brief 5-minute warm-up at the start of Lesson 11 using a physical role-play (half the class are 'antigens' wearing labels, half are 'antibodies' — they 'agglutinate' when they meet their matching type). (2) Were learners able to connect blood group identity back to the driving question about Kipchoge and the comparative transport systems? If the phenomenon connection felt forced or superficial in the Explain phase, consider making the comparison more explicit in Lesson 11 by revisiting the DQB and asking learners to vote on which evidence gathered so far most powerfully answers the driving question. (3) Did the simulation work visually? If colour changes were unclear, adjust the food-colouring concentrations before the next class. Consider adding yellow food colouring for 'Sample B' and orange for 'Sample AB' to make contrasts sharper. (4) Were learners aware of the Kenya National Blood Transfusion Service before this lesson? If not, this is an opportunity to assign a brief home inquiry: 'Find the nearest KNBTS donation centre to your home and note what blood groups they urgently need.' This builds citizenship and connects biology to community health. (5) Note any learners who showed particular insight in the DQB phase — their pink question notes may generate excellent inquiry threads for the summative assessment task.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners added coloured Anti-A and Anti-B indicator solutions to four simulated blood samples and observed colour changes (or no change) to determine the ABO blood group of each sample. They also read a bar chart of Kenyan blood group distribution data from the Kenya National Blood Transfusion Service and annotated a short passage on the Rhesus factor.
What did I learn?	Learners learned that human red blood cells carry surface antigens (A, B, both, or neither) that define ABO blood group, and that plasma contains corresponding antibodies that cause agglutination when mismatched blood is introduced. They learned the rules of transfusion compatibility, that blood group O is the most common in Kenya (~46%) making it the most in-demand at the KNBTS, and that ~94% of Kenyans are Rh-positive. They also learned that insects such as grasshoppers have haemolymph rather than true blood and therefore have no ABO system.
How does this explain the phenomenon?	Learners explained that the molecular identity of red blood cells is a fixed feature of the mammalian transport system that must be precisely matched in medical interventions such as transfusions. They connected this to the driving question by reasoning that the efficiency of Kipchoge's transport system depends not only on heart structure and vessel design but also on the biochemical precision of the blood itself — a level of complexity entirely absent in the open circulatory systems of grasshoppers and other invertebrates studied earlier in the sequence.

LESSON 11 (40 minutes): Diversity of Transport Systems: Comparative Overview	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will synthesise all evidence gathered across lessons 3 to 10 into a single comparative framework that explains why transport systems differ across animal groups and why those differences matter for survival.
Knowledge	Learners can construct a comparative table of transport systems from insects through fish, amphibians, reptiles, and mammals, correctly identifying the type of circulation, heart chambers, transport fluid, and efficiency for each group; learners can articulate at least three evolutionary trends in circulatory complexity and link each trend to a specific functional advantage.
Skills	Learners practise scientific synthesis, annotated diagram construction, evidence-based argumentation, and comparative analysis across biological groups.
Attitudes	Learners appreciate that diversity in biological design reflects millions of years of adaptation to different environmental demands, and that no system is simply inferior — each is suited to the metabolic needs of its organism.
Key Inquiry Question	How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	This is the central EXPLAIN lesson of the sequence. Learners have collected ten lessons worth of evidence and now integrate it into a coherent comparative model that directly answers the driving question. This lesson equips them with the full analytical framework they will need for their final presentations in Lesson 12.
Safety Notes	No hazardous materials are used. Learners working with scissors for cutting and pasting diagram strips should be reminded of standard classroom safety. Ensure learners with visual impairments receive enlarged diagram sheets.

B. LESSON OVERVIEW
Learners open by revisiting the Driving Question Board to identify which questions they can now answer. They then work in groups to construct a full five-row comparative table and annotated diagrammatic model spanning insect, fish, amphibian, reptile, and mammal transport systems. Using evidence cards prepared from their notebooks across previous lessons, groups populate each row collaboratively before sharing and defending their entries in a gallery walk. The teacher facilitates a whole-class synthesis discussion that draws explicit evolutionary trend lines and anchors the comparative model back to Eliud Kipchoge and the anchoring phenomenon. By the end of the lesson every learner has a completed, annotated comparative table that is ready to support their final presentation in Lesson 12.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Circulatory System Structure and Function - Overview]	Learners open their exercise books to the Driving Question Board record they maintain individually. The teacher projects or writes on the board the original driving question: How does the	Teacher says: Before we build our final comparative model today, I want you to spend three minutes alone — no talking — flipping through your notes from lesson 3 all the way to lesson 10. Circle any	Activating prior knowledge through self-assessment of the Driving Question Board; partner talk to surface both confident understanding and remaining gaps	Teacher circulates during the 3-minute silent phase and notes which learners are actively engaging with their DQB records versus sitting passively. During partner talk, teacher listens for

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	structure of an animals transport system determine its ability to survive and thrive? Learners are given 3 minutes to silently read their own DQB cards and circle or tick any question they now feel confident they can answer based on lessons 1 to 10. They then turn to a partner and share: one question they can now answer, and one question they are still uncertain about. Pairs share back to the class and the teacher notes the still-uncertain questions on the board for later attention.	question on your DQB that you now feel you can answer with evidence. After 3 minutes of wait time, teacher says: Turn to your partner. Tell them one question you can now answer and give one piece of evidence that supports your answer. You have 90 seconds. After pairs share, teacher says: I am hearing that some of you are still unsure about how we compare all the systems together. That is exactly what we are resolving today. By the end of this lesson you will be able to answer the driving question in full.	before new synthesis work begins.	misconceptions such as confusing open and closed systems or misattributing double circulation to fish. These are noted for targeted clarification during the Explain phase.
Observe Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive a set of pre-prepared Evidence Cards — small strips of card or paper that the teacher prepared in advance. Each strip contains one factual statement from a previous lesson, for example: Grasshopper haemolymph bathes organs directly and does not carry oxygen; Tilapia blood passes through the heart once per complete circuit; Frog ventricle is not fully divided so oxygenated and deoxygenated blood mix; Reptile ventricle has a partial septum reducing mixing; Mammalian double circulation keeps oxygenated and deoxygenated blood fully separate. There are 20 cards per group covering all five animal groups across five categories: type of circulation, heart chambers, transport fluid, presence of capillaries, and metabolic activity supported. Groups spread the cards on the desk and read them aloud to each other. Each learner also opens their notebook to cross-check: does the card match what they recorded in that lesson? Any card that contradicts their	Teacher distributes Evidence Card sets and says: In your group, read every card aloud — one person reads, others listen. Do not sort yet. Just read and check against your own notes. If a card says something different from what you wrote, set it aside. You have 4 minutes. During the reading, teacher moves to each group and asks quietly: Can you find the card that explains why a grasshopper can survive without a closed system? Listen for learners who locate the card about low metabolic demand and open body cavity. After 4 minutes teacher says: Now hold up any card your group set aside as possibly wrong. Let us look at those together. Teacher briefly addresses each disputed card, clarifying or confirming. This surfaces remaining misconceptions before the synthesis table is built.	Concrete manipulation of Evidence Cards forces learners to physically engage with comparative data across all five animal groups simultaneously, making patterns visible before they are asked to articulate them abstractly.	The pile of disputed cards is a direct formative signal. If multiple groups dispute the same card, the teacher knows that concept needs re-teaching before the table is constructed. Teacher notes group-level gaps and addresses them within the 4-minute debrief.

	notes is placed in a separate pile for class discussion.			
Explain Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives a blank Comparative Table template with five rows (insect, fish, amphibian, reptile, mammal) and six columns (type of circulation, heart structure, transport fluid, separation of oxygenated and deoxygenated blood, capillary network present, example organism from Kenya or East Africa). Groups use their Evidence Cards and notebooks to complete every cell. They must also write one sentence at the bottom of the table called The Trend Statement that describes the overall pattern they see moving from insect to mammal. When tables are complete, each group annotates a simplified diagram strip showing a heart shape for each animal group — they label chambers, shade oxygenated blood red and deoxygenated blood blue, and draw arrows showing direction of blood flow. The teacher then leads a 6-minute whole-class discussion drawing out three explicit evolutionary trends: (1) from open to closed circulatory systems, (2) from single to double circulation, (3) from incomplete to complete separation of oxygenated and deoxygenated blood. The teacher writes each trend on the board and asks learners to confirm it with a specific piece of evidence from their table.	Teacher says: You now have 8 minutes to fill in every cell of the comparative table using your evidence cards and your notes. Trend Statement goes at the bottom — in one sentence, describe the pattern you see as you move from insect to mammal. While groups work, teacher circulates and probes: What does the column heading separation of blood mean? Can you show me in your diagram where mixing happens in the frog? After 8 minutes, teacher brings class together and says: I am going to draw three trend lines on the board. Your job is to stop me if the evidence from your table does not support what I write. Teacher writes Trend 1: As animals become more metabolically active, their circulatory systems become more closed and efficient. Wait 10 seconds. Anyone want to challenge or support that with evidence from their table? Teacher waits 15 seconds for responses. Teacher then writes Trend 2 and Trend 3 in the same way. After all three trends are on the board, teacher says: Now look at Kipchoges data from lesson 1. Which of these three trends explains why his heart can do what it does? Wait 20 seconds for learners to think before taking responses.	The comparative table externalises the synthesis process, making implicit patterns explicit. The whole-class trend discussion uses the claim-evidence-reasoning structure: teacher models making a claim and invites learners to test it with evidence, building scientific argumentation habits.	Teacher collects one completed comparative table per group at the end of the phase and does a rapid scan for three specific cells: the type of circulation for fish (should be single, closed), the separation of blood for amphibians (should be incomplete mixing), and the example organism column (should contain a real Kenyan or East African animal such as Nile tilapia, African bullfrog, or impala). Errors in these cells indicate persistent misconceptions that the teacher addresses at the start of lesson 12.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Circulatory	The class returns to the physical DQB display (or the teacher projects the class DQB). Learners are given two sticky note colours: green for answered and orange for still exploring. Each learner writes	Teacher distributes two sticky notes to each learner and says: Green note — pick one question from our DQB that you can now answer. Write the answer in one sentence on the back. Orange	The two-colour DQB update makes the cumulative evidence-gathering visible to the whole class and gives individual learners a metacognitive checkpoint — they can see how much of the original	The content of green notes is a direct formative check. Teacher notes how many learners write accurate one-sentence answers versus vague or incorrect ones. Learners who place only orange

System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	on a green sticky note one question from the DQB that their comparative table now answers, and writes the one-sentence answer on the back. Learners who still have a question they cannot answer write it on an orange note. The class sticks both colours onto the DQB. The teacher reads several green notes aloud and asks the class to confirm the answer. Orange notes are read and the teacher sorts them into two categories: questions that will be addressed in the lesson 12 presentation phase, and questions that go beyond the scope of this sub-strand but are worth pursuing independently. Learners record in their notebooks: Questions I can now answer with evidence and Questions I am still investigating.	note — write any question that is still open for you. You have 3 minutes, working alone. After 3 minutes, teacher says: Come and place both notes on the board. Teacher scans the green notes rapidly and reads three aloud, asking: Does everyone agree with this answer? What evidence from our table supports it? Teacher then reads two or three orange notes and says: These orange questions are powerful. Some of them will be answered in our final presentations tomorrow. Some of them are questions that scientists are still investigating — and that is the honest answer. Teacher writes the boundary-pushing orange questions in a special column labelled Still wondering — and that is good science.	wonder has been converted into understanding, and that remaining curiosity is valued not penalised.	notes and no green notes are identified for targeted support before lesson 12. The teacher records these names.
Model Building Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 8 minutes each learner works independently to update their personal cumulative model — the annotated diagram they have been building since lesson 3. Today they must add: (1) a side-by-side comparison strip showing all five hearts to scale, (2) colour-coded arrows showing the degree of oxygenated and deoxygenated blood separation in each system, (3) a written annotation connecting the mammalian double circulation to the Kipchoge anchoring phenomenon. The annotation must answer in their own words: Why can Kipchoges heart sustain world-record speeds while a grasshoppers dorsal vessel could not? Learners write this annotation in a box labelled My Explanation of the Phenomenon. They are told this model and annotation will be the core of their lesson 12 presentation. The lesson closes	Teacher says: You have 8 minutes. Update your cumulative model with three things. Write them on the board: (1) five hearts to scale side by side, (2) colour-coded blood flow arrows, (3) annotation box titled My Explanation of the Phenomenon. I will come around. If you finish early, add one real Kenyan animal to each row of your comparison strip. During the 8 minutes the teacher circulates and reads annotations over learners shoulders, stopping to ask quietly: What specific feature of the four-chambered heart makes Kipchoges recovery possible? Push learners to go beyond stating double circulation to explaining that full separation means oxygenated blood reaches muscles at maximum concentration without dilution. With 2 minutes remaining teacher says:	Independent model updating consolidates the group synthesis work into individual understanding. The annotation task requires causal reasoning that links structural features to functional outcomes and ultimately to the anchoring phenomenon, completing the ARES explain cycle.	Teacher collects nothing at this stage but reads annotations during circulation and mentally flags three categories: learners who articulate the structural-functional link clearly, learners who describe structure but do not explain function, and learners who copy from the board without personal reasoning. These categories inform the differentiated support the teacher provides during the lesson 12 presentation preparation.

	with the teacher reading one strong annotation aloud (with the learners permission) as a model for the class.	I am going to read one annotation that I think is ready for lesson 12. This is not the only good one — it is one example. Teacher reads the chosen annotation and says: What did this learner do well? Take 15 seconds to think, then I will ask two people. Wait 15 seconds before taking responses.		

D. TEACHER REFLECTION
After the lesson consider the following: Did learners accurately complete all six columns of the comparative table, or were there persistent errors in specific cells such as the separation of blood column? Were learners able to write Trend Statements that identified a directional pattern rather than just listing facts? How many green DQB notes contained accurate one-sentence answers — if fewer than two thirds of the class produced accurate green notes, the lesson 12 opening should include a brief whole-class review of the three trend lines. Were any learners unable to connect double circulation to the Kipchoge phenomenon in their annotation — if so, consider pairing those learners with stronger peers during the lesson 12 presentation preparation phase. Reflect also on whether the Evidence Cards effectively activated prior learning or whether learners were still relying heavily on teacher prompting to recall information from lessons 3 to 9. If the latter, consider whether the cumulative model-building strategy across the sequence needs more frequent structured retrieval checkpoints in future iterations of this unit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed and physically manipulated Evidence Cards containing data from all previous lessons, noticing that each animal group has a distinct combination of circulation type, heart structure, transport fluid, and blood separation — and that disputed cards revealed remaining gaps in understanding.
What did I learn?	Learners learned that circulatory complexity increases progressively from insects through fish and amphibians to mammals, characterised by three evolutionary trends: open to closed systems, single to double circulation, and incomplete to complete separation of oxygenated and deoxygenated blood; they also confirmed that the mammalian four-chambered heart with full double circulation is what enables elite metabolic performance such as that of Eliud Kipchoge.
How does this explain the phenomenon?	Learners explained why Kipchoges heart sustains world-record speeds — the double circulation delivers fully oxygenated blood to muscles at maximum efficiency without dilution — and why a grasshoppers dorsal vessel cannot do the same, because haemolymph bathes organs directly and does not carry oxygen, making it sufficient only for the grasshoppers low metabolic demands.

LESSON 12 (80 minutes): Final Model Presentations & Celebratory Sensemaking: Answering the Driving Question	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners consolidate, present, and celebrate their full understanding of animal transport systems by delivering final cumulative model presentations and written explanations that directly answer the driving question about Eliud Kipchoge and the grasshopper.
Knowledge	Learners demonstrate understanding of: (a) the structural and functional features of open and closed circulatory systems; (b) the relationship between an organism's complexity, metabolic demand, and transport system design; (c) the structural adaptations of the mammalian heart that enable high-performance circulation; (d) how double circulation in mammals enables efficient separation of oxygenated and deoxygenated blood, supporting sustained aerobic activity and rapid recovery.
Skills	Learners are able to: (a) construct and present a revised cumulative model comparing transport systems across a grasshopper, tilapia, frog, and Eliud Kipchoge (mammal); (b) write a scientific explanation using evidence gathered across all 12 lessons; (c) evaluate peer models using a structured critique protocol; (d) identify questions that remain open for future investigation.
Attitudes	Learners appreciate the elegance of biological design, recognise that scientific understanding is built incrementally through evidence, and value the contributions of all learners in a collaborative knowledge-building community.
Key Inquiry Question	1. Why is transport important in animals? 2. How does the structure of the transport system relate to the complexity of an animal?
Purpose in Storyline	This is the capstone lesson of the 12-lesson unit. Learners have collected evidence through pulse-rate experiments, breathing-rate observations, capillary-refill tests, dissections, diagrams, and readings. They now synthesise ALL of that evidence into a final model and written explanation that answers the driving question: 'How does the structure of an animal's transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper?' The DQB is reviewed, the anchoring phenomenon is revisited, and the class celebrates 12 lessons of collective sensemaking.
Safety Notes	No hazardous materials in this lesson. Ensure psychological safety during presentations — remind learners that all models are valid works-in-progress and critique must be kind, specific, and helpful. If any learner is uncomfortable presenting verbally, allow a poster-walk format as an alternative.

B. LESSON OVERVIEW
In this final lesson of Sub-Strand 3.2, learners deliver their final cumulative model presentations, comparing their Lesson 1 initial models with their fully revised final models. Each group presents a visual model (poster, diagram, or annotated drawing) and a written Final Explanation that directly answers the driving question using evidence from all 12 lessons. The teacher facilitates structured peer feedback using a gallery-walk critique protocol. The class then gathers at the Driving Question Board (DQB) to confirm which questions have been answered and which remain open. The lesson closes with a celebratory whole-class sensemaking discussion that returns to Kipchoge and the anchoring phenomenon, crystallising the unit's big ideas. Kenya-relevant comparisons (grasshopper on the school fence, tilapia in Lake Victoria, Rift Valley frog, and elite Kenyan marathon runner) anchor every phase of the discussion.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** [Circulatory System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the very first image from Lesson 1: Eliud Kipchoge crossing the finish line and the split-screen heart-rate recovery data. They are also handed back their original Lesson 1 model drawings (kept by the teacher since Lesson 1). Working individually for 3 minutes, each learner silently reads their own Lesson 1 model and writes one sentence: 'The biggest thing I did NOT understand in Lesson 1 that I understand now is ___.' Groups then set up their final model posters around the room for the gallery walk. The contrast between their earliest thinking and their current understanding primes learners for reflective, evidence-rich presentations.	Teacher redistributes the original Lesson 1 model drawings collected at the start of the unit. Say: 'Before anyone presents, I want you to hold your very first attempt — the model you drew in Lesson 1, before we had done ANY experiments. Read it quietly. Don't judge it. Just notice.' Allow 2 minutes of silent reading. Then say: 'Now write one sentence on a sticky note: the single biggest thing you did NOT understand in Lesson 1 that you understand now. You have 3 minutes. Go.' [WAIT — 3 minutes of silent writing.] Cold-call 3 learners to share their sticky-note sentences. Record key phrases on the board (e.g., 'double circulation,' 'closed vs open system,' 'why heart rate matters'). Then say: 'These are the ideas you have BUILT over 12 lessons. Today you will show us the full picture. Please set your final model poster at your group's station around the room.'	STRATEGY — Backward Reveal / Anticipatory Contrast: By confronting learners with their Lesson 1 thinking BEFORE the final presentation, the teacher activates metacognitive awareness of learning progression. Research shows that making prior conceptions explicit before consolidation deepens retention and helps learners articulate conceptual change — a key feature of NGSS Practice 6 (Constructing Explanations).	Observable indicator: Each learner produces a sticky note with a substantive conceptual claim (not 'I learned a lot' but a specific idea like 'I now understand why a grasshopper doesn't need a pump because haemolymph just bathes the organs'). Teacher scans sticky notes during setup. Any learner who writes only a vague statement is quietly prompted: 'Can you name the specific concept — open system, double circulation, heart structure?'
Observe Phase 📹 **VIDEO:** [Circulatory System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups rotate around the room in a structured gallery walk (5 minutes per station, 3 stations visited). At each station, learners silently study the group's final cumulative model poster. They use a structured 'Two Stars and a Wondering' sticky-note protocol: they leave two sticky notes praising specific scientific accuracy ('Star: your model correctly shows that the grasshopper has no heart chambers — just a tubular pump') and one sticky note posing a genuine scientific wondering ('Wondering: your model shows double circulation for Kipchoge — but does the diagram show WHERE oxygenated and deoxygenated blood are	Distribute three sticky notes per learner (two yellow 'Star' notes, one blue 'Wondering' note). Say: 'You are not judging your classmates. You are reading their models as a scientist reads a diagram — looking for accuracy, completeness, and clarity. At each station ask yourself: Does this model show the difference between open and closed systems? Does it explain WHY Kipchoge recovers faster than a grasshopper could? Write your stars and wondering and leave them on the poster.' [WAIT — allow 5 minutes per station. Teacher circulates, noting which posters receive the most 'Wondering' notes about double	STRATEGY — Two Stars and a Wondering Gallery Walk: This is a structured peer-review protocol that mirrors scientific peer review (NGSS Practice 7 — Engaging in Argument from Evidence). By requiring learners to cite specific scientific content in their feedback ('your model correctly shows...'), the protocol elevates critique beyond opinion to evidence-based evaluation. The teacher's role is to monitor for scientific accuracy in the feedback itself.	Observable indicator: Star notes cite specific correct science (e.g., 'correctly shows haemolymph in open system,' 'diagram labels left and right ventricles correctly'). Wondering notes pose genuine mechanistic questions ('does your model explain how double circulation prevents mixing of oxygenated and deoxygenated blood?'). Teacher collects all sticky notes after the lesson for written formative evidence of individual understanding.

	separated?'). Learners are reading peer models as evidence: does this model accurately represent what the class has learned across 12 lessons?	circulation or heart structure — these become discussion anchors in Phase 3.] After rotations, say: 'Return to your own station. Read the sticky notes left on YOUR poster. You have 2 minutes to read them — then we begin presentations.'		
Explain Phase 📹 **VIDEO:** Circulatory System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group presents their final cumulative model (3–4 minutes per group) using a structured format: (1) Show Lesson 1 model vs. Final model side-by-side. (2) Explain what changed and WHY, citing specific evidence from at least three lessons (e.g., 'In Lesson 4 we measured pulse rate before and after star jumps and found that...; in Lesson 7 we studied the heart diagram and learned that...; in Lesson 9 we compared circulatory diagrams for the grasshopper, tilapia, frog, and human and found that...'). (3) Directly answer the driving question: 'Eliud Kipchoge can sustain world-record speeds and recover fast BECAUSE... A grasshopper cannot run a marathon BECAUSE...' After all groups present, each learner independently completes their individual Final Written Explanation on a structured template (10 minutes). The written explanation must: use the CER framework (Claim–Evidence–Reasoning); reference at least two organisms studied; cite at least two pieces of experimental evidence from the unit; answer BOTH key inquiry questions.	Before presentations begin, write the following sentence starters on the board: 'Our Lesson 1 model showed ___, but our final model shows ___ because we learned that ___.' / 'The evidence that most changed our thinking was ___.' / 'Kipchoge's circulatory system can do ___ which a grasshopper's system cannot, because ___.' Say: 'Every group must reference the anchoring phenomenon — Kipchoge and the grasshopper — somewhere in their presentation. You have 3–4 minutes. Begin.' [WAIT — allow each group to present fully without interruption.] After each presentation, teacher poses one targeted cold-call question to the presenting group AND one to the listening class. Example targeted questions: 'You showed double circulation — can you point to exactly where oxygenated and deoxygenated blood are separated in your diagram?' / 'Your model shows the grasshopper has an open system — what would happen to Kipchoge's muscles if HIS blood also just bathed his tissues instead of flowing through closed vessels?' / 'Your evidence from the pulse-rate experiment — what did that tell you specifically about how the mammalian heart responds to increased metabolic demand?' After all presentations, say: 'Now close your eyes for 30 seconds	STRATEGY — CER (Claim–Evidence–Reasoning) Written Explanation + L1-vs-Final Model Comparison: The side-by-side comparison of Lesson 1 and final models operationalises NGSS Practice 6 (Constructing Explanations) and makes conceptual change visible. The CER framework requires learners to link specific empirical evidence (pulse-rate data, capillary-refill observations, comparative anatomy diagrams) to their mechanistic claims, preventing vague generalisations. Oral presentation + individual written explanation provides two modalities of sensemaking, catching learners who express understanding verbally but struggle to write it (or vice versa).	Observable indicators for oral presentation: (1) Group correctly identifies at least one structural difference between open and closed systems. (2) Group correctly links double circulation to efficient oxygen delivery. (3) Group uses evidence from at least two named experiments/lessons. Observable indicators for written Final Explanation: (1) Claim directly answers the driving question. (2) Evidence cites specific data (e.g., 'our pulse rate increased from 72 bpm to 138 bpm after exercise, showing that the heart must pump harder to meet increased oxygen demand'). (3) Reasoning explains the mechanism connecting structure to function. Teacher collects all written Final Explanations as summative evidence.

		and think about YOUR answer to the driving question — in your own words. [30-second wait.] Now open your template and write your Final Explanation. You have 10 minutes. This is individual — no talking.' [WAIT — 10 minutes of silent individual writing.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Circulatory System Structure and Function - Overview]() Source: CK-12 [🔍 Search ARES for similar videos]() 📖 **HTML:** [Animal-like Protists (Flexbook)]() Source: CK-12 [🔍 Search ARES for similar readings]()	The full class gathers in front of the Driving Question Board, which has been displayed since Lesson 1 and updated across the unit. The teacher reads aloud each question card on the DQB. Learners vote (hands up or thumbs up/thumbs sideways/thumbs down) on whether each question is: ANSWERED (we have strong evidence), PARTIALLY ANSWERED (we have some evidence but gaps remain), or STILL OPEN (we did not get to this). For each ANSWERED question, a volunteer learner explains in one sentence what evidence answered it. For STILL OPEN questions, learners discuss: 'Is this something Biology can answer? Where would you go to find out?' Learners also identify whether any new questions emerged from the presentations that were NOT on the original DQB. These are added in a different colour as 'New Questions Born from Our Learning.' The DQB thus becomes a living document of the class's collective scientific journey.	Point to the DQB and say: 'This board has been watching us since Lesson 1. Every question on it was asked by someone in THIS room. Let's find out which ones we can now answer.' Read each question card aloud. After each one, say: 'Thumbs up if you think we ANSWERED this with evidence. Thumbs sideways if we only partially answered it. Thumbs down if it's still open.' [WAIT 5 seconds for learners to show thumbs.] For thumbs-up questions, cold-call ONE learner: 'Tell me in one sentence what evidence answered this question.' For thumbs-down questions, say: 'Good — this one stays open. That is not failure. Unanswered questions are how science grows. Can anyone suggest WHERE you might look for the answer — a textbook, a hospital, a laboratory?' After reviewing all cards, say: 'Did anything in today's presentations GENERATE a new question that was NOT on our board?' [WAIT 10 seconds.] Accept 2–3 new questions, write them on cards of a different colour, and post them on the DQB. Say: 'These are the questions you are taking OUT of this classroom and into your future learning. That is exactly how science works.'	STRATEGY — DQB Audit with Thumbs Consensus + Evidence Citation: The DQB audit (adapted from the STEM Teaching Tools / NGSS storyline design) makes the structure of scientific knowledge visible: some questions are answered, some partially, some remain open. Requiring learners to cite evidence (not just say 'yes we learned it') forces them to connect closure of a question to specific empirical work done in the unit. Adding new questions born from learning communicates the epistemic value of productive uncertainty.	Observable indicator: Learners can correctly classify DQB questions as answered/partial/open AND can cite specific evidence for at least one answered question. Teacher listens for misconceptions during the DQB audit — if a learner incorrectly classifies a question (e.g., claims 'why does haemolymph not use haemoglobin' is fully answered when the class only touched on it), teacher probes: 'What specific evidence did we collect that answers that? Or is there a gap?' This reveals depth of understanding vs. surface recall.
Model Building	The teacher displays a large blank 'Class Consensus Model' sheet at	Unroll the large Class Consensus Model sheet. Say: 'We are going to	STRATEGY — Live Consensus Model Construction + Organism	Observable indicators: (1) Consensus model elements added

Phase 📹 **VIDEO:** [Circulatory System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	the front — a four-column poster with headers: GRASSHOPPER | TILAPIA (Lake Victoria) | FROG (Rift Valley wetland) | ELIUD KIPCHOGE (Mammal). Learners contribute one element each (called up in rapid succession) to build the class consensus model live, adding labels, arrows, and annotations. When the model is complete, the teacher reads the driving question aloud one final time: 'How does the structure of an animal's transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper?' The class answers together in a 'choral response' — but with a twist: the teacher asks each table group to answer on behalf of ONE organism. Table 1 speaks as the grasshopper, Table 2 as the tilapia, Table 3 as the frog, Table 4 as Kipchoge. Each group delivers a 30-second 'voice' statement: 'I am the grasshopper. My transport system works for ME because... but I could never run a marathon because...' The lesson ends with the teacher acknowledging the collective work of the class, naming the scientific practices they have used across 12 lessons, and previewing how Sub-Strand 3.3 (Gaseous Exchange and Respiration) will deepen their understanding of what the circulatory system is ultimately DELIVERING — oxygen — to every cell.	build one final model together — the whole class. I will call learners up one at a time. When I call you, you add ONE thing to the model: a label, an arrow, a word, a number. It must be something we learned from evidence — not something you are guessing.' [Rapid-fire cold-call — call up 12–16 learners in succession, each adding one element. Aim for the model to include: open vs closed system labels, haemolymph vs blood labels, single/double circulation arrows, heart chamber counts for each organism, oxygen delivery pathway, Kipchoge's resting heart rate (37 bpm) vs recovery curve.] Once the model is complete, step back and say: 'Look at what this class built. Every single piece of this model came from evidence YOU collected or analysed.' [Pause 5 seconds.] Then say: 'Now — I am going to read the driving question one last time. [Read it slowly and clearly.] Each table, you are going to answer it — but you are going to speak as your organism. Table 1, you are the grasshopper. Table 2, you are the tilapia from Lake Victoria. Table 3, you are the frog from the Rift Valley wetland. Table 4, you are Eliud Kipchoge. You have 90 seconds to prepare your 30-second voice statement. Go.' [WAIT 90 seconds.] Call each table in order. After all four, say: 'And THAT is the answer to the question we have been chasing for 12 lessons. Well done.' Close by saying: 'In our next sub-strand — Gaseous Exchange and Respiration — we will ask: once the blood arrives at the cell carrying all that oxygen, what happens next? The story is not	Voice Role-Play: The rapid-fire consensus model build (NGSS Practice 2 — Developing and Using Models) makes collective knowledge visible and co-owned. No single learner 'owns' the final model; it belongs to the class. The organism voice role-play is a perspective-taking strategy that requires learners to inhabit the biological logic of each system — a form of mechanistic reasoning (NGSS Practice 6) that tests deep understanding. A learner who truly understands open circulation can speak convincingly as a grasshopper; one who has only memorised terms will struggle. This also creates an emotionally memorable close to the unit, increasing retention.	by individual learners are scientifically accurate (teacher corrects in real time if not, turning error into a teaching moment: 'Almost — who can help us fix that label?'). (2) Organism voice statements include a mechanistic explanation ('I am the grasshopper — my haemolymph doesn't carry oxygen in blood vessels; it just surrounds my organs, which is fine because my cells are close enough to tracheal tubes that get oxygen directly — but this means I could NEVER sustain the oxygen demand of a marathon'). (3) Written Final Explanations (collected in Phase 3) are reviewed after class for CER quality — teacher provides written feedback within 2 lessons for use in Sub-Strand 3.3 reflection.

		over. It is just getting deeper.'		

D. TEACHER REFLECTION
After this final lesson, reflect on the following questions before moving to Sub-Strand 3.3: 1. DRIVING QUESTION RESOLUTION: Did the majority of learners directly and accurately answer the driving question in their Final Written Explanation? Were they able to link Kipchoge's double circulation and four-chambered heart to his performance and recovery, and contrast it with the grasshopper's open system? If learners struggled with the mechanistic reasoning (WHY double circulation enables efficiency), plan a brief 10-minute bridge activity at the start of Sub-Strand 3.3 that revisits this before introducing gaseous exchange. 2. DQB AUDIT OUTCOMES: Which questions on the DQB remained OPEN or PARTIALLY ANSWERED? These are your diagnostic signals for gaps in the unit. Common gaps at the end of this unit include: (a) the role of haemoglobin in oxygen transport (only partially addressed in this sub-strand — will be deepened in 3.3); (b) why resting heart rate varies between athletes and non-athletes (requires understanding of stroke volume, which is beyond Grade 10 but worth naming as an 'open question for future learning'); (c) the mechanism of the heartbeat (sinoatrial node, electrical signals — flagged as an extension concept). 3. MODEL PROGRESSION: Compare the Lesson 1 models with the final models. Did learners' models show genuine conceptual development — moving from naive 'blood just moves around' models to mechanistic models that include: open vs. closed system distinction, single vs. double circulation, heart chamber structure, and linkage to organism complexity and metabolic demand? If a significant number of final models still show fundamental misconceptions (e.g., showing a grasshopper with a four-chambered heart, or not distinguishing oxygenated from deoxygenated blood), these must be addressed before Sub-Strand 3.3. 4. KENYA CONTEXT ENGAGEMENT: Did the use of Eliud Kipchoge, the Lake Victoria tilapia, and the Rift Valley frog increase learner engagement and relevance? Note any moments where learners made spontaneous local connections (e.g., a learner mentioning a relative who farms tilapia in Kisumu, or connecting Kipchoge's recovery rate to what they observed in their own pulse-rate data). These are high-value contextual anchors — record them and use them as openers in Sub-Strand 3.3. 5. INDIVIDUAL EQUITY CHECK: Review the written Final Explanations for learners who were quiet during oral presentations. Written evidence often surfaces understanding (or misconceptions) that oral participation masks. Identify any learners whose written CER is weak and schedule a brief individual check-in before the next sub-strand begins. 6. BRIDGE TO SUB-STRAND 3.3: The natural conceptual bridge is: 'The circulatory system delivers oxygen — but how does oxygen GET INTO the blood in the first place? And what happens when it arrives at the cell?' This sets up gaseous exchange (at the lung/gill/trachea interface) and cellular respiration as the next layer of the same story. Consider starting Lesson 1 of Sub-Strand 3.3 by returning to the Kipchoge phenomenon: 'We know HIS heart pumps efficiently. Today we ask — where does the oxygen his blood is carrying actually come from, and what does his body do with it?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed their own Lesson 1 model drawings (redistributed by teacher) alongside their final cumulative models during the gallery walk, noting specific structural and conceptual changes. They also observed peer group final models at three gallery-walk stations, reading diagrams comparing open and closed circulatory systems across the grasshopper, tilapia, frog, and human (Kipchoge). The class observed the live-built Class Consensus Model grow element-by-element as individual learners contributed one evidence-based label or arrow each.

What did I learn?	Learners synthesised and consolidated the following key understandings from across all 12 lessons: (1) Transport systems vary in structure — open systems (grasshopper) use haemolymph that bathes organs directly and require no closed vessels; closed systems (fish, frog, mammal) use blood confined to vessels, enabling higher pressure, faster delivery, and finer control. (2) Complexity of transport system correlates with metabolic demand — more active, larger, or more complex animals require more efficient delivery of oxygen and nutrients and removal of waste products. (3) Double circulation (mammals) separates pulmonary and systemic circuits, ensuring fully oxygenated blood reaches the body's tissues — the structural basis of Kipchoge's ability to sustain high-intensity effort and recover rapidly. (4) The four-chambered mammalian heart is a structural adaptation that prevents mixing of oxygenated and deoxygenated blood, enabling maximum oxygen delivery per heartbeat. (5) Evidence from pulse-rate experiments, breathing-rate observations, and capillary-refill tests all point to the same underlying mechanism: the mammalian circulatory system is tuned to rapidly match oxygen supply to metabolic demand — which is why Kipchoge's heart rate drops back to 37 bpm so quickly after a race.
How does this explain the phenomenon?	By the end of this lesson, learners can explain, using a CER framework and a cumulative visual model, that: the structure of an animal's transport system directly determines its ability to meet metabolic demands and therefore survive and thrive in its environment. A grasshopper's open circulatory system is efficient for a small, low-metabolic-demand organism — haemolymph does not carry oxygen (which is delivered directly to cells via tracheae), so no high-pressure pump is needed. If Eliud Kipchoge had the circulatory system of a grasshopper, he could not deliver enough oxygenated blood to his leg muscles at marathon pace — his muscles would fail within seconds of any intense effort. His four-chambered heart, double circulation, and haemoglobin-rich blood constitute a system capable of delivering 20+ litres of oxygenated blood per minute under race conditions, then rapidly clearing metabolic byproducts and restoring resting conditions — a feat no open circulatory system could achieve.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.